

Evaluation of Vehicle Understeer Gradient Definitions

Thesis

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By

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ABSTRACT

The aim of this thesis is to study current methodologies for evaluating the understeer gradient. The SAE J266 standard gives the typical procedures to calculate the understeer gradient. The goal was to verify certain assumptions made in the formulation of the standard and to investigate differences in the J266 test methods to help predict the understeer gradient accurately. The procedures specified in this standard have been used to calculate the understeer gradients for the experimental and simulation vehicles in different situations.

Field tests for understeer gradient calculations were carried out at the Transportation Research Center (TRC). The test vehicle used was Subaru Outback. As it is very important to validate the results obtained from experimental testing, vehicle dynamics software CarSim was used to simulate these tests. A vehicle model for the 2003 Ford Expedition was built in CarSim and validated using certain quasi-static and dynamic maneuvers. This model was used for simulation of understeer gradient tests in CarSim. This model was selected as the Expedition falls in the same class of vehicles as the Outback and a well validated vehicle model gives us confidence in the simulation results. The analysis was made for the standard understeer measurement methods: namely the constant radius method, constant steer method and the constant speed method. The results

obtained through simulation were compared with the results from actual test data. The effect of steering and suspension compliances and kinematics on understeer gradient calculation has also been studied with the help of a modified Expedition model with the steering and suspension compliances and kinematics set to zero.

The discussion about the theoretical calculation of understeer gradient and the importance of different contributing factors is made using the Ford Expedition as an example. The resulting trends and conclusions from this investigation are discussed in the last chapter. The future work that could be done to take this research further is also discussed.

DEDICATION

Dedicated to my parents and family,
for their unconditional love and support and faith in me

To my friends;
for their support and guidance in all my endeavors and for making tough times easier

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I would like to express my gratitude towards all those who made this thesis possible. I want to take this opportunity to thank my advisors Dr. Dennis A. Guenther and Dr. Gary J. Heydinger for their valuable guidance. They were a source of constant encouragement and support. I especially thank Denny for giving me complete freedom in choosing my classes and giving me the opportunity to explore a variety of topics.

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CHAPTER 1

1 INTRODUCTION

1.1 Motivation

According to the Fatality Analysis Reporting System (FARS), over 3000 billion vehicle miles were travelled in the United States in the year 2007. The number of police reported motor vehicle crashes that occurred in the United States in the year 2007 was over 6 million. Out of these crashes, 30 percent (1.71 million) resulted in an injury and less than 1 percent (37,248) were fatal. Amongst the fatal crashes, 59 percent involved only one vehicle [7]. Given the high number of single vehicle crashes, it is very important to analyze the handling characteristics of vehicles.

The research and development in the field of vehicle safety systems has made the study of vehicle stability very important. Understeer gradient has been an important metric in analyzing vehicle handling and stability issues.

NHTSA (National Highway Traffic Safety Administration) has conducted some studies which relate the probability of loss of control accidents with the change in understeer gradient [6]. In a study related to tire failures, it was observed that loss of control is very strongly linked with reduction in vehicle understeer gradient. The research

also discussed the limiting conditions on vehicle handling and stability in terms of speed and steer angles. The speed steer boundaries were seen to increase with an increase in understeer gradient. Thus, understanding the trends and effects of the understeer gradient is crucial in analyzing vehicle stability.

The focus of this thesis is to understand the effects of different parameters on vehicle understeer gradient. The standard test procedures for the calculation of the understeer gradient have been used for the purpose of analysis. These test procedures have been specified in the ‘Steady-State Directional Control Test Procedures for Passenger Cars and Light Trucks’ of J266 standard as the constant radius test, constant steer test and constant speed test. The comparison of results from these standard tests gives information about changes in understeer gradient according to various inputs.

1.2 Simulation with CarSim

The role of simulation in vehicle dynamics has become more important with the development of powerful computational techniques which give reliable and realistic results. Actual testing of vehicles typically involves high cost and time. Simulation can give quick results which can be used to guide test programs and validate experimental results. A commercially available vehicle dynamics software package, CarSim, was extensively used for this research. CarSim is a vehicle dynamics software which can simulate the dynamic behavior of racecars, passenger cars, light trucks, and utility vehicles. CarSim vehicle model for a 2003 Ford Expedition and was used in this work to simulate the understeer tests.

CarSim has a Graphical User Interface which can be used to give inputs, specify test procedures and model vehicle parameters. Figure (1) shows a screen shot of the CarSim Graphical User Interface.

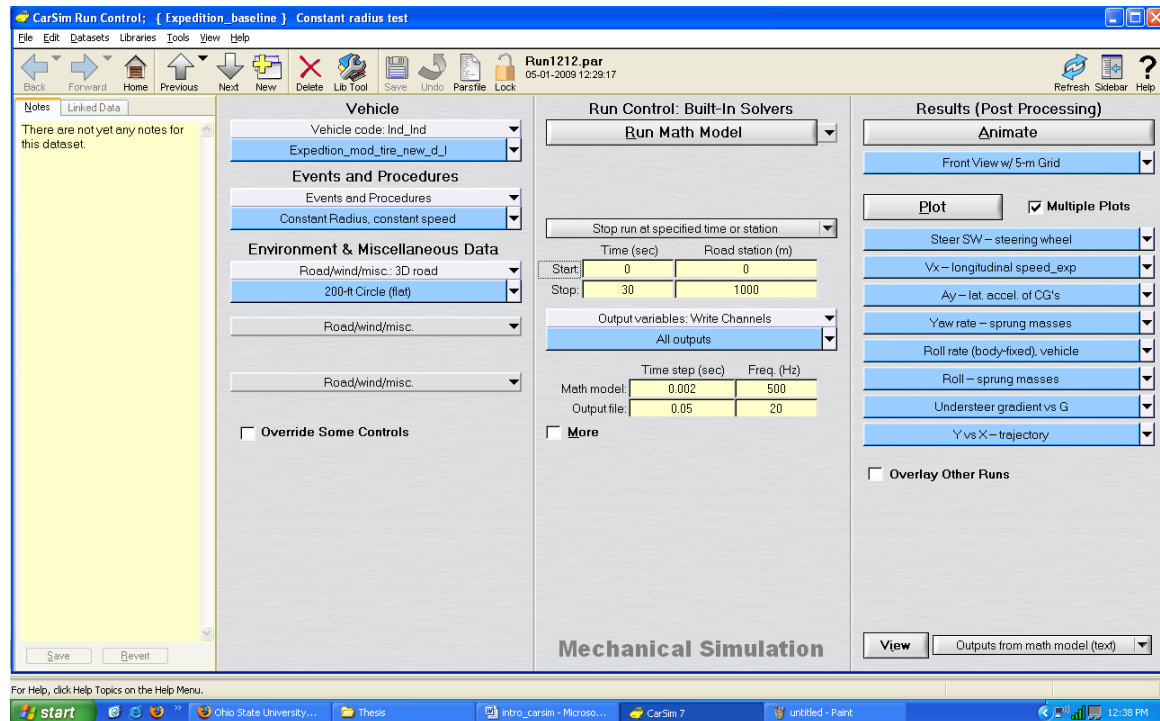


Figure 1: CarSim Screen Shot

CarSim math models:

CarSim has three main control inputs. They are steering, braking and speed. Steering, braking, throttle, gear shifting and speed profiles can be specified with open loop or closed loop control.

Environmental Inputs:

Three dimensional (3D) road surfaces can be defined and modeled in CarSim by specifying the horizontal, vertical, cross- elevation geometry and other required details. Friction between tire and ground can also be specified. CarSim also includes aerodynamic and wind inputs.

3D Vehicle Dynamics:

Full nonlinear 3D motions of rigid bodies can be analyzed with CarSim math models. Nonlinear tables of measurable data are used to describe detailed nonlinear tire models, nonlinear spring models and other nonlinear component models.

Tire Models:

The shear forces and moments applied to the tires from the 3D road surface can be reproduced in CarSim. CarSim tire models have the capability to reproduce full nonlinear behavior of rolling tires.

Import and Export Variables:

Arrays of import and export variables are used to communicate with other software. CarSim 7 models have over 230 potential import variables and over 600 available output variables.

MATLAB/Simulink:

Simulink provides a graphical user interface for building math models as block diagrams. The CarSim vehicle model is stored as a Simulink S-Function. Inputs for the CarSim simulation can be given through MATLAB/Simulink. Also, the output files from CarSim simulations can be converted into MATLAB data files.

1.3 Experimental Analysis

An important part of this thesis is the experimental analysis of the understeer gradient and comparison of trends with simulation results. For this purpose, field tests were carried out at the Transportation Research Center (TRC) [8]. The data collected from these tests was used for analysis and calculation of the understeer gradient. The test vehicle used was a 2006 Subaru Outback.

1.4 Thesis Outline

Chapter 1 contains the introduction and the outline of the thesis and discusses the motivation behind the research. Chapter 2 explains the theory behind development of understeer gradient starting with the fundamentals of dynamics. Chapter 3 focuses on the use of simulation in the research. Modeling and validation of a 2003 Ford Expedition is described in detail in this chapter. The first part of the chapter talks about building a

vehicle model in CarSim. The validation of the vehicle model using standard static and dynamic tests has been explained in the second part of this chapter.

Chapter 4 discusses the results from the CarSim simulations for the understeer gradient. The 2003 Ford Expedition which is an SUV similar to the 2006 Subaru Outback was chosen as the test vehicle. The well validated CarSim model for the 2003 Ford Expedition from chapter 3 was used for these tests.

Chapter 5 discusses the results from the experimental tests for understeer gradient that were carried out at TRC. The test vehicle used for this purpose was a 2006 Subaru Outback.

The discussion of the results from the understeer tests is included in Chapter 6. Chapter 7 discusses the conclusions drawn from the results of the research work. Future work that can be done in this area has also been discussed.

CHAPTER 2

2 UNDERSTEER GRADIENT THEORY

2.1 Overview

The understeer gradient has been an important performance metric in analyzing the handling behavior of vehicles. A study in vehicle handling comprises analyzing the controllability of the vehicle. Thus, it is important to analyze the dynamic response of the vehicle system to disturbances.

To study vehicle handling and stability we have to consider that the driver and the automobile system are inseparably connected in a closed loop system. The action of the driver to get desired motion from the vehicle is a consequence of environmental inputs, driver motive, and feedback from the automobile system.

2.2 Bicycle Model

To analyze and understand handling behavior, a very simplified vehicle model is used. The very basic representation of the vehicle is in the form of a model known as the ‘bicycle model’. In this model, longitudinal and lateral load transfer and rolling and pitching motions of the vehicle are neglected. The aerodynamic and suspension

compliance effects are also neglected. Also, the simplified model considers only linear range of tire response which works well for studying the normal handling behavior.

The degrees of freedom for this simplified vehicle model are taken as lateral velocity ' v ' and the yaw velocity ' r '. The system input is taken as the front wheel steer angle ' δ '.

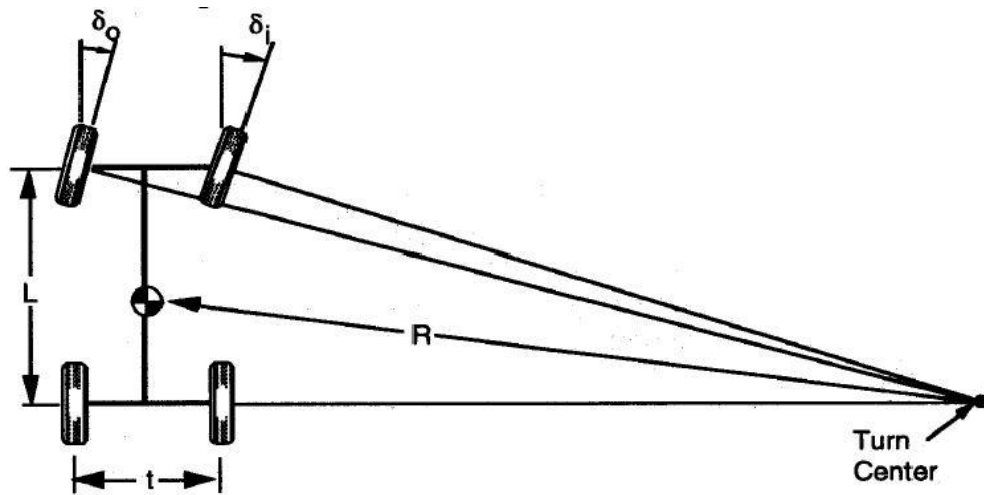


Figure 2: Slow Speed Cornering [1]

2.3 Very slow speed cornering

In the case of slow speed cornering, the lateral accelerations and slip angles are very low. If small angle approximations are used, the steering angle required for the turn is equal to the ratio of the wheelbase ' L ' and the radius of the turn ' R '. This is known as the Ackermann steering angle ' δ '.

$$\delta = \frac{L}{R} \quad (2.1)$$

2.4 High Speed Cornering

At higher speeds, the lateral acceleration during cornering increases and it cannot be neglected. This gives rise to slip angles at the wheels and the tires produce lateral forces in order to counteract the lateral acceleration.

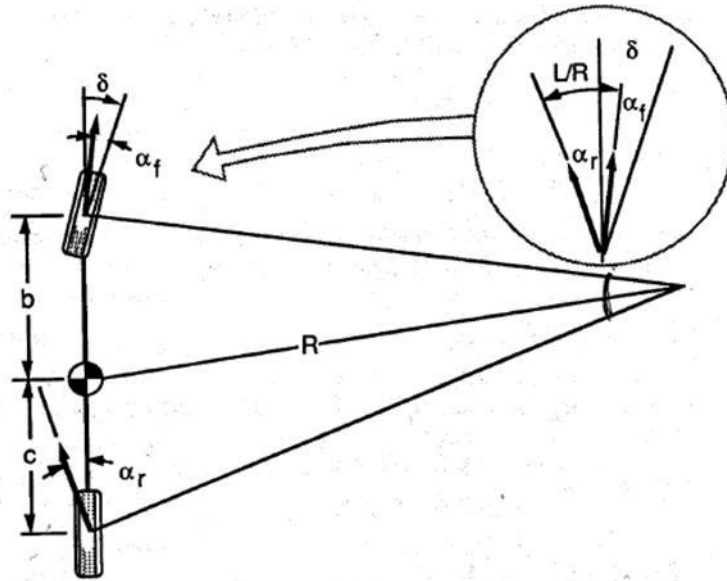


Figure 3: High Speed Cornering [1]

The slip angle can be defined as the angle between the direction of heading of the tire and its direction of travel (direction of the velocity vector). The cornering force and slip angle have a linear relationship at low slip angles.

$$\alpha = \frac{F_y}{C_\alpha} \quad (2.2)$$

where ‘ C_α ’ is the slope of the cornering force versus slip angle curve and is known as cornering stiffness.

Also, from the geometry of Figure (3),

$$\delta = \frac{L}{R} + \alpha_f - \alpha_r \quad (2.3)$$

Or

$$\delta = 57.3 \frac{L}{R} + \alpha_f - \alpha_r \text{ in degrees} \quad (2.4)$$

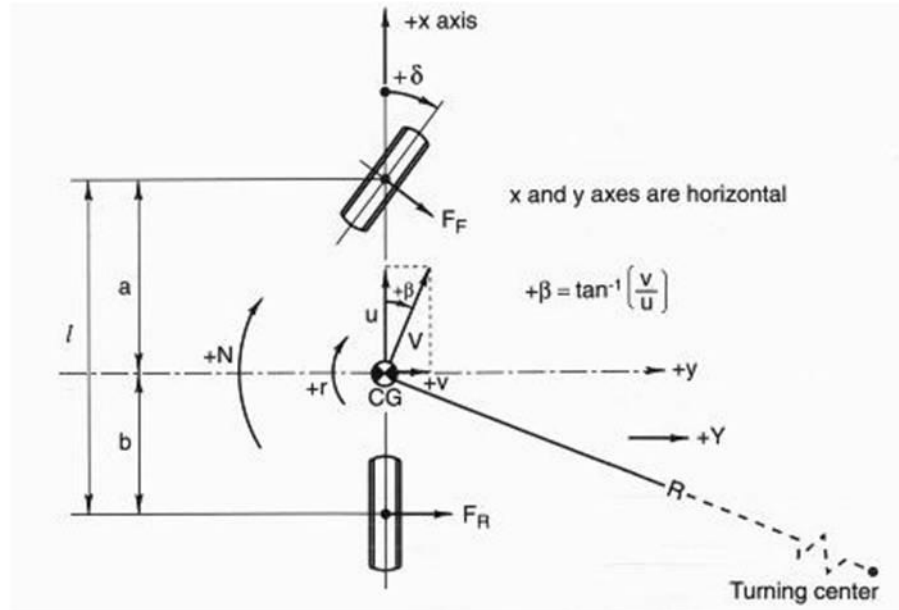


Figure 4: High Speed Cornering (bicycle model) [2]

Consider the vehicle negotiating a turn. Let ‘ V ’ be the total velocity, ‘ u ’ be the forward velocity component and ‘ v ’ the lateral velocity component. The total lateral acceleration is a resultant of the acceleration due to change in lateral velocity and the acceleration resulting from path curvature. If the center of gravity experiences a lateral velocity ‘ v ’, the front and the rear axles will also experience a lateral velocity equal to ‘ v ’. If there is a positive yaw velocity ‘ r ’ at the center of gravity, the lateral velocity at the front axle will be ‘ $a \cdot r$ ’ and that at the rear axle will be ‘ $-b \cdot r$ ’.

Assuming small angle approximation, the vehicle body slip angle is given by

$$\beta = \frac{v}{V}$$

(2.5)

The rear slip angle is

$$\alpha_R = \frac{v - br}{V} = \beta - \frac{br}{V} \quad (2.6)$$

The front slip angle is a resultant of the body slip angle ‘ β ’, yaw velocity ‘ r ’, and the steering angle ‘ δ ’

$$\alpha_F = \frac{v + ar}{V} - \delta = \beta + \frac{ar}{V} - \delta \quad (2.7)$$

During slow speed cornering, a neutral steer car will require the same steering angle equal to the Ackermann steering angle. An understeer vehicle will require more steering angle than the Ackermann steering angle and an oversteer vehicle will require less steering angle than the Ackermann.

From the tire cornering force characteristics, considering the linear region, we have,

$$F_y = C_\alpha \alpha \quad (2.8)$$

Where, C_α is the cornering stiffness

$$\alpha = \frac{F_y}{C_\alpha} \quad (2.9)$$

Therefore, load on front axle

$$W_f = \frac{Wb}{L} \quad (2.10)$$

Load on the rear axle

$$W_r = \frac{Wa}{L} \quad (2.11)$$

The sum of forces in the lateral direction gives

$$\sum F_y = F_{yf} + F_{yr} = M \frac{V^2}{R} \quad (2.12)$$

The moment equilibrium about the Z axis gives

$$F_{yf} a - F_{yr} b = 0 \quad (2.13)$$

Where, ‘a’ is the distance of the front axle from center of gravity and ‘b’ is the distance of the rear axle from the center of gravity.

Thus, solving equations (2.12) and (2.13) for the tire forces,

$$F_{yr} = \frac{W_r V^2}{gR} \quad (2.14)$$

$$F_{yf} = \frac{W_f V^2}{gR} \quad (2.15)$$

where $M = \frac{W}{g}$

Substituting $\alpha_f = \frac{F_{yf}}{C_{\alpha f}}$ and $\alpha_r = \frac{F_{yr}}{C_{\alpha r}}$ in equation (2.4),

$$\delta = 57.3 \frac{L}{R} + \frac{W_f V^2}{C_{\alpha f} g R} - \frac{W_r V^2}{C_{\alpha r} g R} \quad (2.16)$$

$$\delta = 57.3 \frac{L}{R} + K a_y \quad (2.17)$$

where ‘ K ’ is the understeer gradient.

$$K = \frac{W_f}{g C_{\alpha f}} - \frac{W_r}{g C_{\alpha r}} \quad (2.18)$$

where $a_y = \frac{V^2}{R}$

2.5 Understeer Gradient and Vehicle Stability

We have three conditions for understeer gradient ‘ K ’, positive, zero and negative. The vehicle is understeer when ‘ K ’ is positive, neutral steer when ‘ K ’ is zero and oversteer when ‘ K ’ is less than zero.

For a constant radius turn, for understeer condition, ‘ K ’ is positive and the required steer angle increases with increase in speed to negotiate the curve. The vehicle turns out of the curve. For a neutral steer condition, ‘ K ’ is zero and the steering angle required remains the same with increase in speed and is equal to the Ackermann steering angle. For an oversteer vehicle, ‘ K ’ is negative and the steering angle decreases with increasing speed. Therefore, the vehicle turns in to the curve.

Characteristic speed is defined for understeer vehicles. As the speed of the vehicle is increased, the steering angle required to negotiate the turn will increase. The speed at which this steer angle becomes twice the Ackermann steer angle is known as characteristic speed.

Therefore, from equation (2.17),

$$2(57.3 \frac{L}{R}) = 57.3 \frac{L}{R} + K a_y \quad (2.19)$$

$$\text{Therefore, } K \frac{V^2}{Rg} = 57.3 \frac{L}{R} \quad (2.20)$$

$$V = \sqrt{\frac{57.3 L g}{K}} \quad (2.21)$$

Equation (2.21) gives the characteristic speed.

Critical speed is defined for oversteer vehicles as the speed at which the vehicle becomes unstable. The speed at which the required steering input becomes zero is known as the critical speed. ‘ K ’ being negative for oversteer vehicles, the critical speed is given by the following expression

$$V = \sqrt{\frac{-57.3 L g}{K}} \quad (2.22)$$

Another measure of steady state handling behavior is the static margin. Static margin of a vehicle is defined as the distance between the neutral steer point and the center of gravity of the vehicle. The neutral steer point of the vehicle is a point at which the application of a lateral force produces no yaw velocity.

The static margin for a vehicle is positive if the neutral steer point is behind the center of gravity and negative if it is in front of the center of gravity. It is zero if the neutral steer point coincides with the center of gravity. The vehicle is understeer when the static margin is positive; oversteer when it is negative and neutral steer when it is zero.

2.6 Stability Analysis

According to the Routh's stability criterion, for stability of first and second order systems, the coefficient of the characteristic equation must be positive.

From the force equilibrium in the Y direction,

$$M \frac{V^2}{R} = F_{yf} + F_{yr} \quad (2.23)$$

From the moment equilibrium about the Z axis,

$$\sum M_z = I_z \dot{r} = aF_{yf} - bF_{yr} \quad (2.24)$$

But, the total lateral acceleration is $a_y = \dot{v} + rV$

Therefore, from equation (2.23),

$$M(\dot{v} + rV) = -C_f \alpha_f - C_r \alpha_r \quad (2.25)$$

Since $F_{yf} = -C_f \alpha_f$ $F_{yf} = -C_f \alpha_f$ and $F_{yr} = -C_r \alpha_r$

Substituting equations (2.6) and (2.7),

$$M(\dot{v} + rV) = -C_f\left(\frac{v + ar}{V} - \delta\right) - C_r\left(\frac{v - br}{V}\right) \quad (2.26)$$

From equation (2.24),

$$I_z \dot{r} = -aC_f\left(\frac{v + ar}{V} - \delta\right) + bC_r\left(\frac{v - br}{V}\right) \quad (2.27)$$

Since here the steering angle is the input, taking the δ terms on one side,

$$\begin{aligned} C_f \delta &= M(\dot{v} + rV) + C_f\left(\frac{v + ar}{V}\right) + C_r\left(\frac{v - br}{V}\right) \\ \therefore C_f \delta &= M(\dot{v} + rV) + (C_f + C_r)\frac{v}{V} + (aC_f - bC_r)\frac{r}{V} \end{aligned} \quad (2.28)$$

Also,

$$\begin{aligned} aC_f \delta &= I_z \dot{r} + aC_f\left(\frac{v + ar}{V}\right) - bC_r\left(\frac{v - br}{V}\right) \\ \therefore aC_f \delta &= I_z \dot{r} + (aC_f - bC_r)\frac{v}{V} + (a^2C_f + b^2C_r)\frac{r}{V} \end{aligned} \quad (2.29)$$

Using the Laplace operator ‘ s ’, Equations (2.28) and (2.29) can be rewritten in a system matrix form,

$$\begin{bmatrix} Ms + \frac{(C_f + C_r)}{V} & MV + \frac{(aC_f - bC_r)}{V} \\ \frac{(aC_f - bC_r)}{V} & I_z s + \frac{(a^2C_f + b^2C_r)}{V} \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} = \begin{bmatrix} C_f \\ C_r \end{bmatrix} \delta \quad (2.30)$$

The characteristic equation of the system can be analyzed to determine the stability of the system according to Routh's criterion. The characteristic equation of the system is the determinant of the above matrix which is

$$MI_z s^2 + \left[\frac{M(a^2 C_f + b^2 C_r)}{V} + I_z \frac{(C_f + C_r)}{V} \right] s + (a+b)^2 \frac{C_f C_r}{V^2} + M(bC_r - aC_f) = 0 \quad (2.31)$$

According to Routh's stability criterion, first and second order systems are stable if all the coefficients of the characteristic equation are positive. The coefficients of 's' and 's²' terms are always positive. Therefore, for stability,

$$(a+b)^2 \frac{C_f C_r}{V^2} + M(bC_r - aC_f) > 0 \quad (2.32)$$

If $bC_r > aC_f$, the stability criterion is always satisfied at varying speeds. This is true for an understeer vehicle.

If $bC_r < aC_f$, the second term in Equation (2.32) becomes negative. The limit of stability is when

$$(a+b)^2 \frac{C_f C_r}{V^2} + M(bC_r - aC_f) = 0 \quad (2.33)$$

This is true for oversteer vehicles.

The critical speed which is the speed at which vehicle loses stability and is therefore

$$V = \sqrt{\frac{(a+b)^2 C_f C_r}{M(aC_f - bC_r)}} \quad (2.34)$$

From Equation (2.22),

$$V = \sqrt{\frac{Lg}{K}} \text{ for } K \text{ in radians/g}$$

From Equation (2.18),

$$K = \frac{W_f}{gC_f} - \frac{W_r}{gC_r}$$

$$\text{But, } \frac{W_f}{g} = \frac{bM}{L} \text{ and } \frac{W_r}{g} = \frac{aM}{L} \text{ where } L = a + b$$

Therefore,

$$V = \sqrt{\frac{Lg}{\left(\frac{bM}{C_f L} - \frac{aM}{C_r L}\right)}}$$

Therefore,

$$V = \sqrt{\frac{L^2 C_f C_r}{M(bC_r - aC_f)}} \text{ which gives the same critical speed as Equation (2.34)}$$

2.7 Yaw Rate and Lateral Acceleration Response

The response of the system to the steer angle input can be analyzed through transfer function analysis. From Equation (2.30) using Cramer's rule, the lateral velocity function is obtained as,

$$\frac{v}{\delta}(s) = \frac{\begin{vmatrix} C_f & MV + \frac{(aC_f - bC_r)}{V} \\ aC_f & \frac{Izs + (a^2C_f + b^2C_r)}{V} \end{vmatrix}}{|C.E.}|} \quad (2.35)$$

The yaw rate transfer function is obtained as

$$\frac{r}{\delta}(s) = \frac{\begin{vmatrix} Ms + \frac{(C_f + C_r)}{V} & C_f \\ aC_f & \frac{I_z s + (a^2 C_f + b^2 C_r)}{V} \end{vmatrix}}{|C.E.}| \quad (2.36)$$

where C.E. is the characteristic equation.

The order of these transfer functions is

$$\frac{v}{\delta}(s) = \frac{1^{st} \text{ order}}{2^{nd} \text{ order}} \quad \text{and} \quad \frac{r}{\delta}(s) = \frac{1^{st} \text{ order}}{2^{nd} \text{ order}}$$

Thus, in both cases, the frequency response magnitude decreases as frequency of steering input increases. Also, in both cases, the phase angle approaches -90 degrees as frequency of steering input increases.

The lateral acceleration response can be found by combining the above transfer functions since

$$a_y = sv + rV \quad (2.37)$$

$$\therefore \frac{a_y}{\delta} = s \frac{v}{\delta} + \frac{r}{\delta} V \quad (2.38)$$

Substituting the values from equations (2.35) and (2.36),

$$\frac{a_y}{\delta}(s) = \frac{I_z C_f s^2 + \frac{(a+b)}{V} b C_f C_r s + (a+b) C_f C_r}{C.E.} \quad (2.39)$$

The order of this transfer function is

$$a_y(\delta) = \frac{2^{nd} \text{ order}}{2^{nd} \text{ order}}$$

The magnitude of lateral acceleration frequency response becomes constant and the phase angle approaches 0 degrees as frequency of steering input increases.

CHAPTER 3

3 MODELING AND VALIDATION OF 2003 FORD EXPEDITION USING CARSIM

3.1 Overview

The focus of this chapter is modeling of the 2003 Ford Expedition and its validation using CarSim. A vital component of this study is the use of simulation. The use of simulation in vehicle dynamics has numerous advantages. It reduces the time and cost of development and testing. It makes repetitive testing of vehicles possible by eliminating the human error in executing complex maneuvers and risk factor in potentially hazardous maneuvers. Simulations are often used to guide test programs.

The 2003 Ford Expedition was modeled in CarSim for the purpose of vehicle dynamic analysis. CarSim has a collection of a large number of datasets linked together. To model the 2003 Ford Expedition, dataset for a generic SUV was used with suitable modifications.

The weight, inertia, center of gravity location of the vehicle and the suspension and steering components was determined from measurements on an actual Expedition. These measurements were made at SEA, Ltd. using the Vehicle Inertia Measurement Facility (VIMF), in-house suspension testing equipment and other test equipments. The data from bounce and roll tests was used to determine the suspension and steering system

characteristics. The measured data was in accordance with the SAE coordinate system. It was suitably converted for use in the CarSim coordinate system. CarSim follows the SI (Système Internationale) coordinate system. In the SAE coordinate system, the X axis points forward, Y axis points to the right and the Z axis points downwards. In the CarSim coordinate system, X axis points forward, Y axis points to the left and the Z axis points upwards.

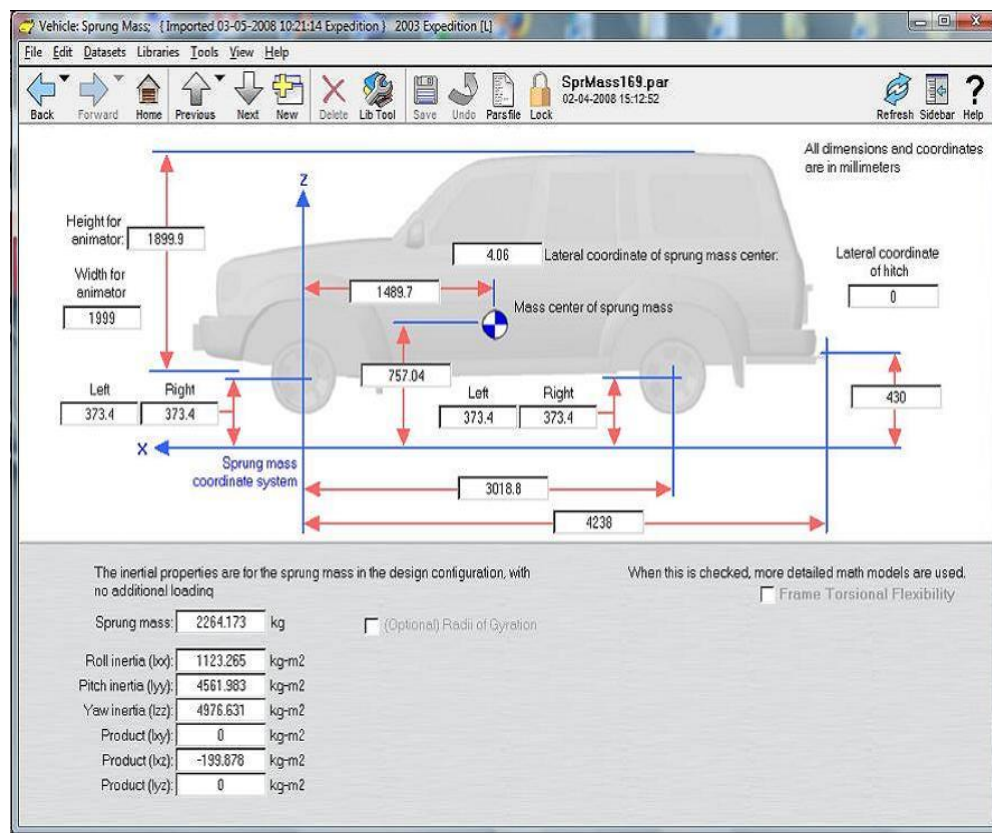


Figure 5: Vehicle Sprung Mass Model

Figure (5) is a screen shot of the CarSim sprung mass model for Expedition. The sprung mass, inertia, center of gravity location and other geometric parameters are specified here using the data obtained from parameter determination. The sprung mass properties are for unladen condition. The conversion from SAE to CarSim coordinate system results into a negative I_{xz} product of inertia.

3.2 Modeling

The experimental data was used to build a vehicle model for 2003 Ford Expedition in CarSim. The 2003 Ford Expedition XLT is a four door sports utility vehicle. It is equipped with a 5.4L V8 engine, automatic transmission and 4WD. It has Continental Contitrac SUV P265/70R17 113S M+S tires, the tire pressures being 35 psi for both front and rear tires.

The vehicle sprung mass model was built with the measured values for mass, inertias, center of gravity location, etc. The front and rear suspension model, steering system model and the tire model are also based on measured data. The measurements were made for components such as front and rear upper and lower control arms, front steering knuckle and rear knuckle, front and rear propeller shafts, rear lateral link, steering link.

For modeling the braking system and the powertrain, comparable generic models available in CarSim were used. A 5.0 L 238KW engine with a four wheel drive was selected for the 2003 Expedition powertrain. As braking is not an important part of this

study, the powertrain, brake and aerodynamic coefficients used were generic for a typical car of this size. The braking performance measured during field tests was matched by adjusting the ratios of brake torque to wheel cylinder pressure.

3.2.1 Suspension System

The CarSim model for the suspension system was created using measurement data. The 2003 Ford Expedition has short-arm long-arm coil over shock independent front suspension and stabilizer bar. The values for parameters like roll center height, track width, spin inertia, squat/lift ratio, compliance coefficients were used to build the model.

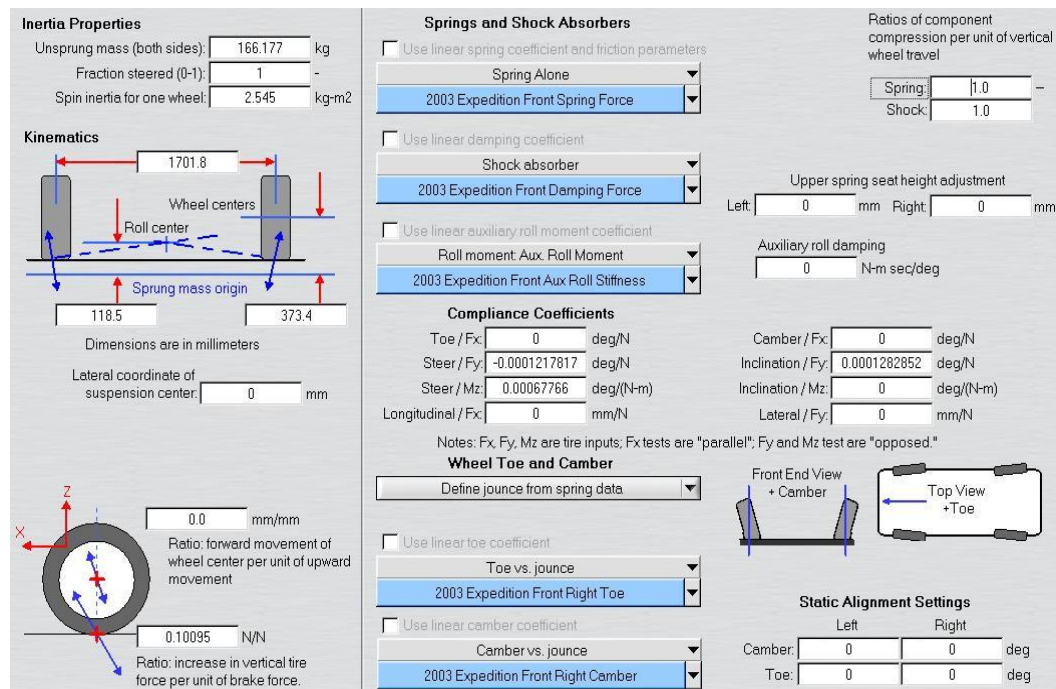


Figure 6: Front Suspension Model

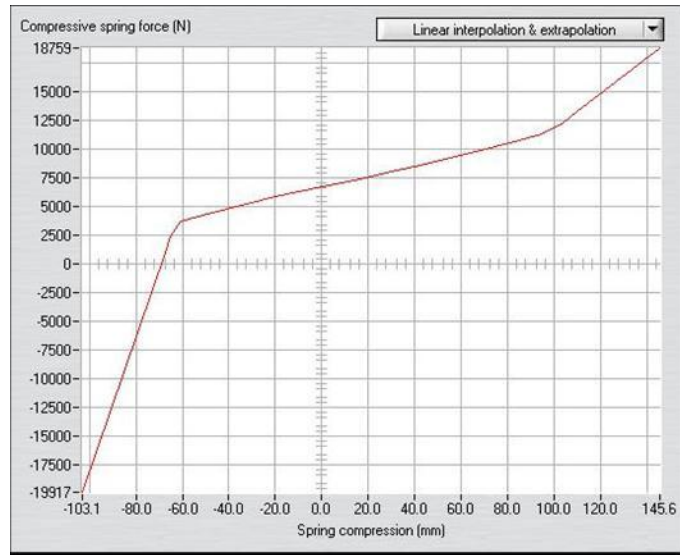


Figure 7: Front Suspension Spring Rate

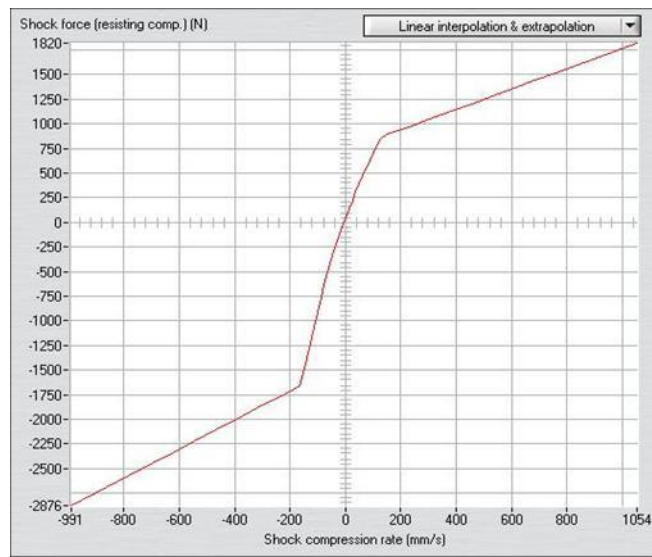


Figure 8: Front Damper Characteristics

Figure (6), Figure (7) and Figure (8) are screen shots from the CarSim suspension model. In CarSim, compression is considered positive while extension is considered negative.

The overall roll stiffness, suspension stiffness and tire stiffness measurement data was used to calculate the auxiliary roll stiffness. The auxiliary roll stiffness and the suspension stiffness are in parallel and their combination is in series with the tire stiffness. Equation 3.1 gives the formula used to calculate the auxiliary roll stiffness.

$$K_{aux} = \frac{K_{\phi}K_w \frac{T_s^2}{2} + K_{\phi}K_T \frac{T_r^2}{2} - K_wK_T T_s^2 \frac{T_r^2}{4}}{K_T \frac{T_r^2}{2} - K_{\phi}} \quad (3.1)$$

Where K_{ϕ} : Overall roll stiffness

K_w : Suspension stiffness

K_T : Tire stiffness

T_s : Suspension spring lateral distance

T_r : Track width

3.2.2 Steering System

An approximate linear relationship between handwheel and roadwheel steer angle was used with a nominal steering ratio of 19.7 deg/deg since tests showed that the linear relationship holds true for the range of $\pm 360^\circ$. Also, other parameters like caster angle, kingpin angle, lateral offset were modeled based on measurements. Steering compliances were calculated from the measured data and used in the model.

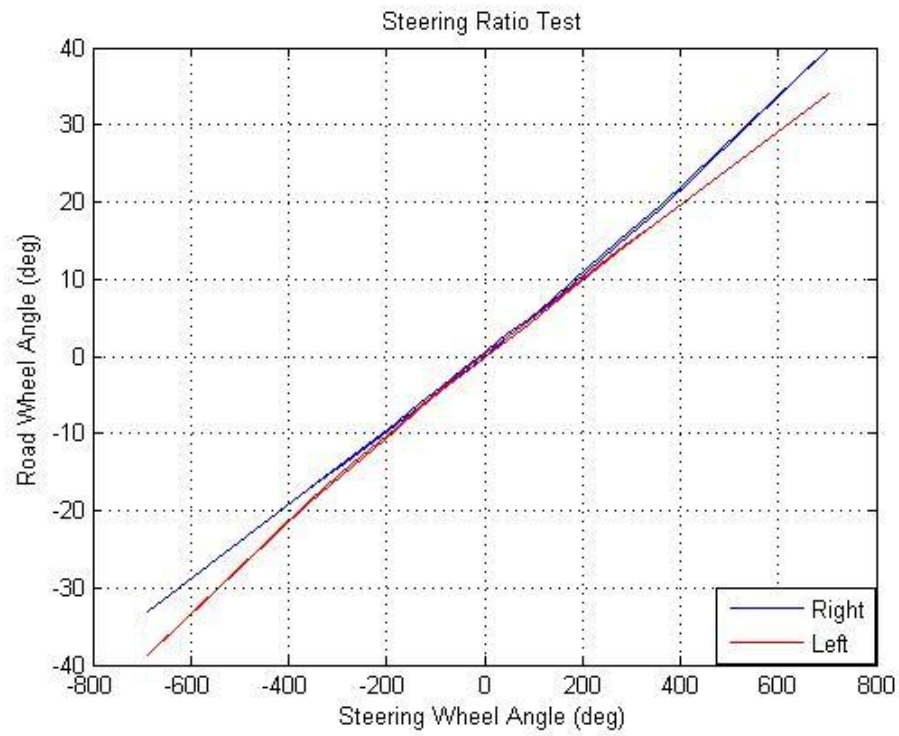


Figure 9: Steering Ratio Test

3.2.3 Tire Model

Experimental tire test data was used to model the tire characteristics in CarSim through the look up table feature. A generic value was used for the rolling resistance.

Vertical Force

☐ Use tire force table

Effective rolling radius: mm

Spring rate: N/mm

Maximum allowed force: N

Tire vertical force is proportional to the tire radial deflection, and characterised by a spring rate. The effective rolling radius relates forward speed to angular spin.

Animator Settings

Tire width: mm

The effective rolling radius is also used to scale the animated wheel.

☒ Use custom animator description

Animator: Shape Group

Sports Car

The animator description should be for a wheel on the left-hand side of the vehicle. (It is automatically flipped when used on the right-hand side.)

Rolling Resistance

$F_{x_rr} = F_z * R_{r_surf} * (R_{r_c} + R_{r_v} * V_x)$

Rr_c: -

Rr_v: h/km

The X, Y, Z axes define the tire/ground coordinate system

Shear Forces and Moments

Model Option

Internal Tire Model

Longitudinal force: Tire: Fx

Expedition P265S70R17 Fx

Lateral force: Tire: Fy

Expedition P265S70R17 Fy_new

Aligning moment: Tire: Mz

2003 Expedition P265S70R17 Mz

Camber thrust

2003 Expedition FyGama

Dynamic Properties

Tire spin moment of inertia (added to the spin inertia of the wheel): kg-m2

Tire Lag

Tire force or moment

L for Fx: mm

L for Fy and Mz: mm

The models use modified equations to simulate tire lag at speeds below this threshold.

Cut-off speed: km/h

Figure 10: Tire Model- Right Front Tire

The tire model always adds a row for zero slip and a column for zero load.
 Calculation of combined slip effects involves peak values at each Fz.
 The data must include peak values for results to be valid.

Tire/ground friction coefficient for this data: -

Figure 11: Lateral Tire Force versus Slip Angle

28

Figure (11) shows a plot of tire lateral forces versus slip angle for different vertical loads. The tire was tested for lateral slip angles up to 28 degrees. The model was to be used for simulating extreme maneuvers wherein the lateral slip angles would exceed this value. The use of the measurement data up to angles of 28 degrees caused an unrealistic decay in the lateral forces at higher slip angles. In order to represent the tire saturation, the lateral force at higher slip angles was maintained constant.

3.3 Model Validation

A well validated model is very important to ensure reliability of the results. It is also useful for studying the effects of new components or system modifications on vehicle performance. The accuracy of simulation depends upon the realism of the model. Hence, it is important to have a truly representative and well validated vehicle model.

Quasi static and dynamic tests were used in order to validate the model. Quasi static tests included Bounce and Roll tests and dynamic tests included ‘Slowly Increasing Steer’ and ‘Sine with Dwell’ tests.

3.3.1 Quasi Static Tests

The quasi static tests consisted of the bounce and roll tests. The standard suspension kinematics and compliance (K & C) tests available in CarSim were used to simulate the tests. The simulation results were compared with the experimental data. These tests were mainly used for validating the suspension and steering kinematics and compliances. The quasi static bounce and roll tests were conducted at SEA, Ltd [3, 4].

Bounce tests:

This test is performed by giving vertical motion to the chassis. The camber angle, steer angle, suspension deflections and tire vertical deflections were measured for different vertical loads. The parameters of interest are bounce camber characteristics, bounce steer characteristics, suspension spring rate and tire stiffness characteristics.

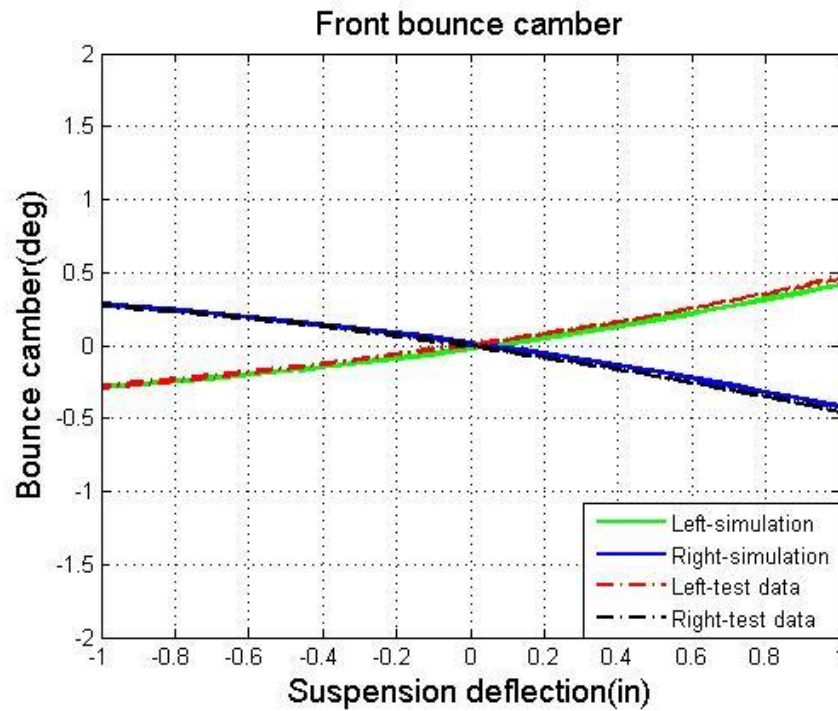


Figure 12: Front Bounce Camber Comparison

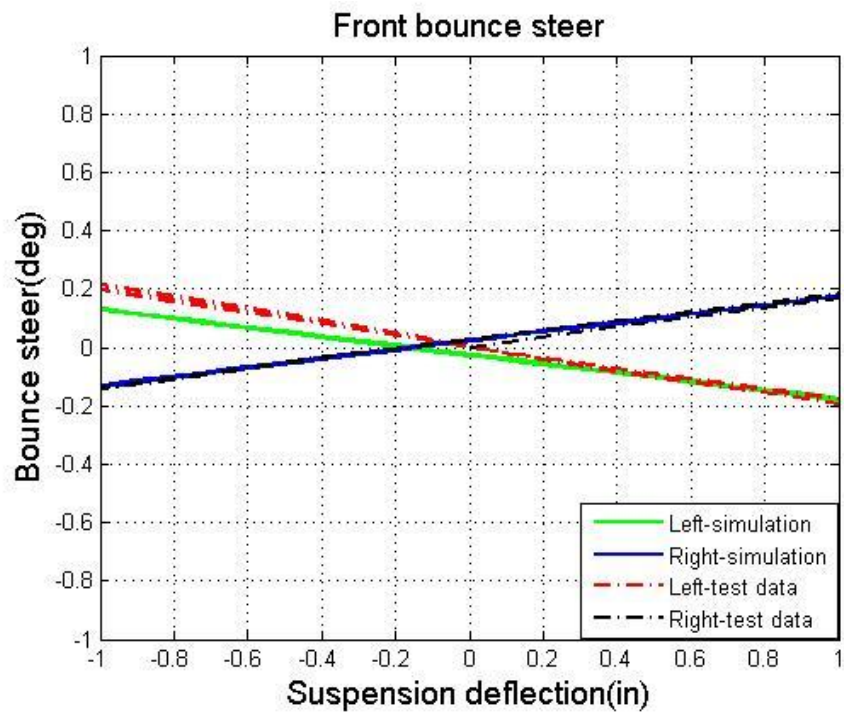


Figure 13: Front Bounce Steer Comparison

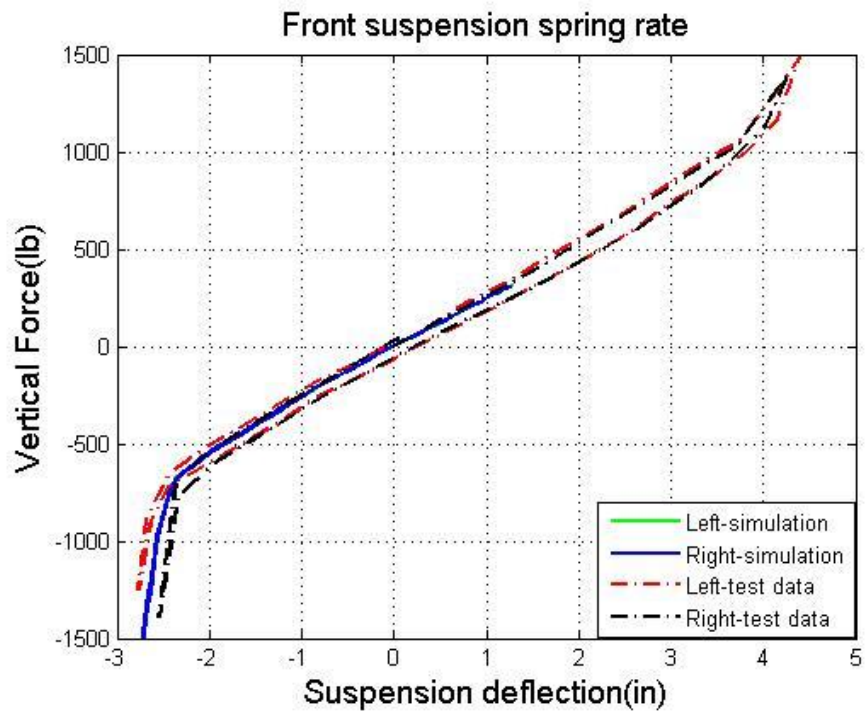


Figure 14: Front Suspension Spring Rate Comparison

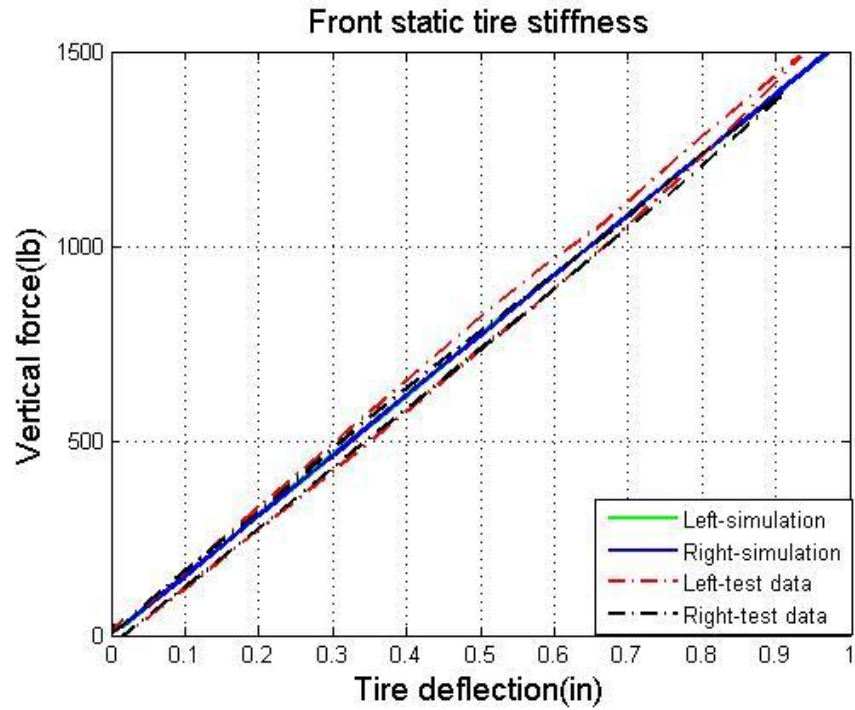


Figure 15: Front Static Tire Stiffness

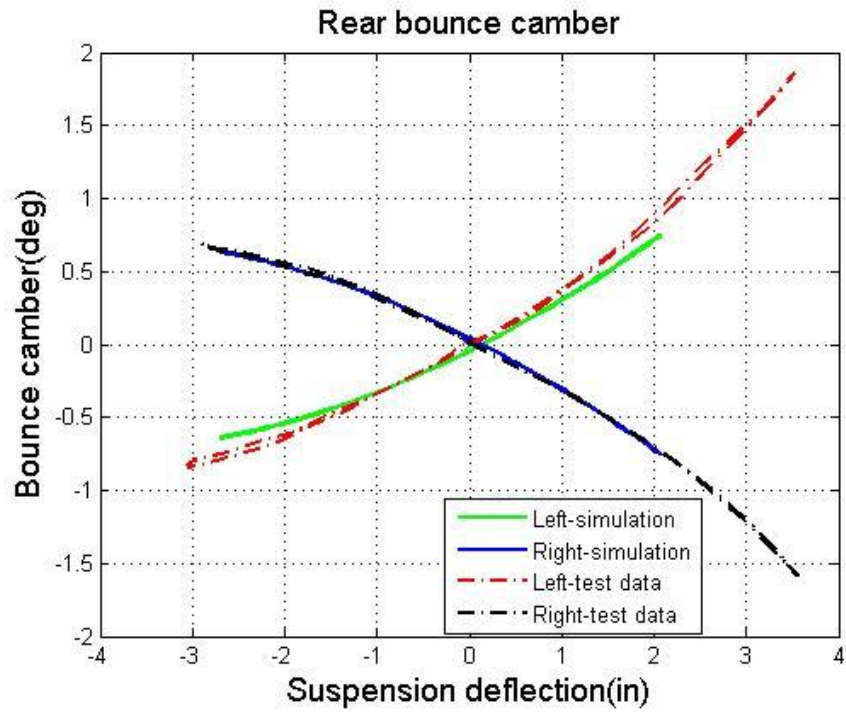


Figure 16: Rear Bounce Camber Comparison

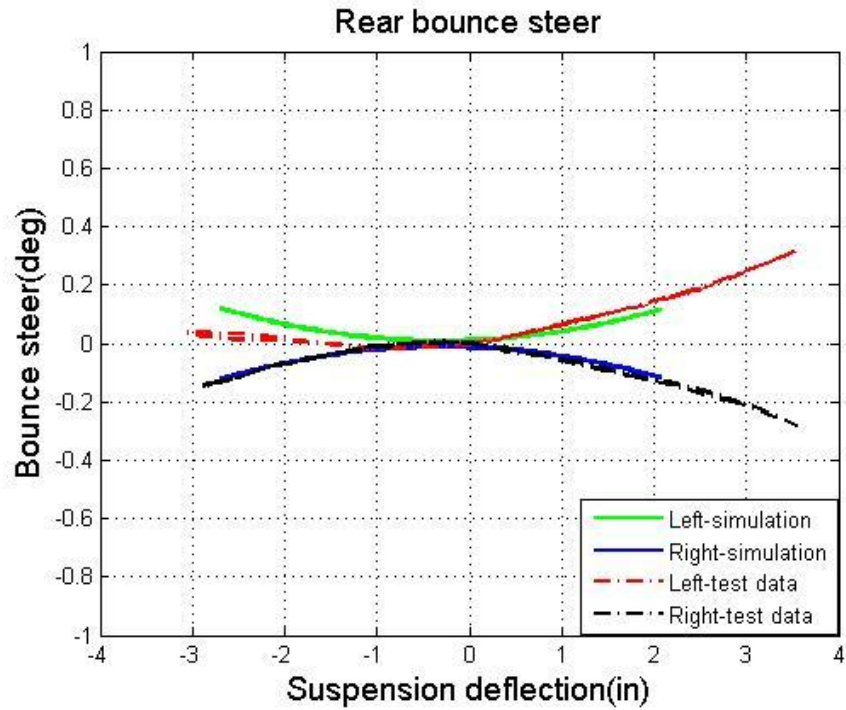


Figure 17: Rear Bounce Steer Comparison

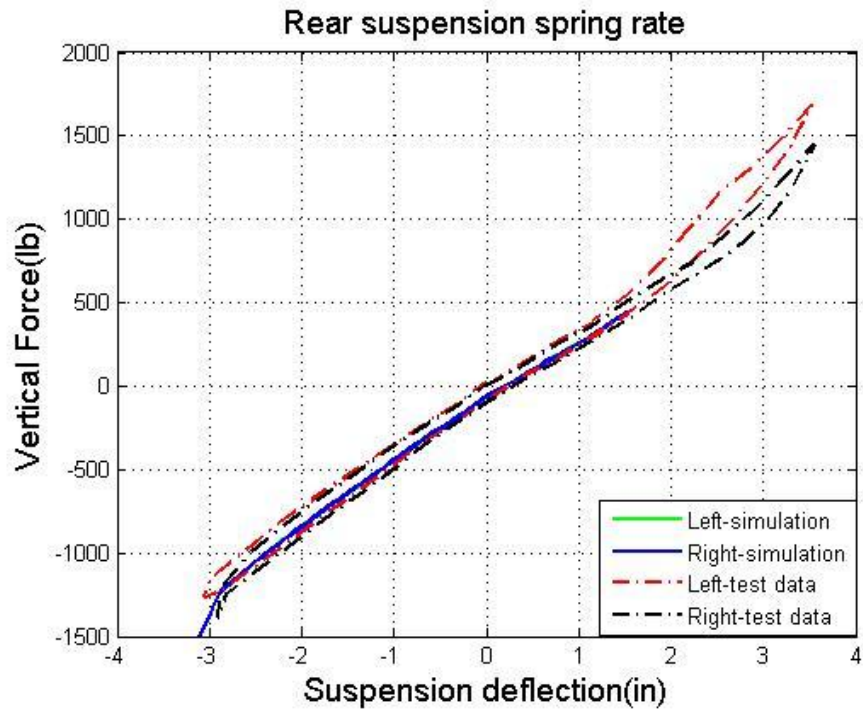


Figure 18: Rear Suspension Spring Rate Comparison

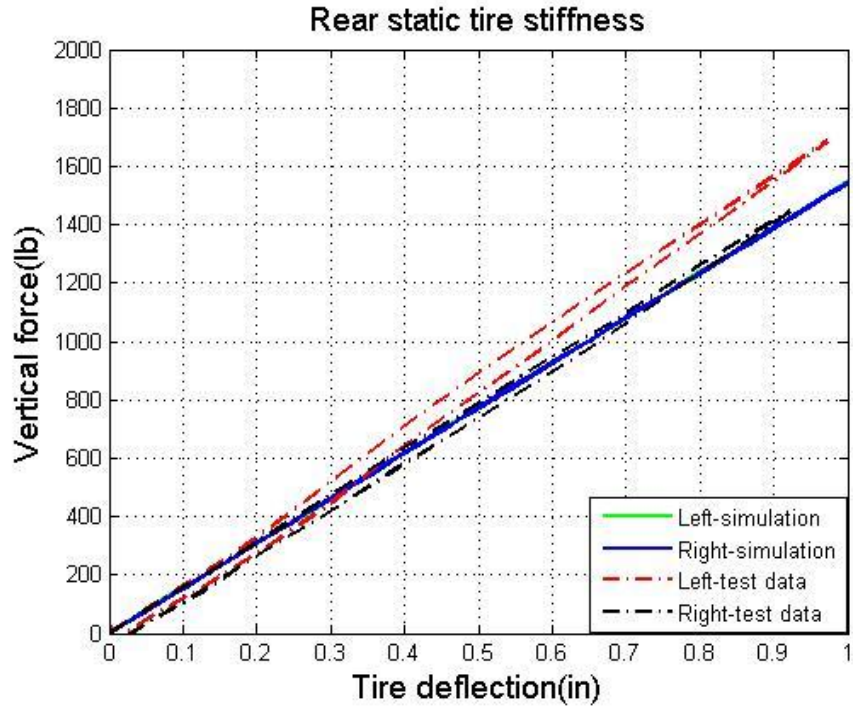


Figure 19: Rear Static Tire Stiffness Comparison

The front and rear bounce test characteristics for the simulation and experimental data were compared. The simulation results were seen to be in good agreement with the experimental data well.

Roll tests:

This test is performed by giving roll motion to the chassis. The response of the CarSim model was compared with the experimental data using parameters like camber angle, steer angle, roll moment and roll angle. The parameters of interest include roll camber characteristics, roll steer characteristics and overall roll stiffness.

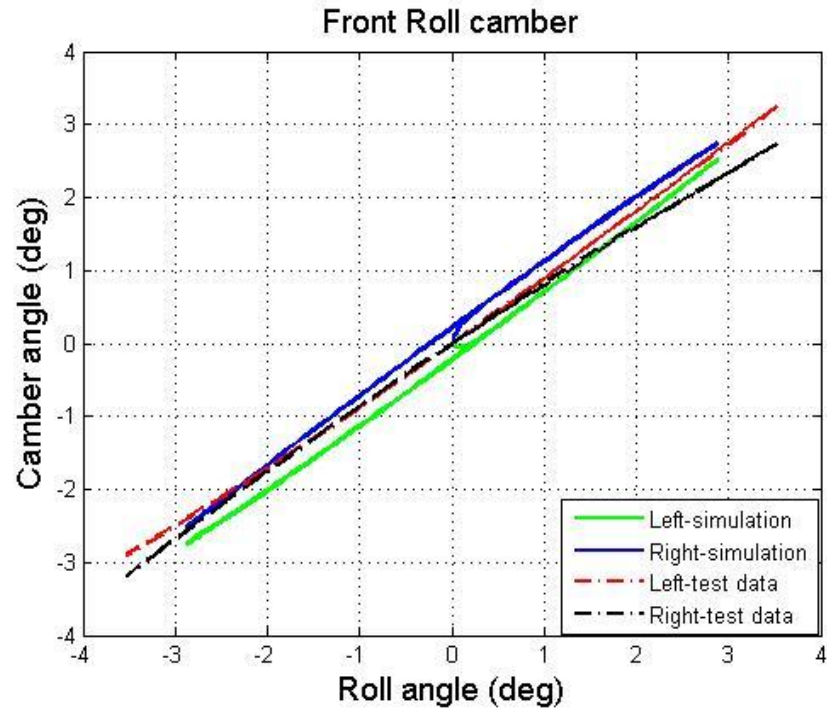


Figure 20: Front Roll Camber Comparison

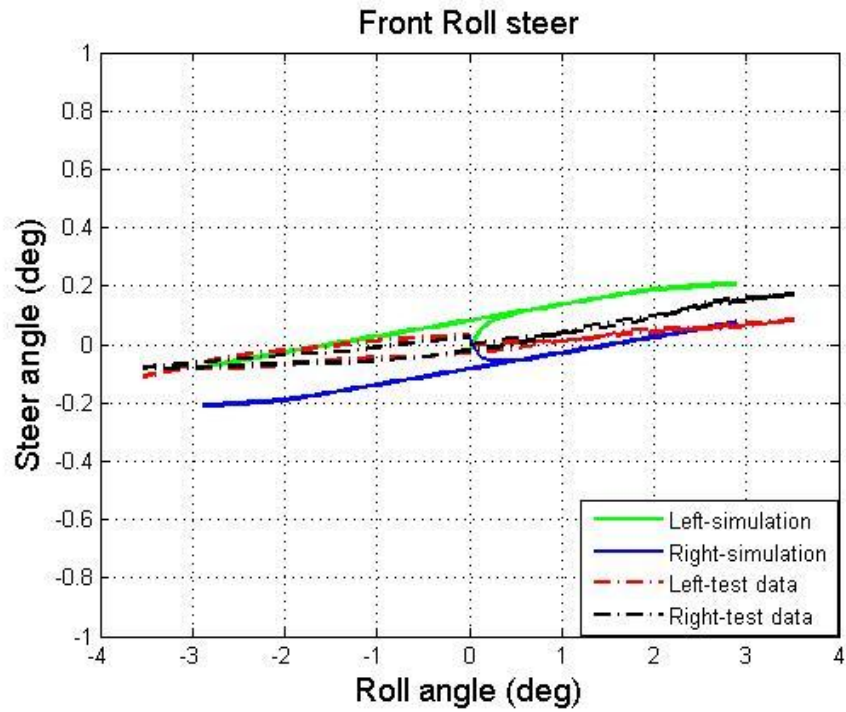


Figure 21: Front Roll Steer Comparison

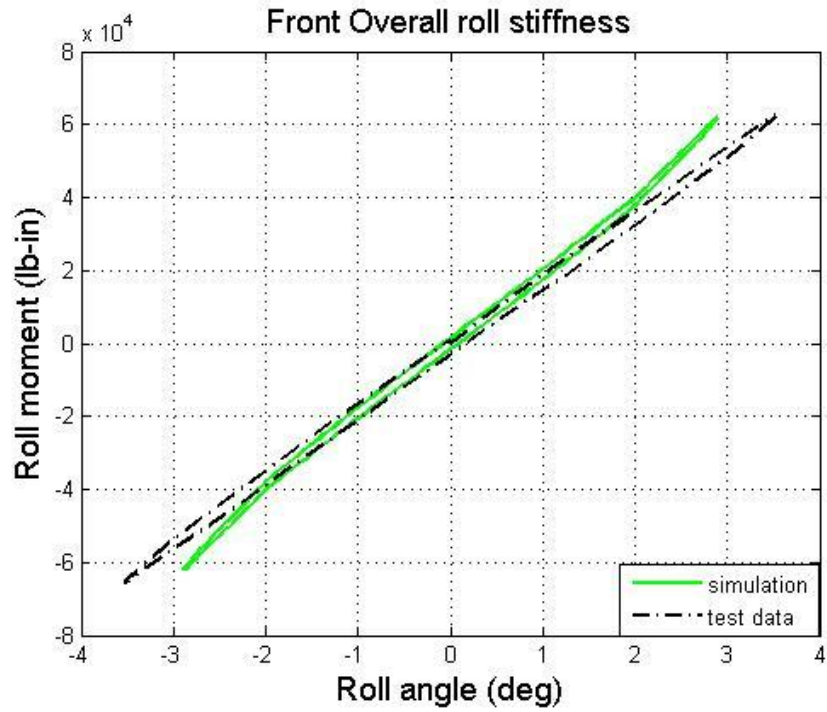


Figure 22: Front Overall Roll Stiffness Comparison

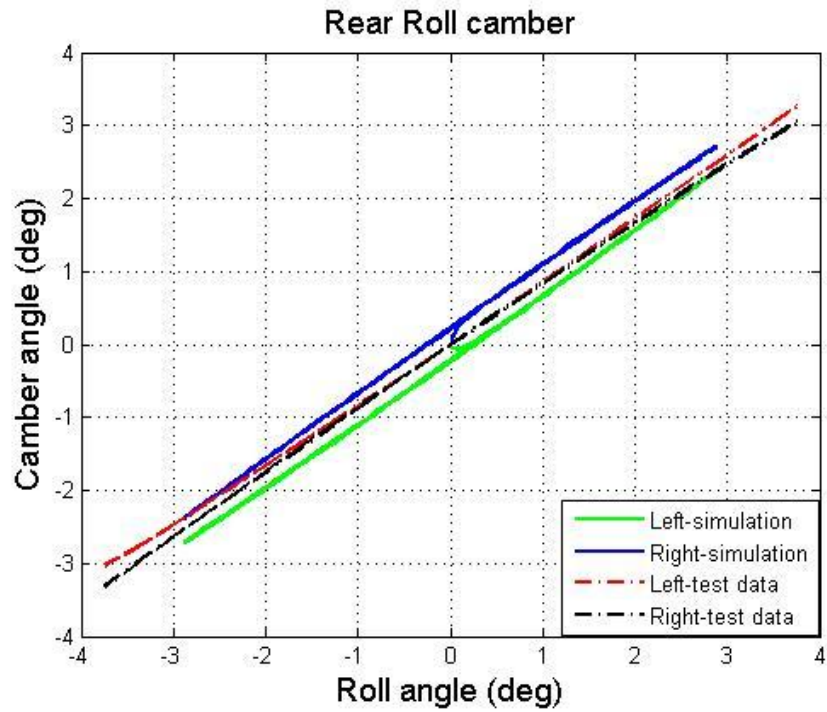


Figure 23: Rear Roll Camber Comparison

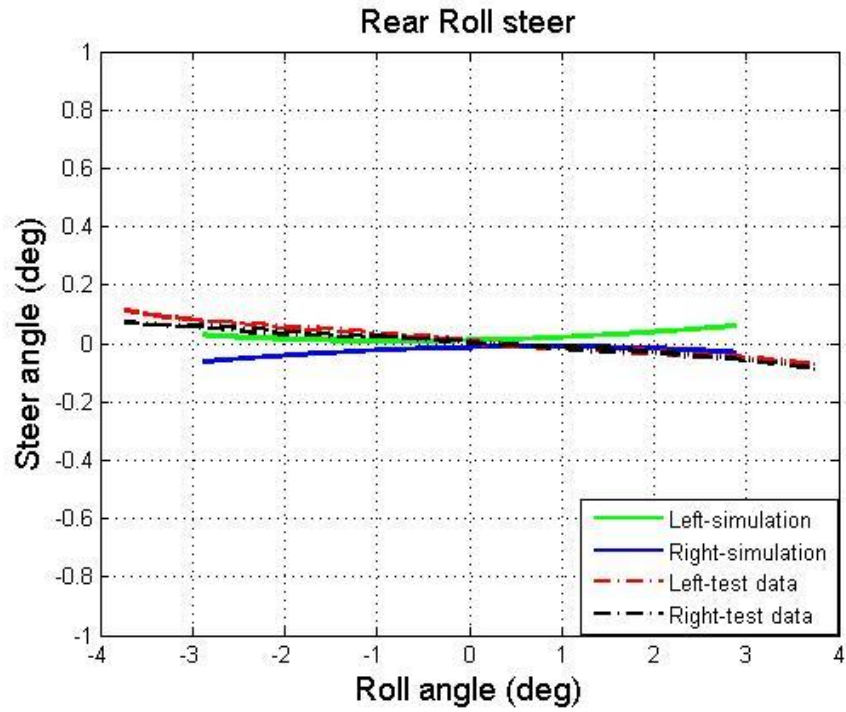


Figure 24: Rear Roll Steer Comparison

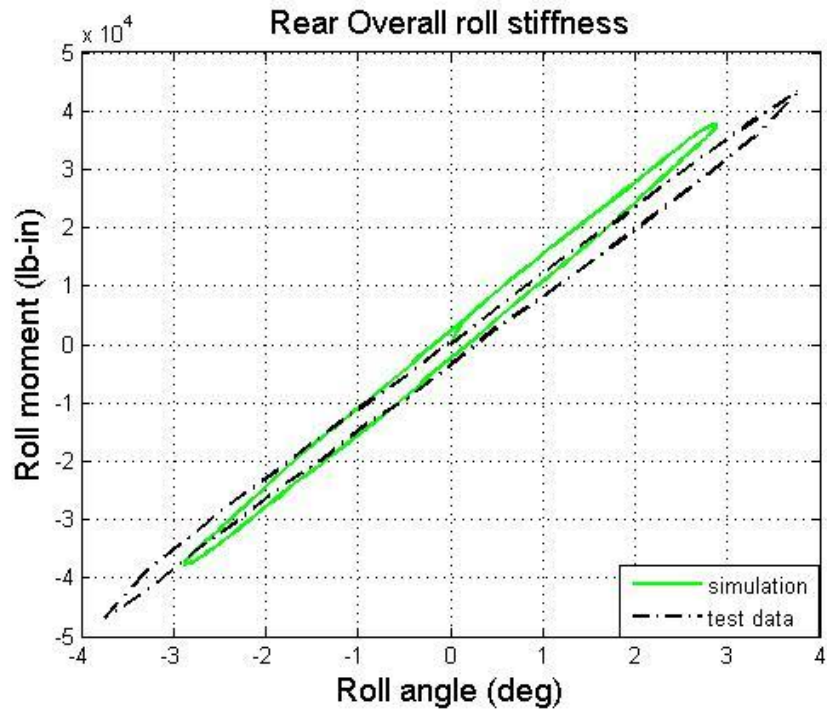


Figure 25: Rear Overall Roll Stiffness Comparison

The simulation results and experimental data are compared for the bounce and roll tests. The simulation results show a good correlation with the experimental data.

3.3.2 Dynamic Tests

Dynamic tests used for validating the model include the ‘Sine with Dwell’ and ‘Slowly Increasing Steer’. These dynamic tests were conducted by the Vehicle Research and Test Center (VRTC) for the 2003 Ford Expedition. Field test data from these tests was used to validate the CarSim model.

Sine with Dwell:

The sine with dwell maneuver is an important maneuver for analyzing vehicle stability. The maneuver comprises of a sinusoidal steering input of 0.7 Hz frequency and 500 milliseconds dwell after the third quarter cycle. The vehicle is allowed to coast down with throttle off from a speed of 80 km/h for this maneuver.

The following figures show the comparison of experimental and simulation results for a 120 degree sine with dwell maneuver.

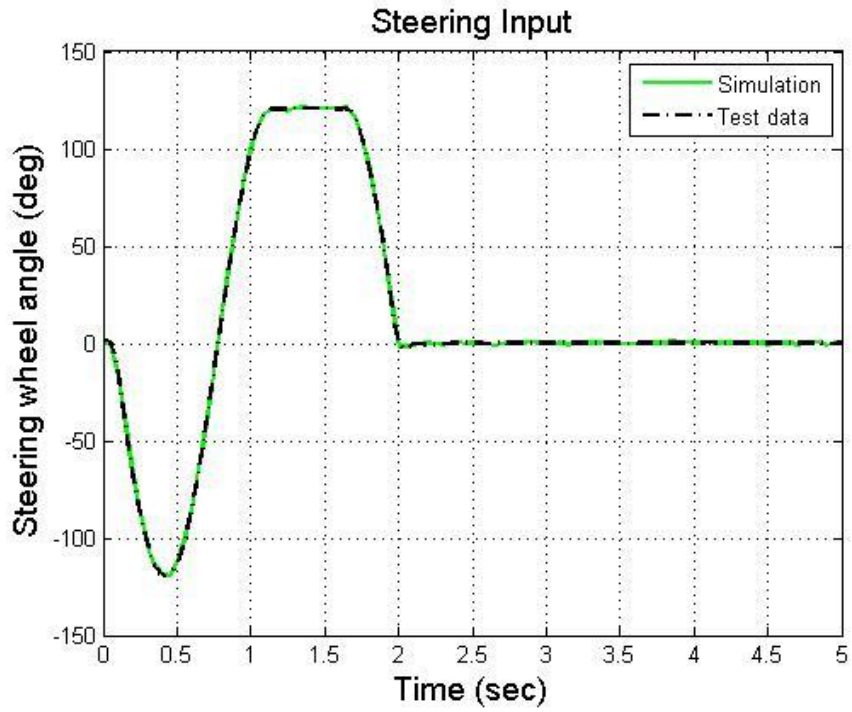


Figure 26: Steering Wheel Angle Comparison

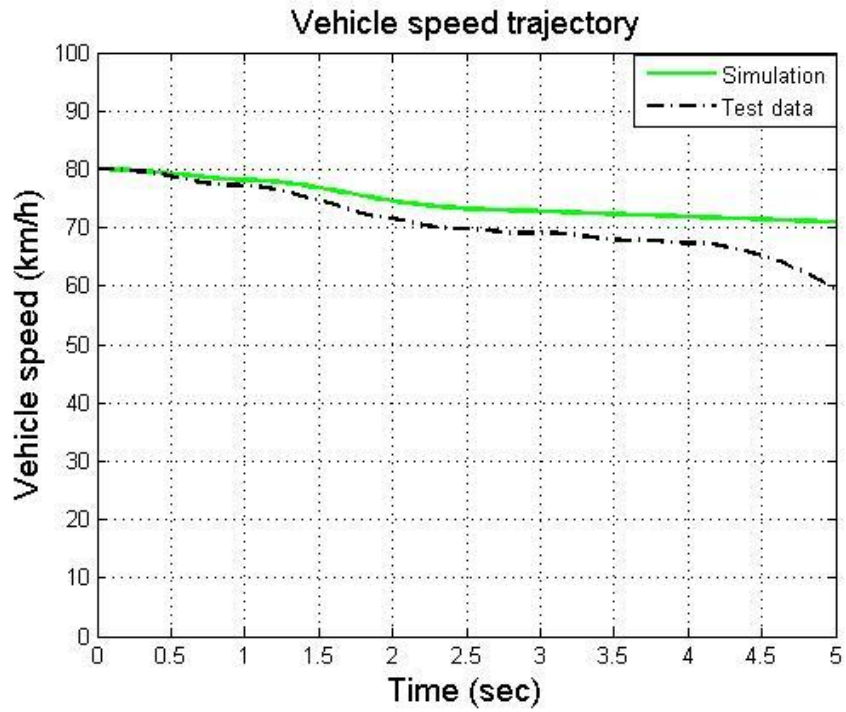


Figure 27: Vehicle Speed Comparison

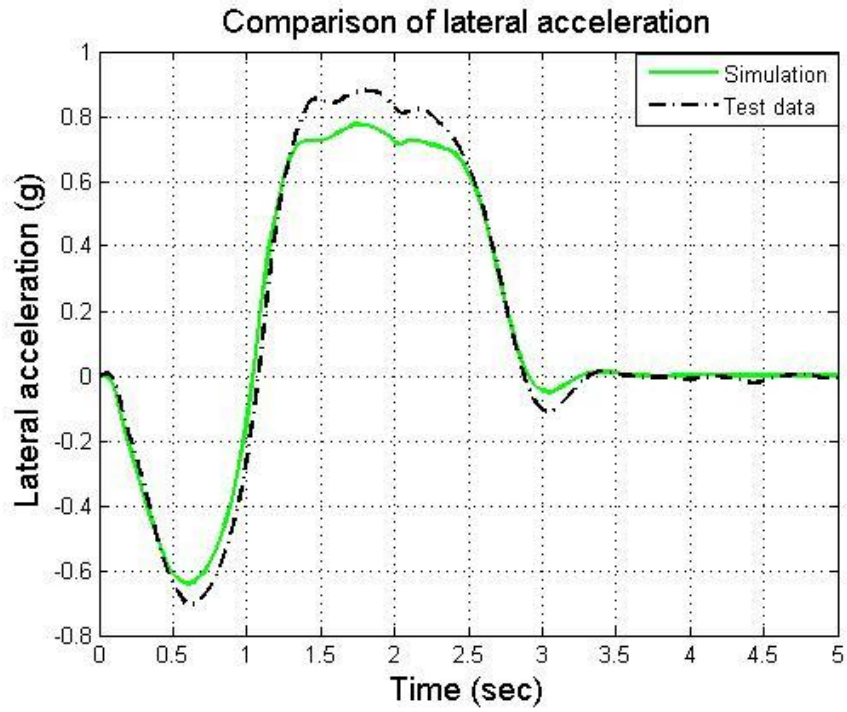


Figure 28: Lateral Acceleration Comparison

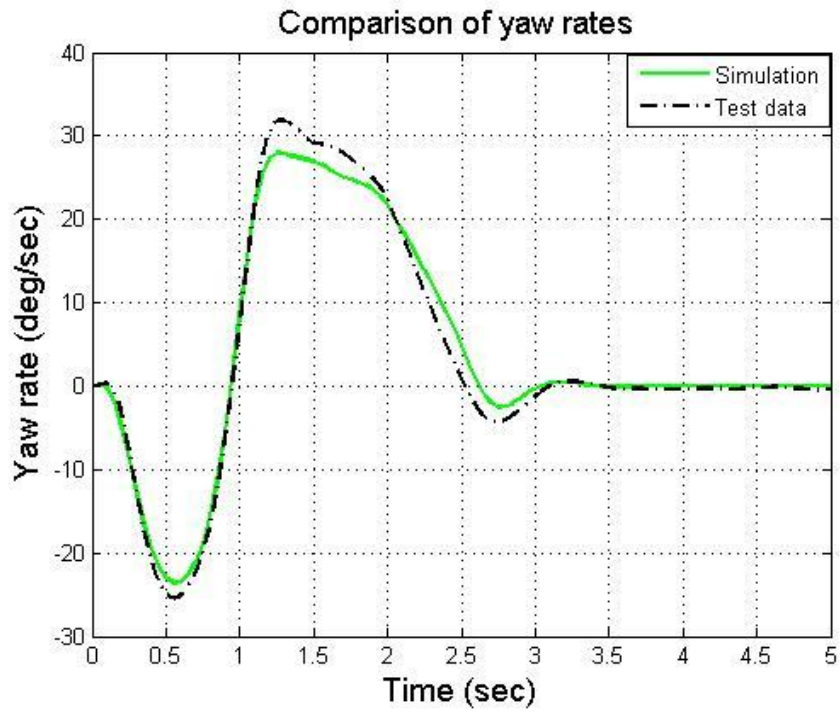


Figure 29: Yaw Rate Comparison

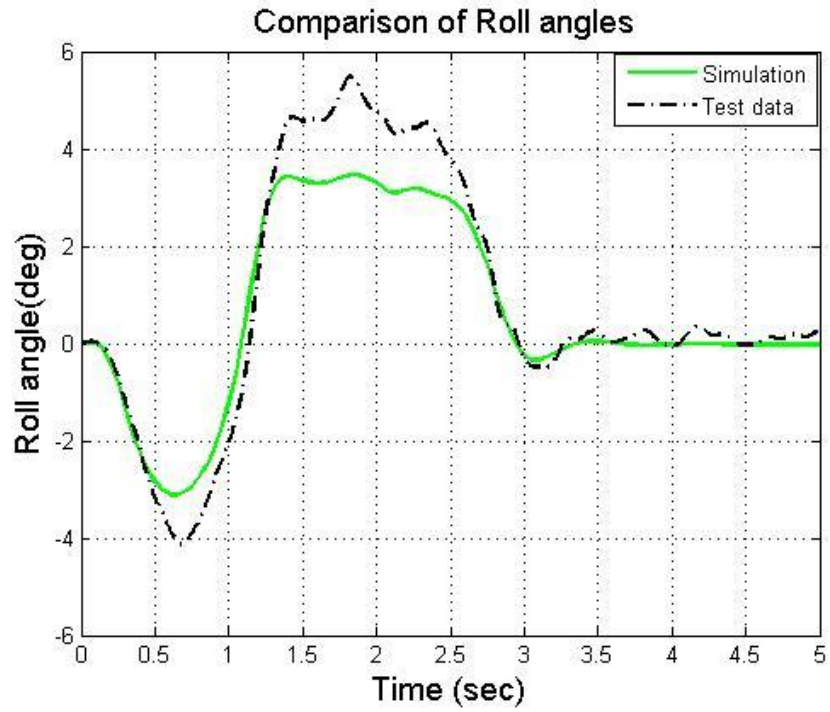


Figure 30: Roll Angle Comparison

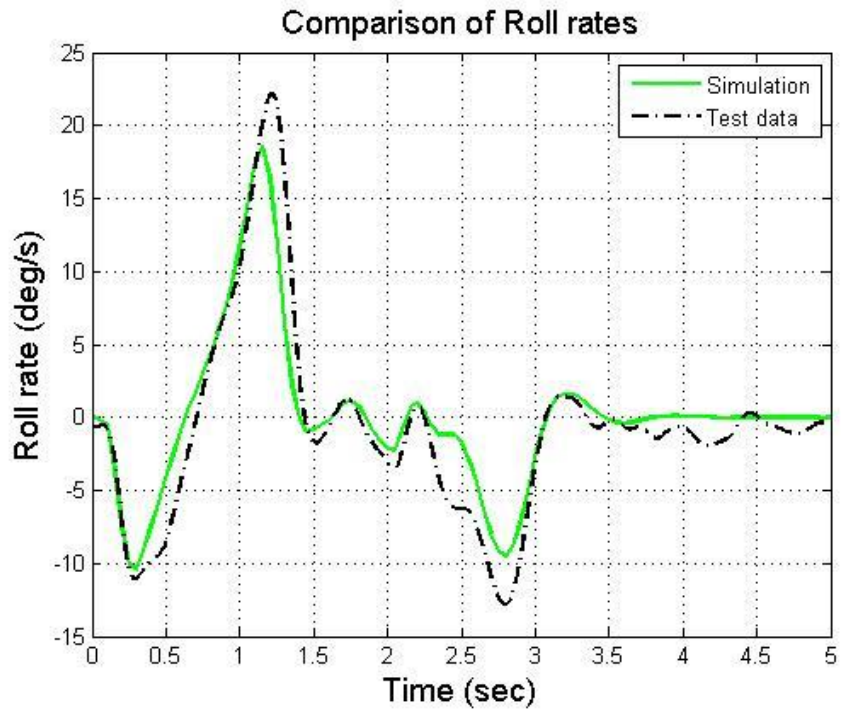


Figure 31: Roll Rate Comparison

The vehicle responses from CarSim tests in terms of variables like lateral acceleration, yaw rate and roll rate are seen to match well with the experimental results. The roll angles predicted by the simulation are lower than the experimental values. The unmodeled compliances in the suspension system and phenomena like deformation of the tire sidewall could result in higher roll angles in the field data.

Slowly Increasing Steer:

This test consists of a steering input at a constant steering rate to get to a stipulated steering angle. The speed is maintained constant at approximately 80 km/h. The parameters of interest are the lateral acceleration, yaw rate, roll angle and roll rate.

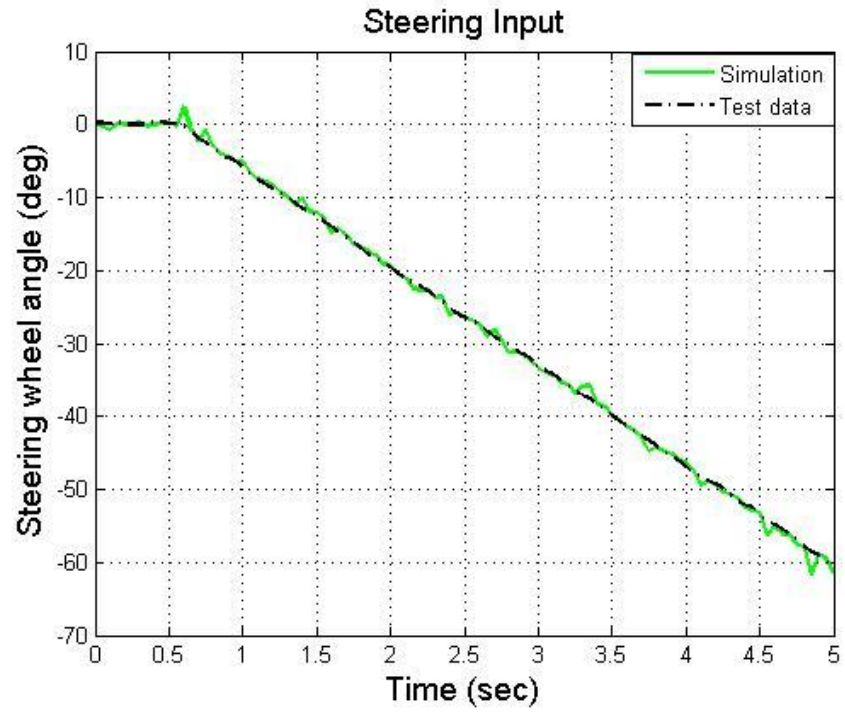


Figure 32: Steering Wheel Angle Comparison

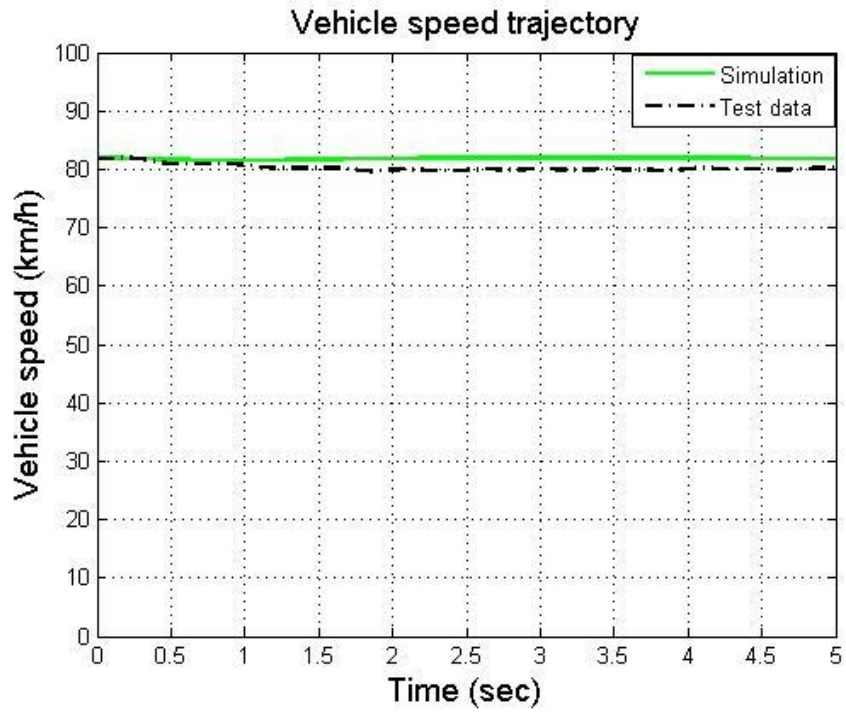


Figure 33: Vehicle Speed Comparison

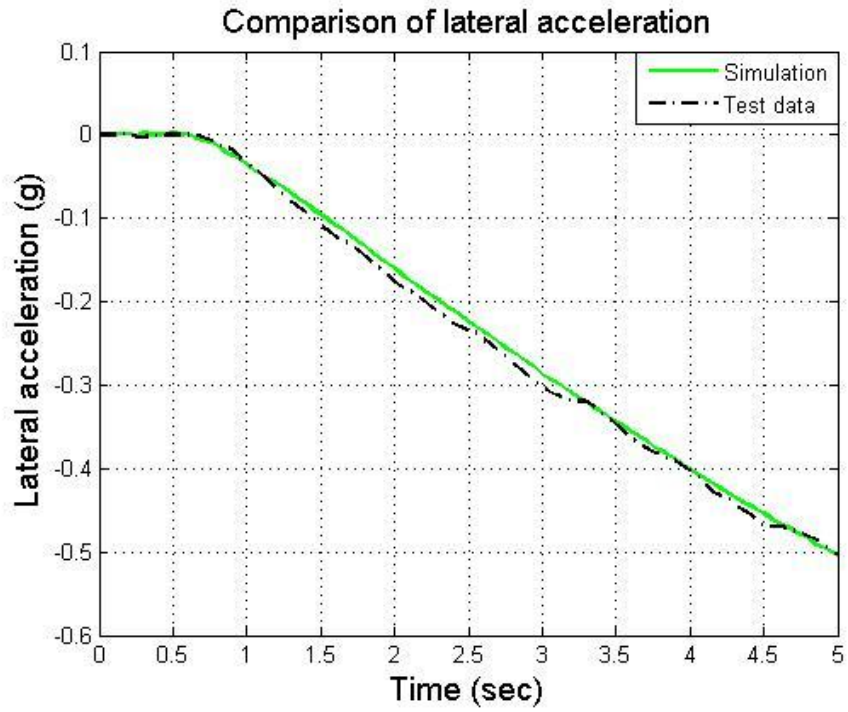


Figure 34: Lateral Acceleration Comparison

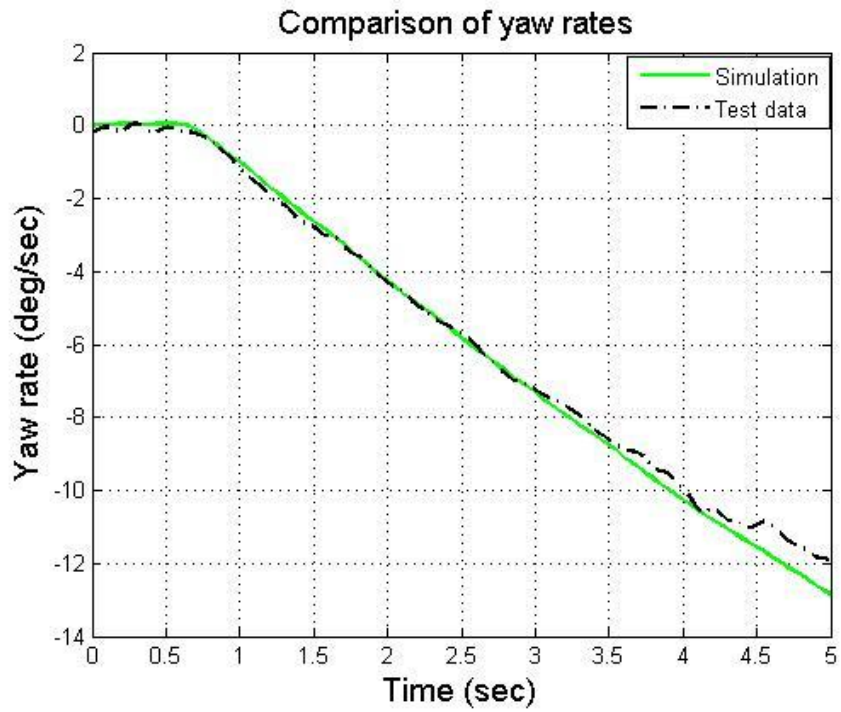


Figure 35: Yaw Rate Comparison

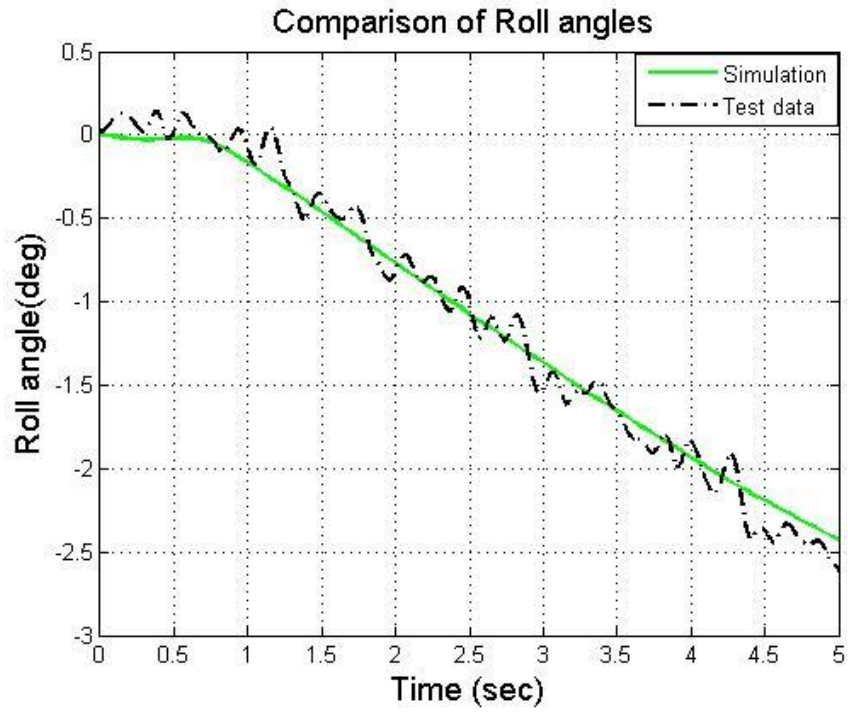


Figure 36: Roll Angle Comparison

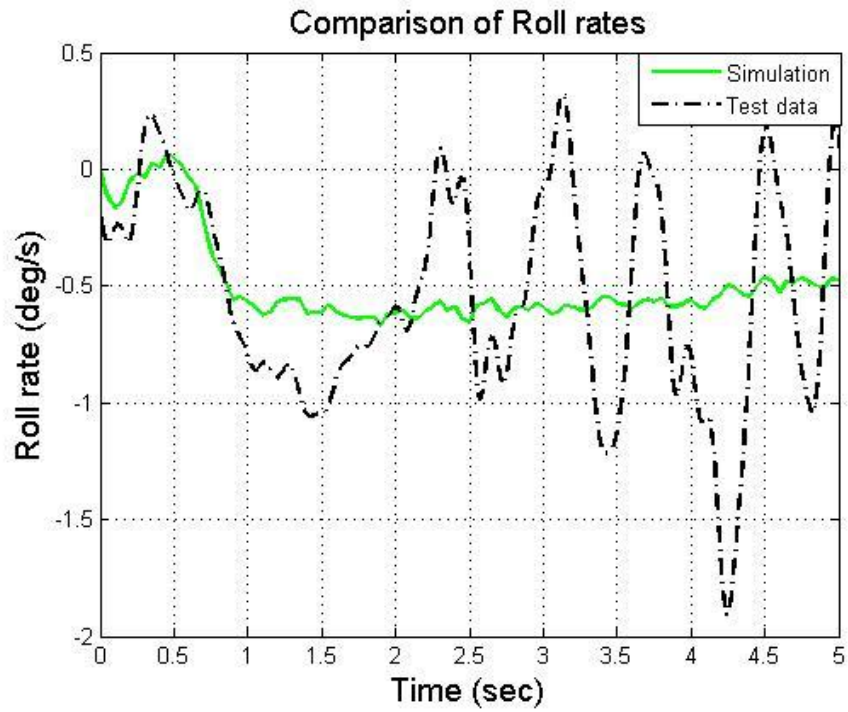


Figure 37: Roll Rate Comparison

The results from the Slowly Increasing Steer test are shown in Figure (32) - Figure (37).

A good correlation is obtained between the experimental and simulation data.

CHAPTER 4

4 SIMULATION WITH CARSIM

4.1 Background

The 2003 Ford Expedition vehicle model from Chapter 3 was used to simulate the understeer gradient tests in CarSim. These tests included the constant radius test, constant steer test and constant speed test. SAE J266 gives the consistent test procedure for determining steady state directional control properties for passenger cars. The calculations for the understeer gradient have been made as per the SAE J266 standard [9].

4.1.1 Constant Radius Test

The vehicle is driven over a path of constant radius at successively higher speeds as a part of the test procedure. The data is recorded when a steady state is achieved. The minimum radius for the test is recommended to be 30 m. There are two variations of this test, discrete and continuous. In the first one, the vehicle is driven at successively increasing constant speeds and data is taken for steady state conditions for each speed. The second method involves increasing the speed at a slow constant rate and recording the data continuously as the lateral acceleration increases in a near steady state manner.

For both the discrete and continuous test procedures, the recommended interval for recording the data is 0.05 g. For the continuous test procedure, the rate of increase of speed should not increase 1.5 km/h per second.

The understeer gradient is given by Equation (4.1)

$$K = \frac{d\delta}{d(a/g)} - \frac{d}{d(a/g)} \left(57.3 \frac{L}{R} \right) \quad (4.1)$$

Where δ = Steer angle

a = Lateral Acceleration

g = Acceleration due to gravity

L = Wheelbase of the vehicle

R = Radius of the turn

The understeer gradient for the constant radius test is equal to the steer angle gradient since the Ackermann steer angle gradient $\left(\frac{d}{d(a/g)} \frac{L}{R} \right)$ is zero.

Therefore,

$$K = \frac{d\delta}{d(a/g)} \quad (4.2)$$

Thus, the understeer gradient for a constant radius test is given by the steer angle gradient.

4.1.2 Constant Steer Angle Test

The vehicle is driven at different speeds at a constant steering wheel angle as a part of this test procedure. There are discrete and continuous versions of this test. In the

discrete test, the steer angle is held constant and the vehicle is driven at a constant speed until steady state conditions are achieved. In the continuous test, the steering is held constant while the speed is increased continuously until the limit of control. The initial minimum path radius for the continuous test is 20 m according to SAE J266.

Since the steer angle is constant in this test, the steer angle gradient is zero. Therefore, since the wheelbase 'L' is constant,

$$K = 57.3L \frac{d(1/R)}{d(a/g)} \quad (4.3)$$

Thus, the understeer gradient is found from the path curvature gradient for the constant steer angle test. This relationship can be simplified from the steady state kinematic relation, we have

$$V = rR / 57.3 \quad (4.4)$$

Where r is the yaw velocity

Therefore,

$$K = L \frac{d(r/V)}{d(a/g)} \quad (4.5)$$

4.1.3 Constant Speed Test

The vehicle is driven at constant speed and at a range of steering wheel angle amplitudes as a part of the test procedure. The data is collected for a steady state response at constant speed.

We have the relationship

$$a = V^2 / R$$

The speed V is constant, therefore,

$$K = \frac{d\delta}{d(a/g)} - \frac{d}{d(a/g)} \left(57.3Lg \frac{(a/g)}{V^2} \right) \quad (4.6)$$

$$K = \frac{d\delta}{d(a/g)} - \frac{57.3Lg}{V^2} \quad (4.7)$$

Thus, the understeer gradient for the constant speed test is found from the steer angle gradient and the Ackermann steer angle gradient.

4.2 2003 Ford Expedition

The following section discusses the results obtained for the understeer gradient tests from the CarSim simulations. The values of the required variables for calculating the understeer gradient were obtained from the CarSim simulation results. The relationships in Equations (4.1) to (4.7) were used calculate the understeer gradient.

4.2.1 Constant Radius Test

The test was carried out for constant radius values of 30 m and 60 m. A slowly increasing speed profile was used with a constant rate of speed increase of 0.36 km/h per second (approximately 0.1 m/s per second). The data obtained from these CarSim tests was used to make understeer gradient calculations.

The analysis was carried out for the lateral acceleration range of 0.05 g to 0.3 g. The CarSim tests were run until steady state lateral acceleration values were attained. The speed profile used for the CarSim constant radius test is shown in Figure (38).

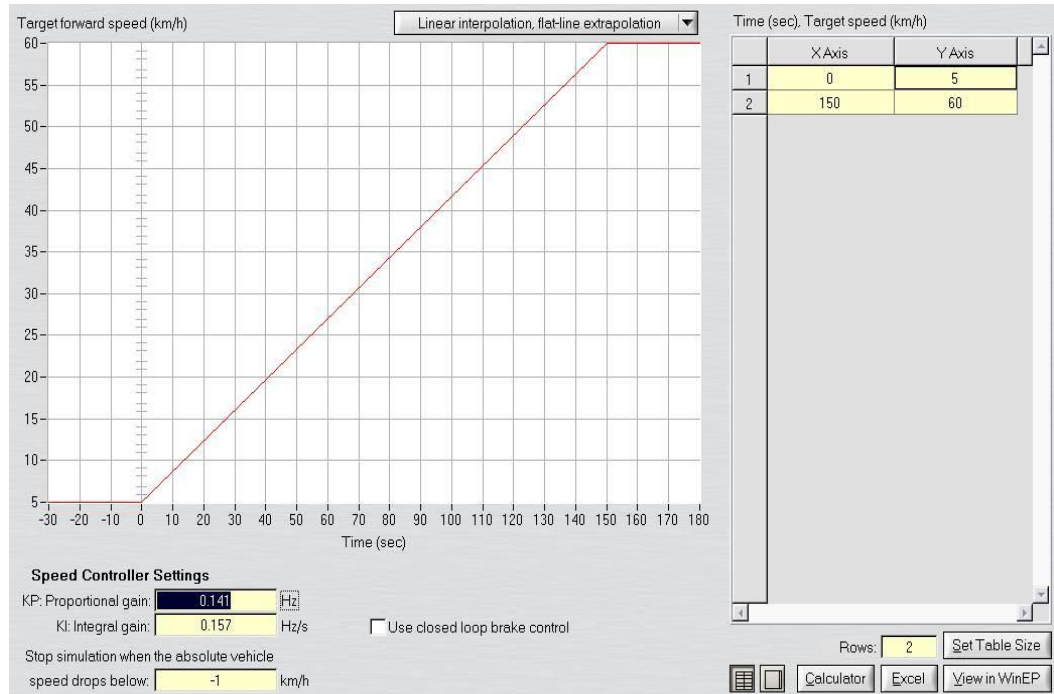


Figure 38: Speed Profile – Constant Radius Test

The understeer gradient was calculated using steer angle calculated from the steering wheel angles divided by steering ratio as well as directly from the road wheel angles. The understeer gradient values calculated using both these methods are significantly different. Table (1) gives the values for the constant radius test.

From Steering wheel angle K (deg/g)		From Roadwheel angle K (deg/g)	
30 m	60 m	30 m	60 m
1.91	1.95	0.25	0.31

Table 1: Constant Radius Test Understeer Gradient (Simulation)

The understeer gradient values obtained for the constant radius test are seen to increase with increase in the radius. There is also a significant difference in the understeer gradient values obtained for the steering wheel angle divided by the steering ratio and the roadwheel angle. The steer angles from the steering wheel angles divided by the steering ratio are different from the road wheel steer angles. Figure (40) shows these differences.

Figure (39) shows the plot of steer angle versus lateral acceleration for the 30 m radius. The steering input starting at zero lateral acceleration has been shown. The nature of the steering input can be seen from Figure (39). Since, until 0.05 g lateral acceleration, the steering profile almost coincides for both the steering wheel angle divided by steering ratio and roadwheel angles, we neglect the data till 0.05 g lateral acceleration for the analysis. For the analysis, since we only consider the lateral acceleration range from 0.05 g to 0.3 g, the following plots show the steer angle data only for this range.

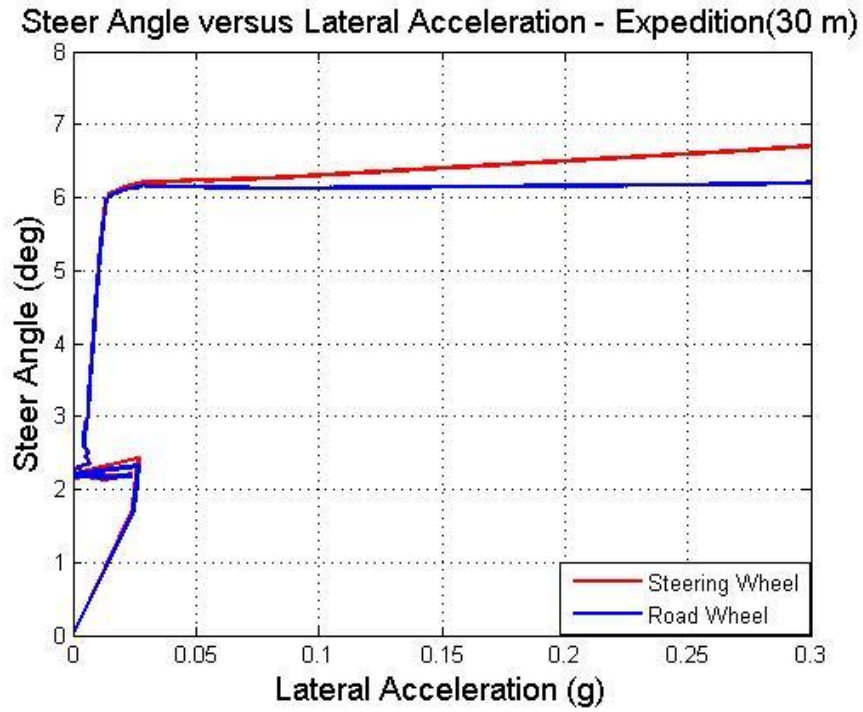


Figure 39: Steer Angle versus Lateral Acceleration

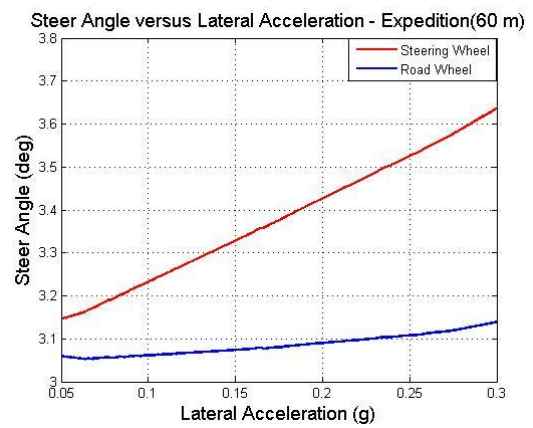
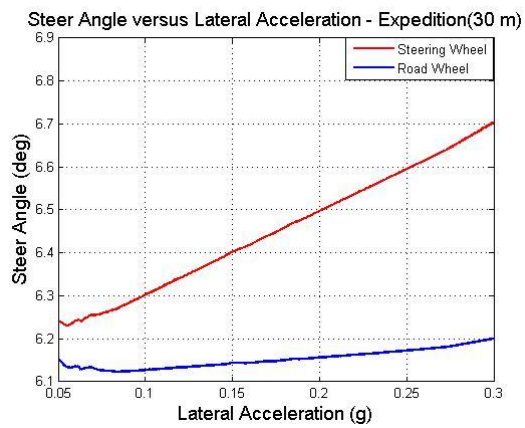


Figure 40: Steer Angle versus Lateral Acceleration

From Figure (40) we observe that at the lateral acceleration of 0.3 g, the difference between the steer angle given by the steering wheel angle divided by the steering ratio and the roadwheel angle is about 0.5 degree. Figure (41) shows the comparison of lateral acceleration versus time for the two constant radii.

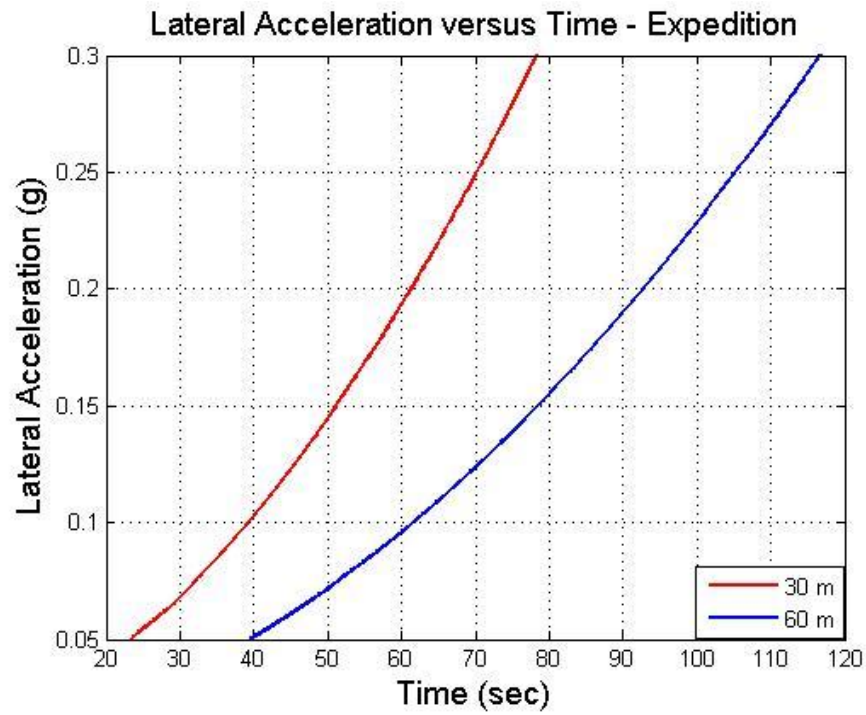


Figure 41: Lateral Acceleration versus Time Comparison

4.2.2 Constant Steer Angle Test

This test was carried out for constant hand wheel steer angle values of 45 degrees, 60 degrees, 90 degrees and 120 degrees. A slowly increasing speed profile with a rate of speed increase of 0.625 km/h per second (approximately 0.173 m/s per second) was used. The speed profile in Figure (42) was used for this method. The analysis was made for a lateral acceleration range of 0.05 g to 0.3 g. The understeer gradient values obtained for this test method are given in Table (2).

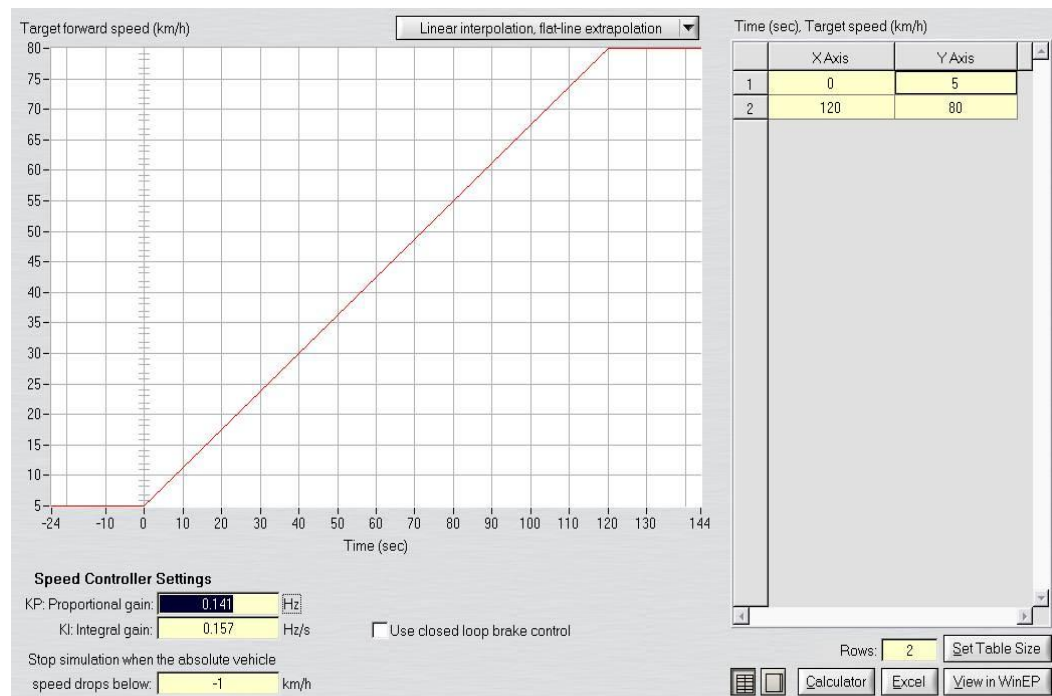


Figure 42: Speed Profile – Constant Steer Test

K (deg/g)			
45 deg	60 deg	90 deg	120 deg
1.89	1.88	1.86	1.82

Table 2: Constant Steer Test Understeer Gradient (Simulation)

The understeer gradient values obtained for the constant steer test are seen to decrease slightly with an increase in the value of the steer angle. Figure (43) shows the comparison of path curvature versus lateral acceleration for different steer angle values. Figure (44) and (45) show the comparison of rate of lateral acceleration and yaw rate respectively.

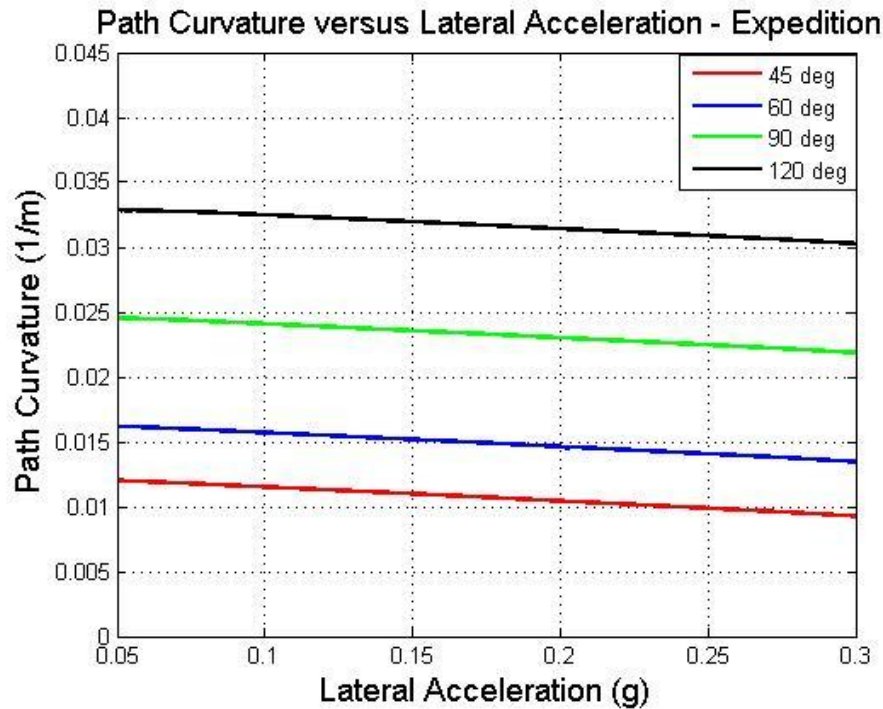


Figure 43: Path Curvature versus Lateral Acceleration

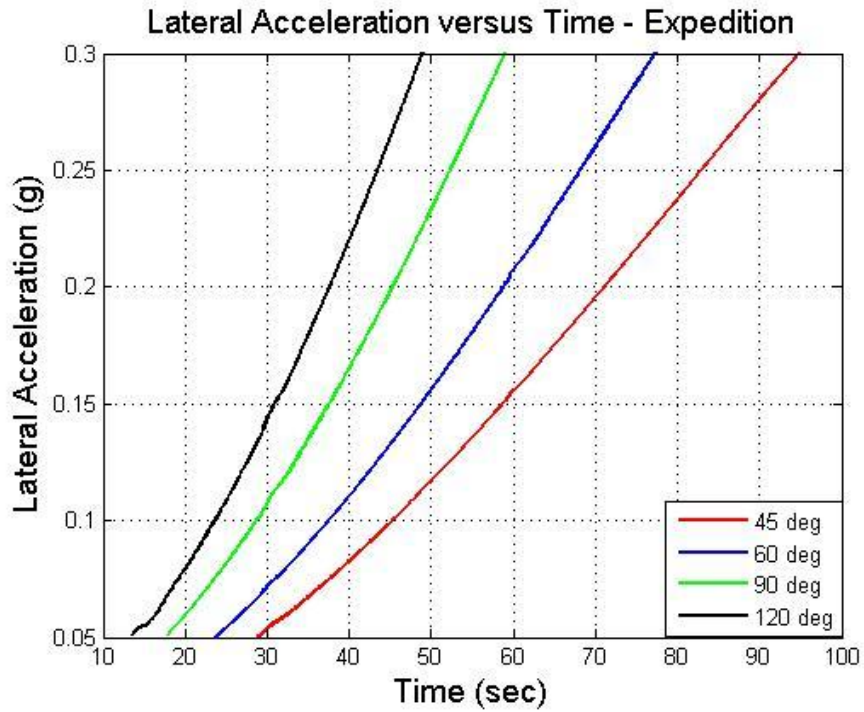


Figure 44: Lateral Acceleration versus Time Comparison

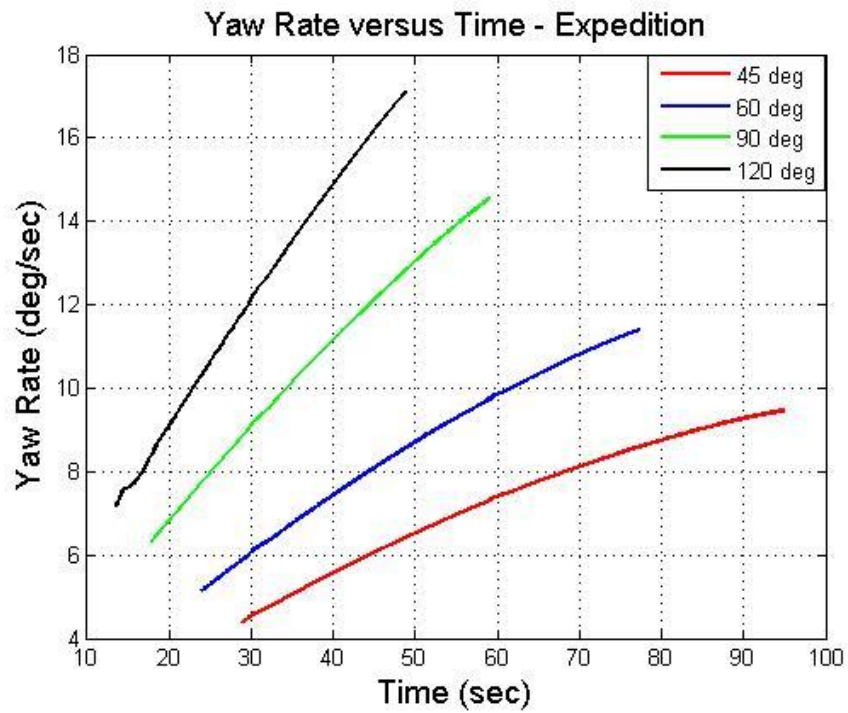


Figure 45: Yaw Rate versus Time Comparison

The steer angles from the steering wheel angles divided by steering ratio were plotted in comparison with the road wheel steer angles. The constant steer test is executed by locking the steering wheel at a desired angle. It was observed that, as per the test requirement, the steer angle calculated from the steering wheel angle and steering ratio remains constant through the test. But the road wheel angles do not remain constant through the test. The plots in Figure (46) show these results.

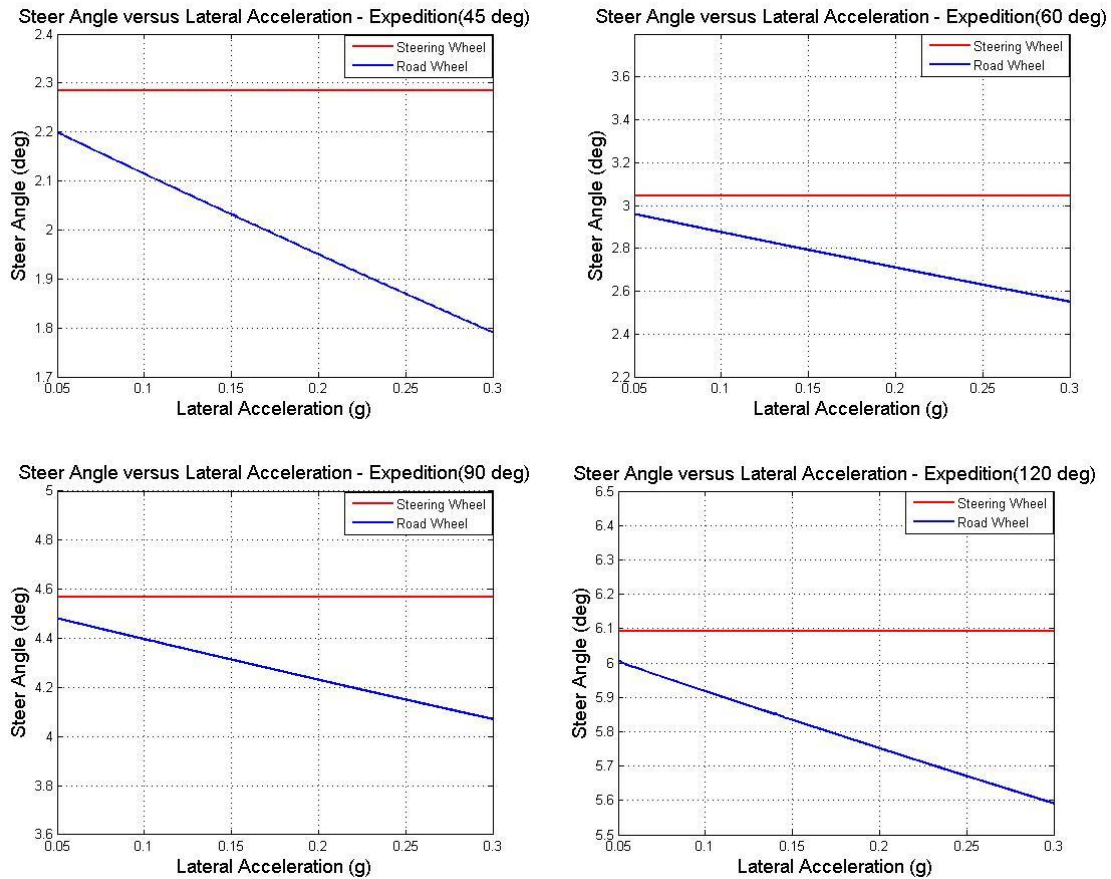


Figure 46: Steer Angle versus Lateral acceleration Comparison

4.2.3 Constant Speed Test

The speed is held constant for this test. A steering rate of 5 deg/sec was used. The test was run for constant speed values of 9 m/s, 13 m/s and 18 m/s. The analysis was made for a lateral acceleration range of 0.05 g to 0.3 g. The steering wheel angle profile used is shown in Figure (47). The understeer gradient values for this test using steering wheel angles divided by steering ratio and the road wheel angle are given in Table (3).

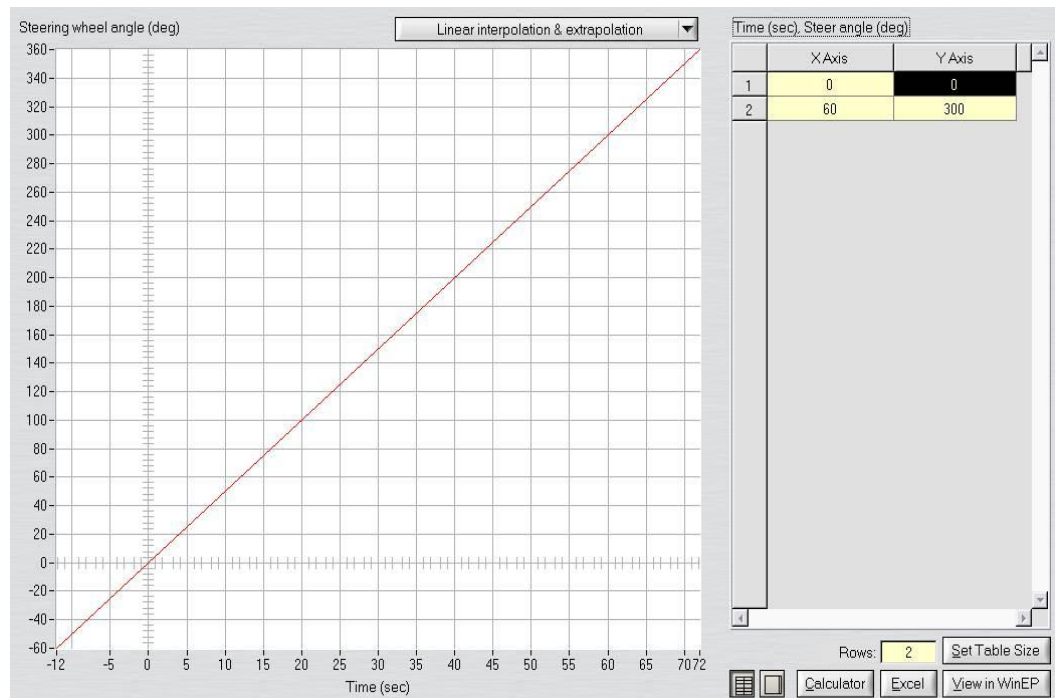


Figure 47: Steer Profile – Constant Speed Test

Using Steering Wheel Angle K (deg/g)			Using Roadwheel Angle K (deg/g)		
9 m/s	13 m/s	18 m/s	9 m/s	13 m/s	18 m/s
2.98	2.48	2.23	1.31	0.83	0.59

Table 3: Constant Speed Test Understeer Gradient (Simulation)

The understeer gradient values obtained from the constant speed test are shown in Table (3). These values are seen to decrease with an increase in the value of the speed. There was a difference between the values of understeer gradient obtained from the steering wheel angle divided by steering ratio and the roadwheel angle.

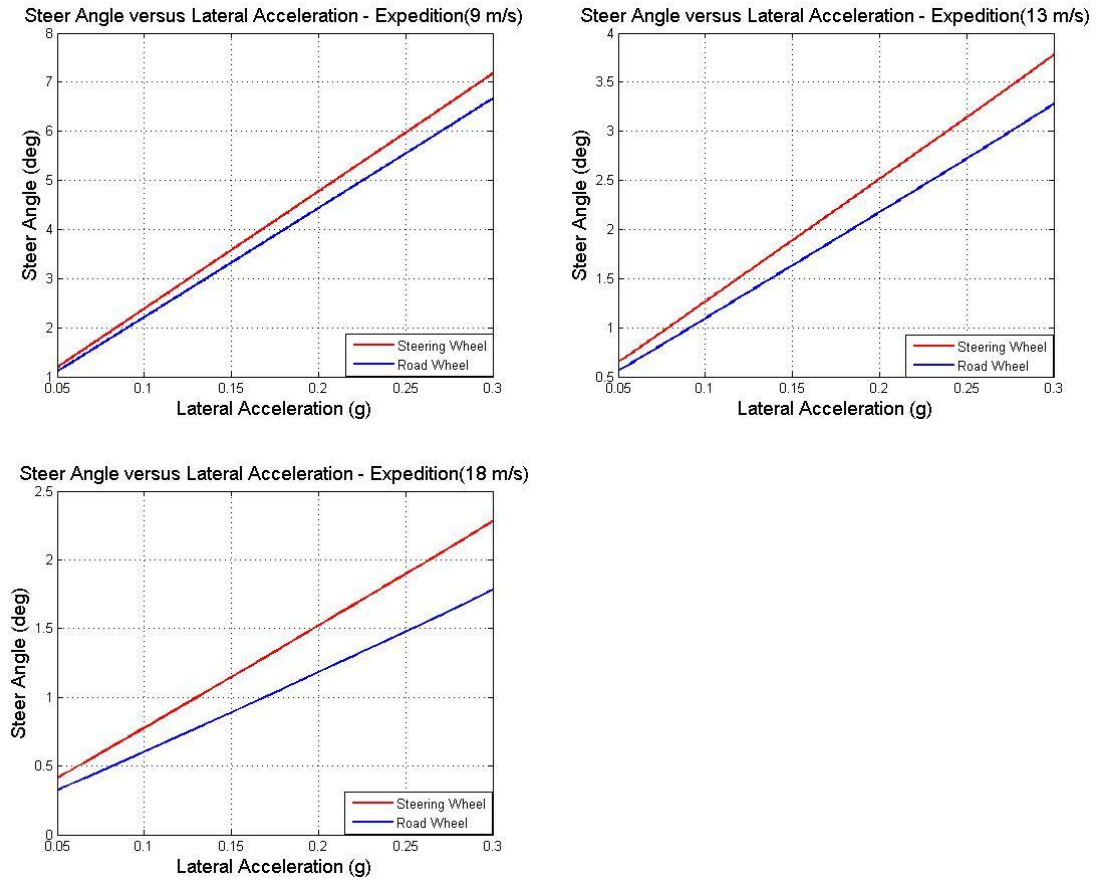


Figure 48: Steer Angle versus Lateral Acceleration

Figure (49) shows the comparison of steer angle versus lateral acceleration. Figure (50) shows the comparison of lateral acceleration versus time for the three constant speeds.

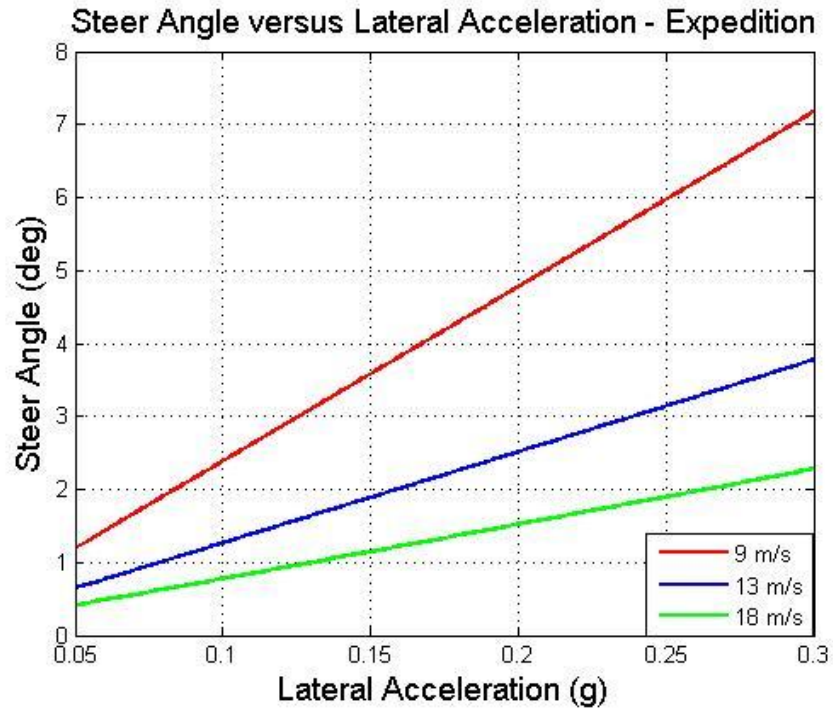


Figure 49: Steer Angle versus Lateral Acceleration Comparison

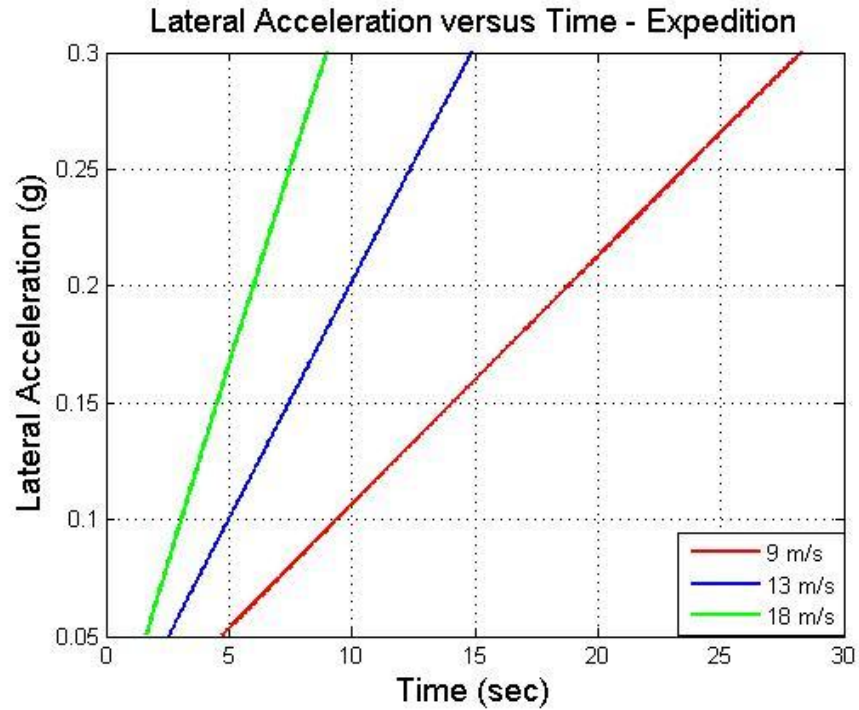


Figure 50: Lateral Acceleration versus Time Comparison

From all the understeer gradient tests, the plots comparing the characteristics for steering wheel angle divided by steering ratio with roadwheel angle show a constant difference between the steer angle values at 0.03 g. This difference in the steer angle values is observed to be about 0.5 degrees for all the tests.

4.3 Expedition Tests with Zero Compliances

The standard understeer tests were run for Expedition by setting all of the steering and suspension steer compliances and kinematics to zero. These included the steering system compliance (the column compliance), all the compliance coefficients for the suspension system and the toe-jounce and camber-jounce. The other inputs to the system were exactly the same as that for tests with the original Expedition model.

For this configuration, the understeer gradient calculations gave the same results for the steer angle from the steering wheel angle divided by steering ratio and direct road wheel angle. For this system with all steering compliances set to zero, the roadwheel angle was observed to be the same as the steering wheel angle divided by the steering ratio.

The understeer gradient values obtained for the zero compliance model were observed to be closer to the values obtained for direct road wheel angle calculations for the original Expedition model. The following sections show the results from the tests.

4.3.1 Constant Radius Test

The constant radius test was carried out for constant radius values of 30 m and 60 m. The understeer gradient values for the test are given in Table (4). These values are seen to increase with an increase in the value of the radius.

Figure (51) shows that the steer angles obtained from the steering wheel angles divided by steering ratio coincides with the roadwheel angles. This confirms that the steering and suspension compliance and kinematics were the reason for the difference in the two different measurements in CarSim.

K (deg/g)	
R = 30 m	R = 60 m
0.51	0.53

Table 4: Constant Radius Test Understeer Gradient (Zero Compliances)

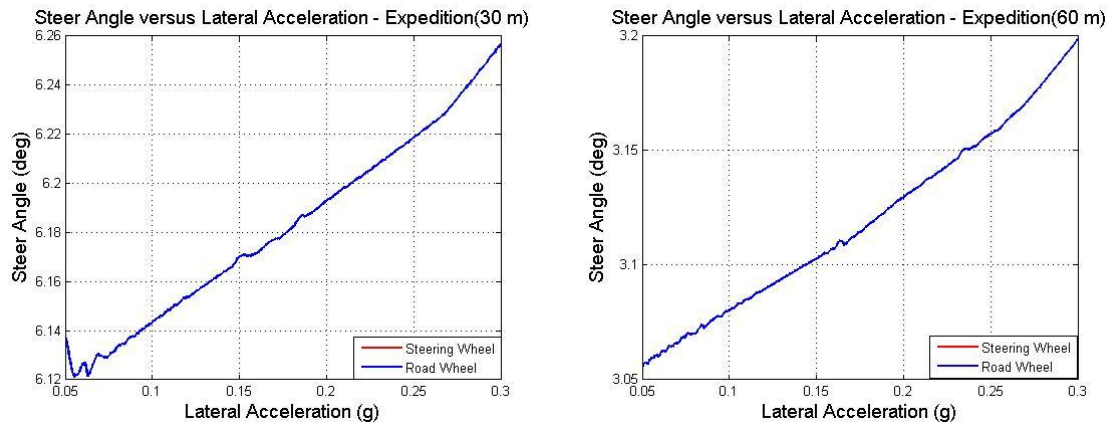


Figure 51: Steer Angle versus Lateral Acceleration

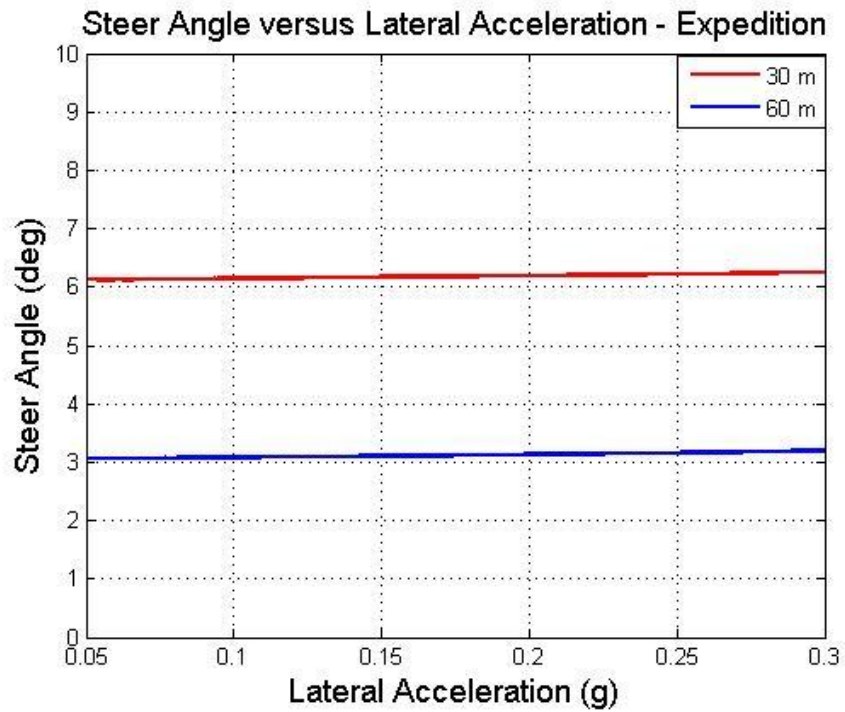


Figure 52: Steer Angle versus Lateral Acceleration Comparison

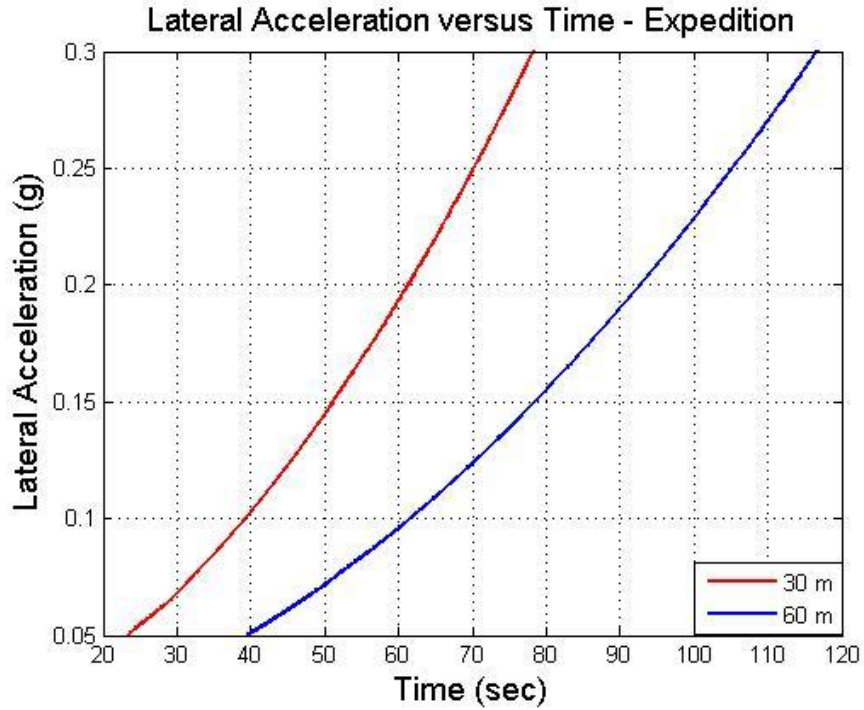


Figure 53: Lateral Acceleration versus Time Comparison

4.3.2 Constant Steer Test

The constant steer test was carried out for constant steer angle values of 45 deg, 60 deg, 90 deg and 120 degrees. The understeer gradient values obtained for this test are given in Table (5). These values are seen to decrease with an increase in the value of the steer angle. Figure (54) shows the comparison of steer angles from steering wheel to road wheel angles for different constant steer angles.

K (deg/g)			
45 deg	60 deg	90 deg	120 deg
0.54	0.53	0.51	0.48

Table 5: Constant Steer Test Understeer Gradient (Zero Compliances)

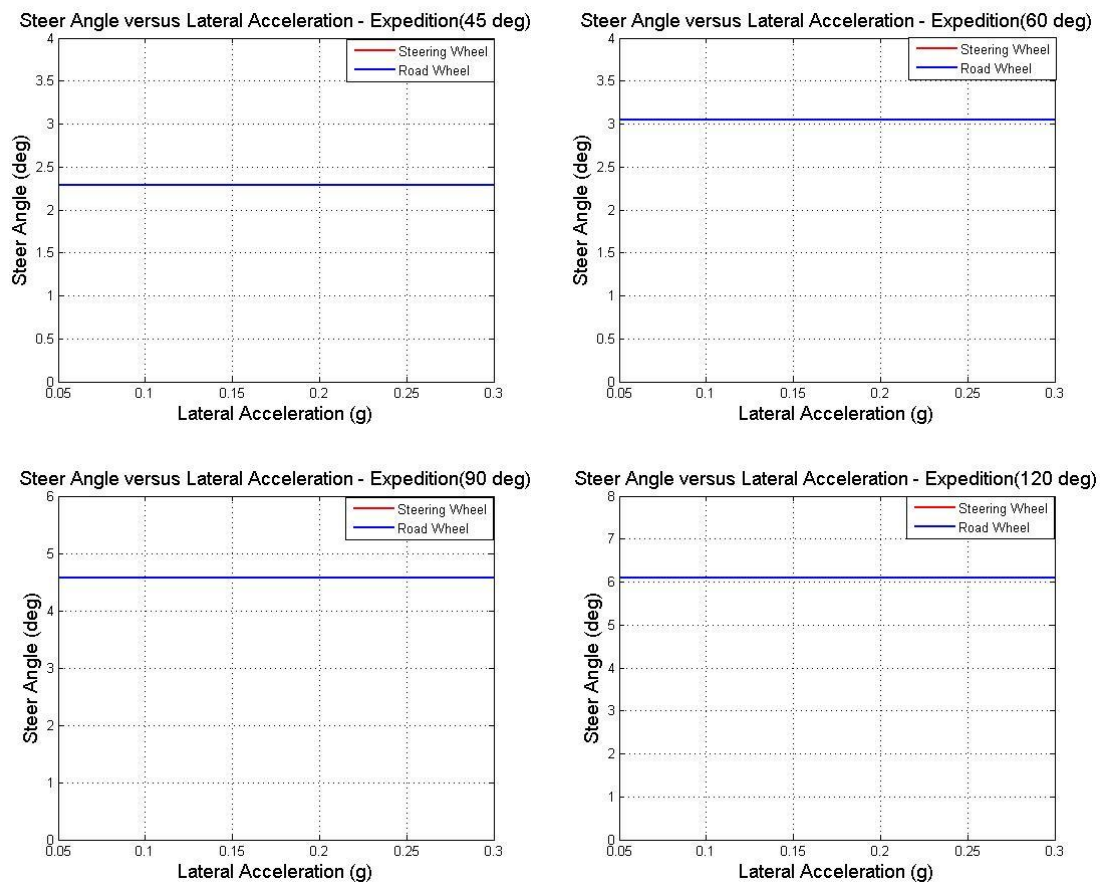


Figure 54: Steer Angle versus Lateral Acceleration

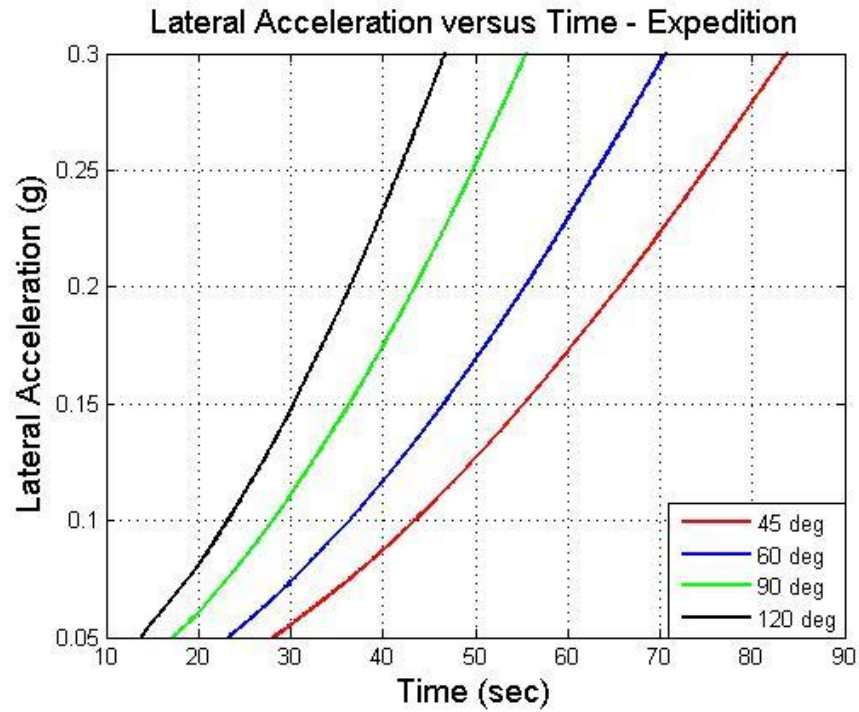


Figure 55: Lateral Acceleration versus Time Comparison

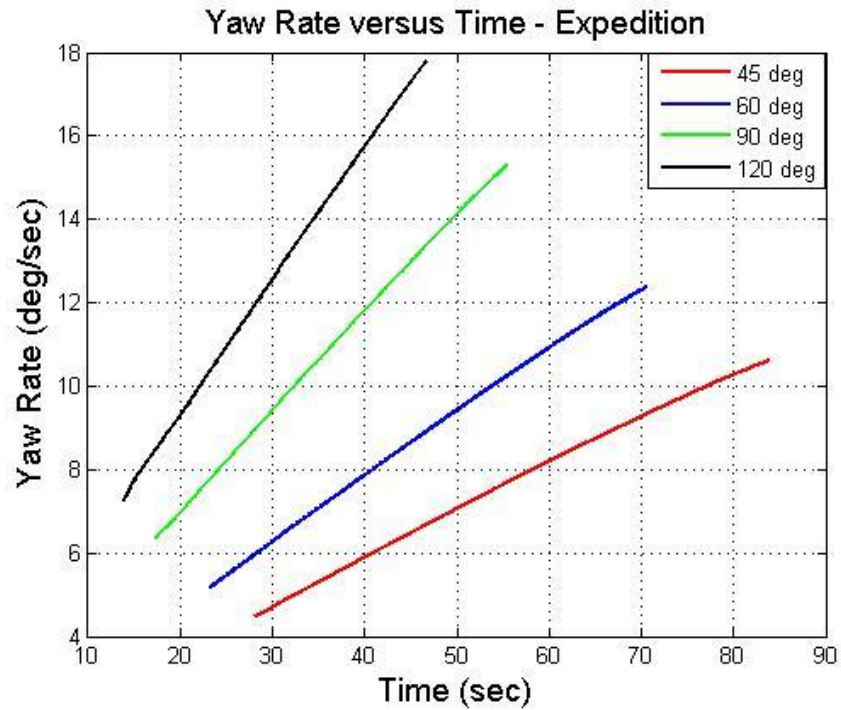


Figure 56: Yaw Rate versus Time Comparison

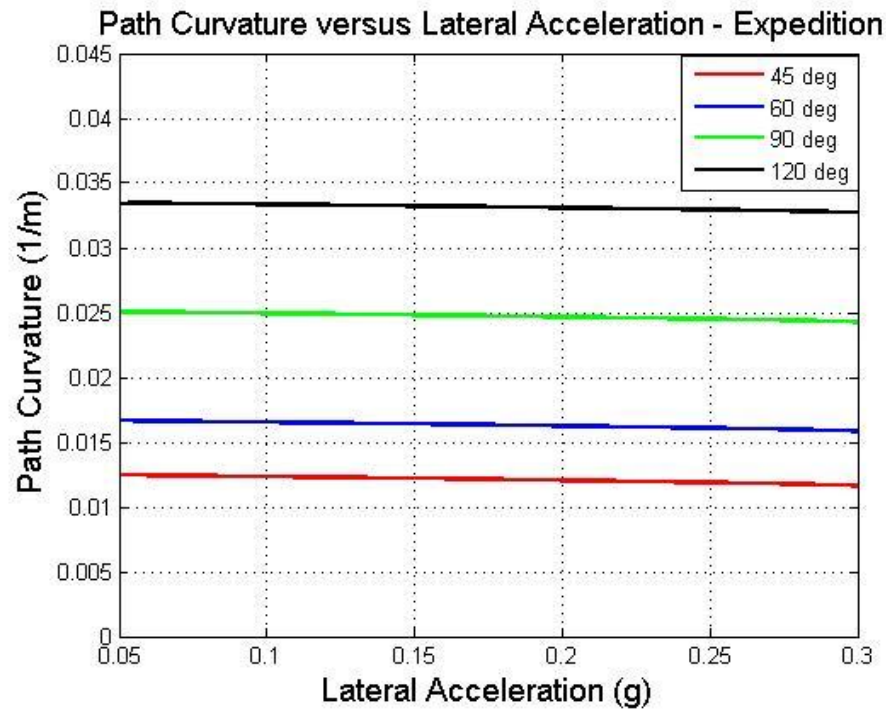


Figure 57: Path Curvature versus Lateral Acceleration Comparison

4.3.3 Constant Speed Test

The constant speed test was carried out for constant speed values of 9 m/s, 13 m/s, and 18 m/s. Table (6) shows the understeer values obtained for this test. These values are seen to decrease with an increase in the values of speed. Also, the comparison between the steer angles from steering wheel and road wheel is shown in Figure (58) for different constant speeds.

K (deg/g)		
V = 9 m/s	V = 13 m/s	V = 18 m/s
1.49	1.02	0.79

Table 6: Constant Speed Test Understeer Gradient (Zero Compliances)

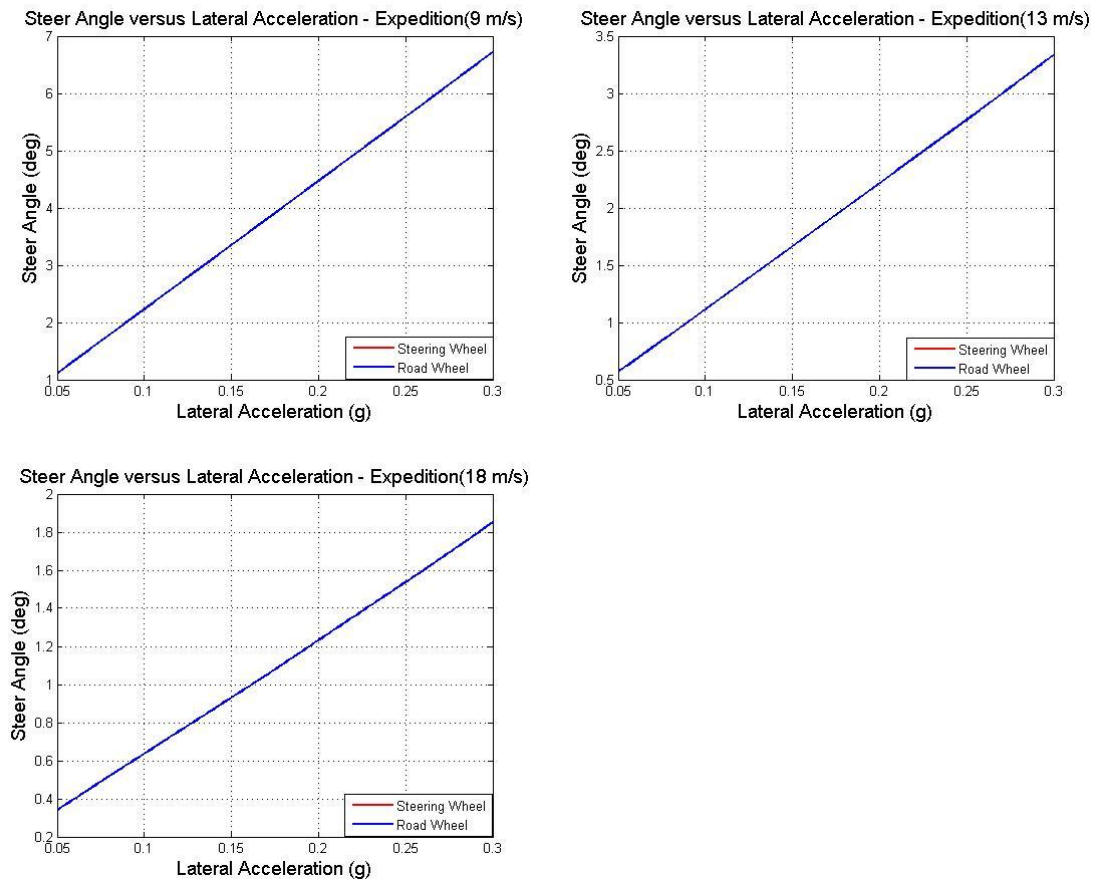


Figure 58: Steer Angle versus Lateral Acceleration

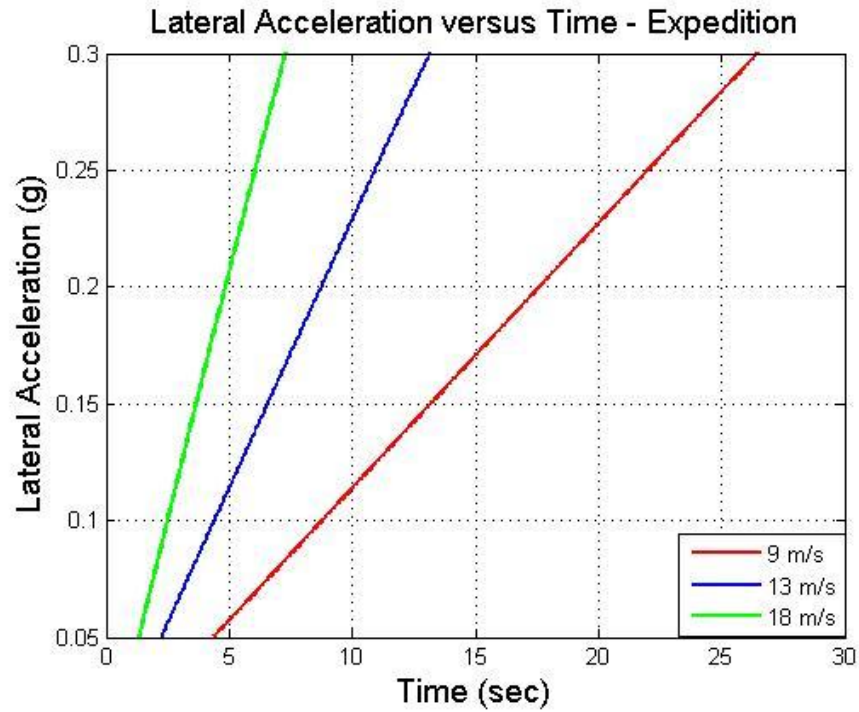


Figure 59: Lateral Acceleration versus Time Comparison

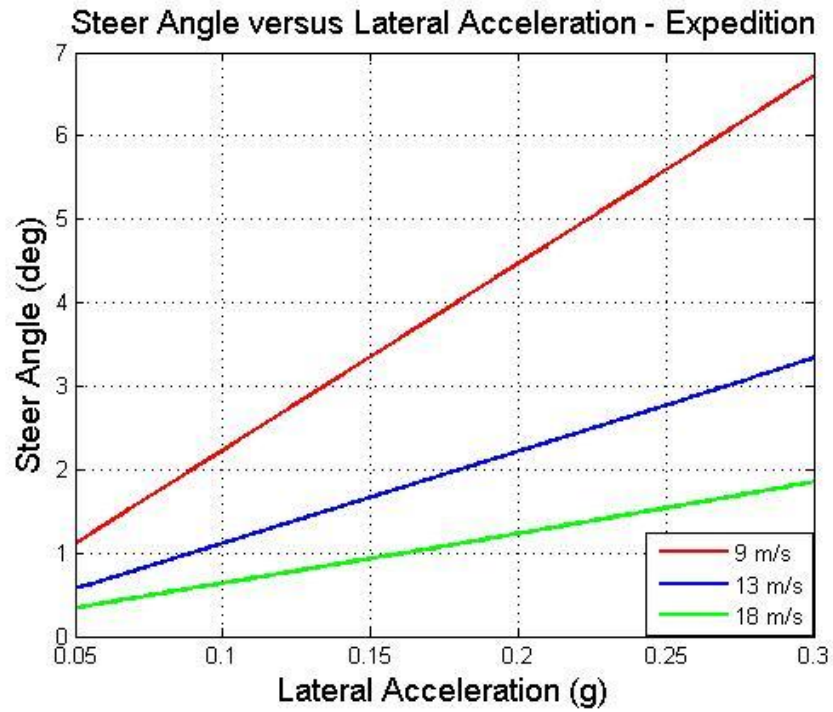


Figure 60: Steer Angle versus Lateral Acceleration

From the comparison of the understeer gradient values obtained from all of the understeer tests, we note that the understeer gradient values obtained for the original Expedition model using roadwheel angles are different from those obtained for the zero compliance Expedition model. To understand the cause of this difference, we need to investigate further the vehicle response and development of slip angles and lateral acceleration during these maneuvers.

4.4 Understeer Gradient from Slowly Increasing Steer Test

The slowly increasing steer maneuver with a steering rate of 13.5 deg/sec is one of the standard maneuvers used by NHTSA to determine the handling characteristics of the vehicle. The constant speed test for the speed of 13 m/s was run for the original Expedition model with all compliances for a steering rate of 13.5 deg/sec. The understeer gradient values obtained for this steering rate were compared with the values obtained for a steering rate of 5 deg/sec. The following table gives the comparison.

For V = 13 m/s	From Steering Wheel angle K (deg/g)	From Roadwheel angle K (deg/g)
13.5 deg/sec	2.52	0.87
5 deg/sec	2.48	0.83

Table 7: Slowly Increasing Steer Understeer Gradient

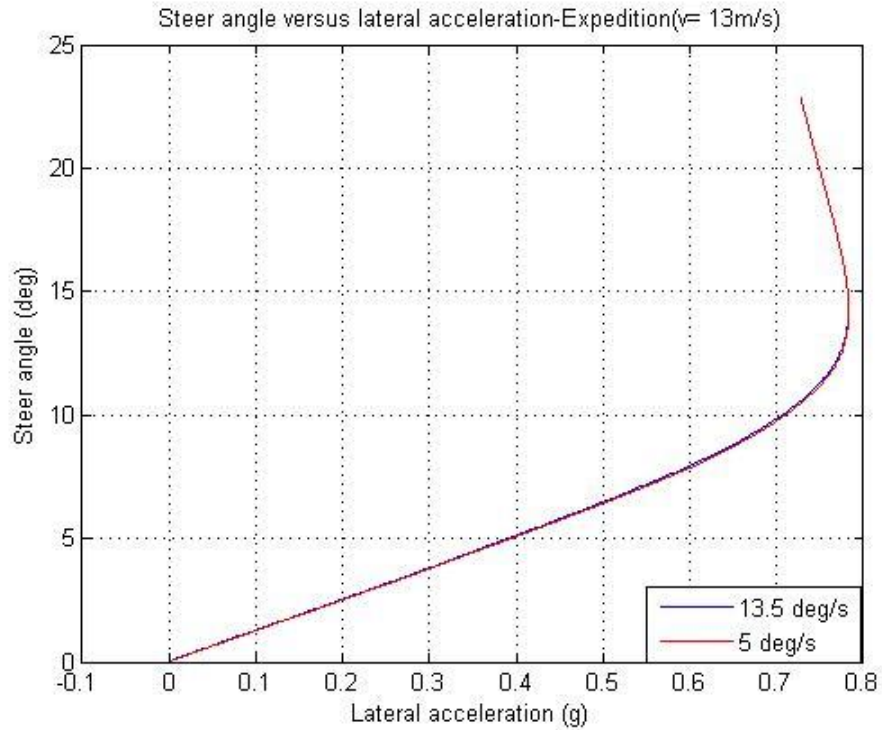


Figure 61: Steer Angle versus Lateral Acceleration

The steering angle versus lateral acceleration curve for the two tests coincides and the understeer gradient values obtained from them are close to each other. The understeer gradient values from both, steering wheel angle divided by the steering ratio and the roadwheel angles show the same trend.

CHAPTER 5

5 EXPERIMENTAL ANALYSIS

5.1 Background

The experimental analysis for understeer gradient was made using a 2006 Subaru Outback. The tests on this vehicle were carried out at the test facility at the Transportation Research Center (TRC). The facility's Vehicle Dynamics Area (VDA), a 50-acre asphalt test pad was used. These tests were carried out with the help of the Automated Driver [11, 12]. The test methods included constant radius test, constant steering angle test, and constant speed test.

The field test data was taken under dry conditions. This high quality accelerometer data with good repeatability was measured with the help of RT3002 GPS IMU (Inertia Measurement Unit) system [12]. The test procedures described in SAE J266 were used for the analysis of test data and calculation of understeer gradient. Note that all of the experimental test understeer gradient calculations are based on the steering wheel angle divided by the steering ratio. This is the most typical procedure used today in most industries and by NHTSA. There was no measurement made of the roadwheel steer angles during the experimental tests and this too is typical for most organizations making

understeer gradient measurements. The following section gives description of the test results from the field tests.

5.2 2006 Subaru Outback

A 2006 Subaru Outback with a wheelbase of 2.67 m and a steering ratio of 16.5 was used as the test vehicle. The following sections give the results from the understeer gradient tests.

5.2.1 Constant Radius Test

A speed profile with a rate of 0.1 m/s per sec (approximately 0.36 km/h per second) was used which for these tests. The constant radius values were 30 m and 60 m. Table (8) gives the understeer gradient values obtained from the field test data for each test and Table (9) gives the average values. Figure (62) and Figure (63) show the plot of steer angle versus lateral acceleration from the field test for constant radii of 30 m and 60 m respectively. Tests were conducted in both the clockwise (CW) and counterclockwise (CCW) direction.

	K (deg/g) CW		K (deg/g) CCW	
R = 30 m	3.03	3.12	3.04	2.96
R = 60 m	3.08	3.26	3.33	3.24

Table 8: Constant Radius Test Understeer Gradient (Experimental)

Average values:

K (deg/g) R = 30 m	K (deg/g) R = 60 m
3.04	3.23

Table 9: Constant Radius Test Average ‘K’ values

The average understeer gradient values obtained from these test measurements are shown in Table (8). These values are seen to increase with an increase in the test radius.

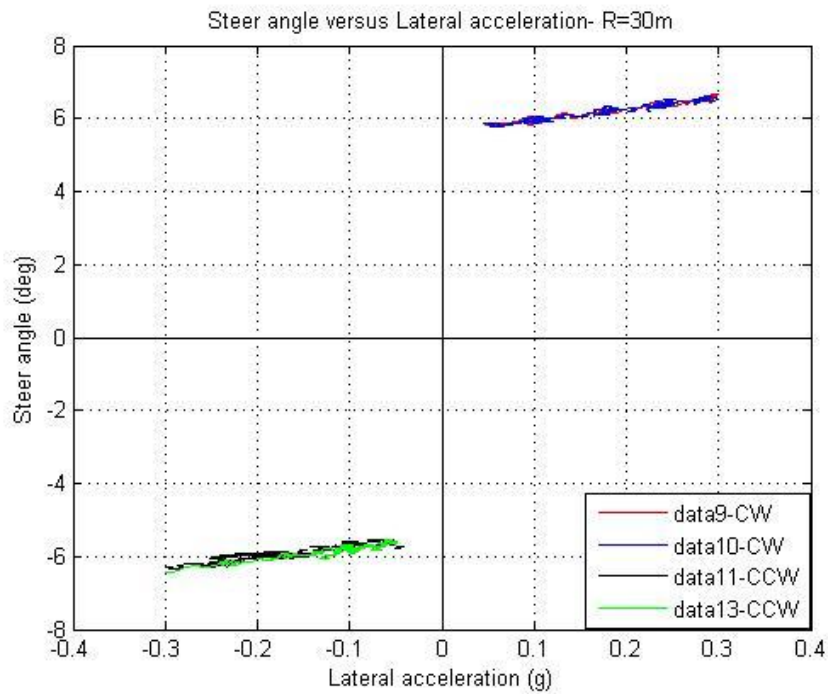


Figure 62: Steer Angle versus Lateral Acceleration (R=30m)

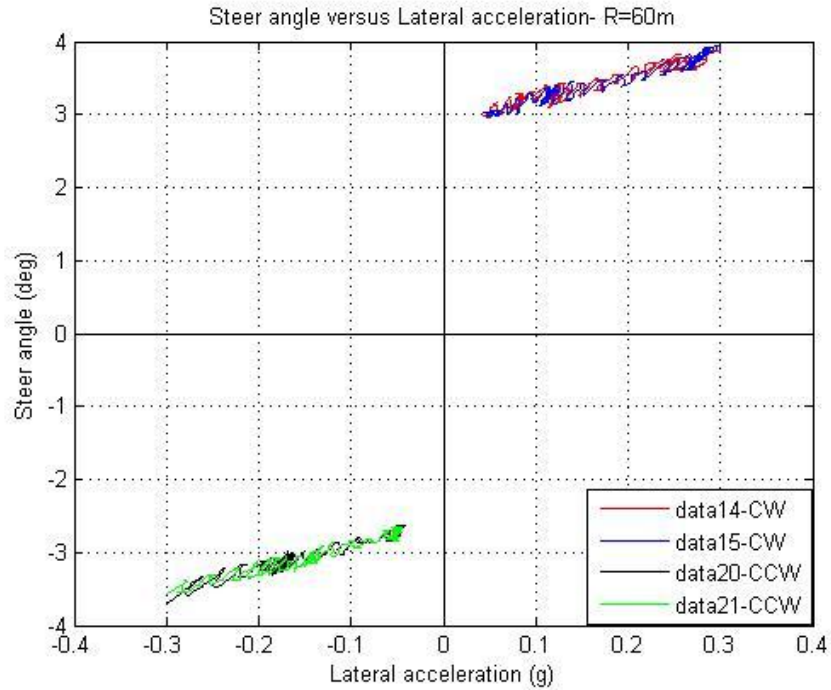


Figure 63: Steer Angle versus Lateral Acceleration (R=60m)

5.2.2 Constant Steer Angle Test

A speed profile with a rate of 0.625 km/h per second was used for the tests. The test was carried out for constant steer angle values of 60 degree, 90 degree and 120 degrees. Table (10) shows the values of the understeer gradients obtained for each test and the average values are given in Table (11). Figures (64), (65) and (66) show the plots of path curvature versus lateral acceleration for constant steer values of 60 degree, 90 degree and 120 degrees respectively.

	K (deg/g) CW		K (deg/g) CCW	
60 degree	2.83	3.02	2.86	2.84
90 degree	2.69	2.78	3.05	2.7
120 degree	2.48	2.47	2.73	2.37

Table 10: Constant Steer Test Understeer Gradient (Experimental)

Average values:

K (deg/g) 60 degree	K (deg/g) 90 degree	K (deg/g) 120 degree
2.89	2.81	2.51

Table 11: Constant Steer Test Average 'K' values

The average understeer gradient values obtained from these tests are shown in Table (11). These values are seen to decrease with an increase in the steer angle value.

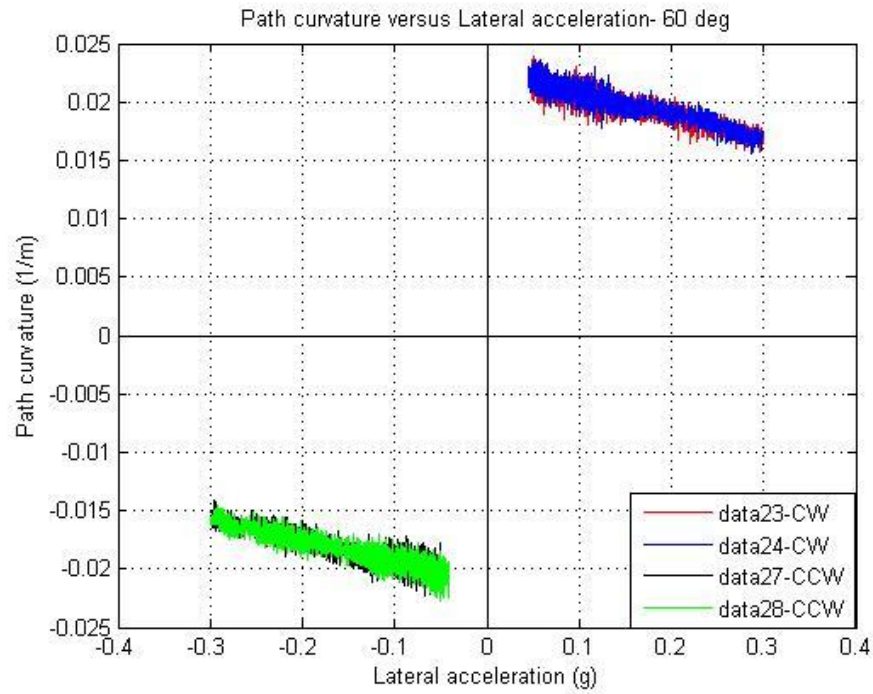


Figure 64: Path Curvature versus Lateral Acceleration (60 deg)

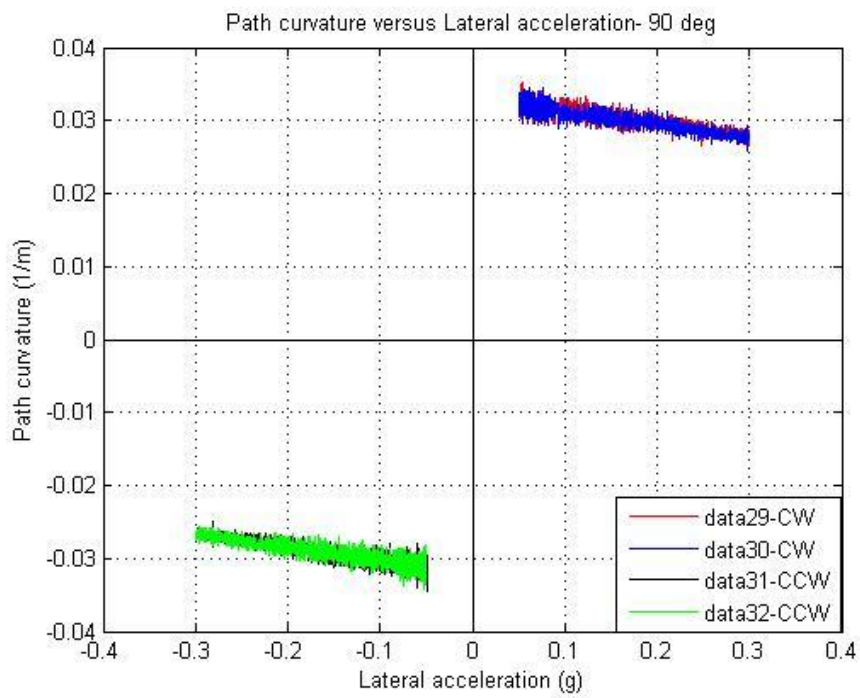


Figure 65: Path Curvature versus Lateral Acceleration (90 deg)

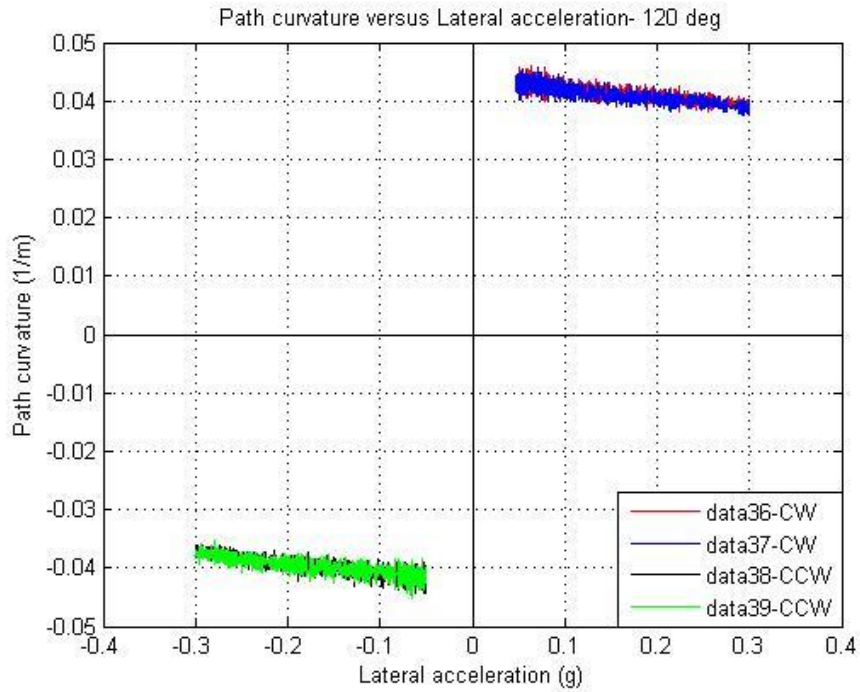


Figure 66: Path Curvature versus Lateral Acceleration (120 deg)

5.2.3 Constant Speed Test

A steer profile with a steering rate of 5 deg/sec was used for this test. The test was carried out for constant speed values of 9 m/s, 13 m/s and 18 m/s. Table (12) gives the understeer gradient values obtained for each test whereas Table (13) gives the average values. Figures (67), (68) and (69) show the plot of steer angle versus lateral acceleration for constant speeds of 9 m/s, 13 m/s and 18 m/s.

	K (deg/g) CW		K (deg/g) CCW	
V = 9 m/s	5.05	5.71	3.28	6.77
V = 13 m/s	4.38	3.99	4.26	3.33
V = 18 m/s	3.27	3.59	3.45	3.49

Table 12: Constant Speed Test Understeer Gradient (Experimental)

Average values:

K (deg/g) V = 9 m/s	K (deg/g) V = 13 m/s	K (deg/g) V = 18 m/s
5.2	3.99	3.45

Table 13: Constant Speed Test Average 'K' values

The average understeer gradient values obtained for these tests are shown in Table (13). These values are seen to decrease with an increase in the values of speed.

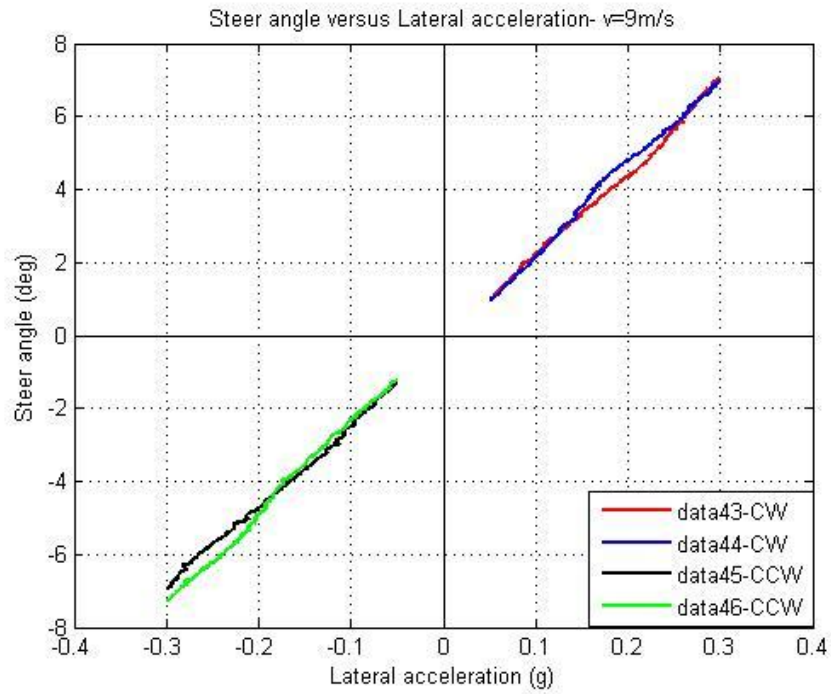


Figure 67: Steer Angle versus Lateral Acceleration ($V = 9 \text{ m/s}$)

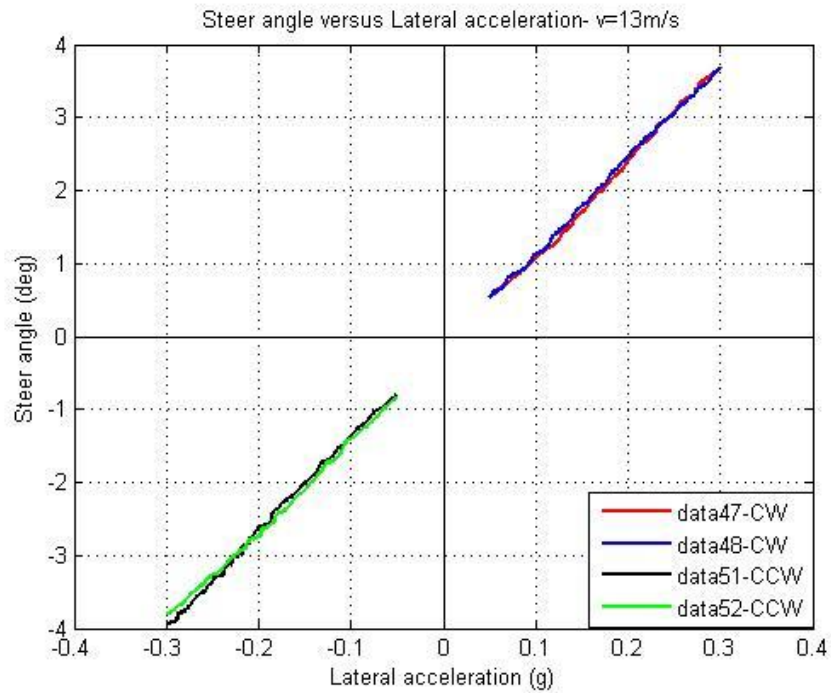


Figure 68: Steer Angle versus Lateral Acceleration ($V = 13 \text{ m/s}$)

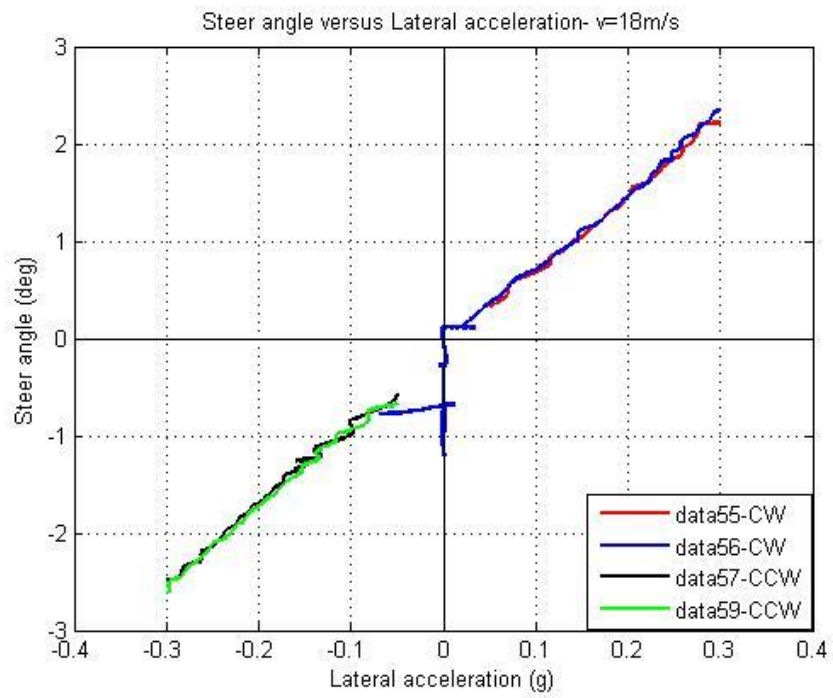


Figure 69: Steer Angle versus Lateral Acceleration ($V = 18 \text{ m/s}$)

CHAPTER 6

6 RESULTS AND DISCUSSION

6.1 Background

This chapter discusses the results obtained from the experimental and simulation understeer tests. A comparative study of the average understeer gradient values from the three tests has been made. Also, the contributions to understeer gradient from different have been calculated. The 2003 Ford Expedition is used as a model for these calculations due to the availability of the measurement data for computing the required parameters. The recommendations to this work have also been discussed in the last section.

6.2 Constant Radius Test

The results obtained from the experimental and simulation constant radius tests are discussed in this section.

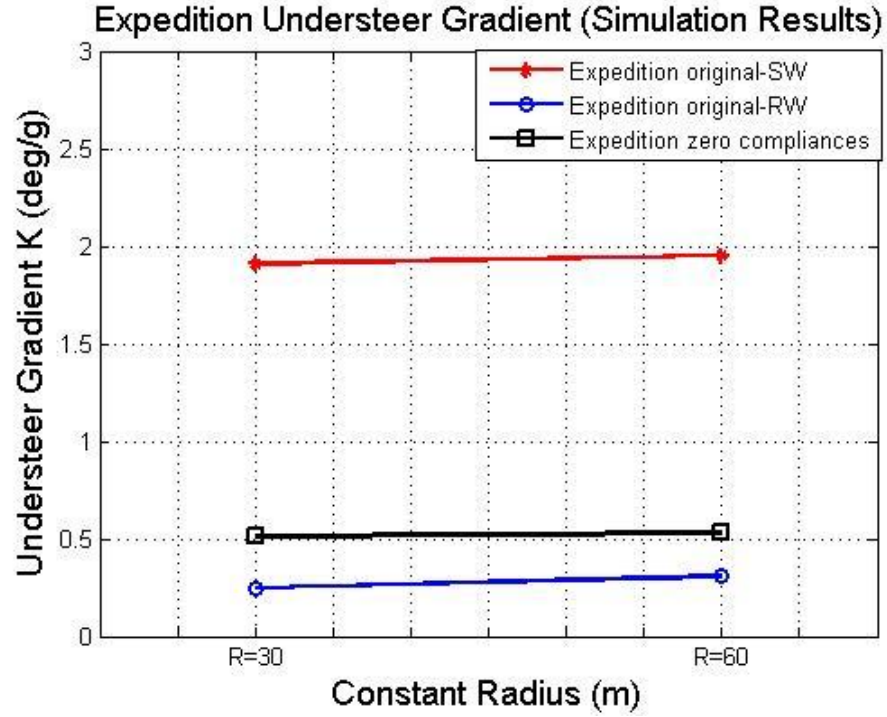


Figure 70: Expedition Understeer Gradient (Constant Radius Test)

Figure (70) shows the understeer gradient values for the constant radius test. The plot shows the data for the simulation results for Ford Expedition for radii of 30 m and 60 m. There are three sets of data in the plot. The first is the understeer values calculated from the steering wheel angles for the original Expedition model. The second one is the understeer values calculated from road wheel angles for the original Expedition model. The third is the understeer values calculated for Expedition model with zero compliances. These three sets of result show that there is an increase in the value of understeer gradient with increase in the value of the radius of the curve.

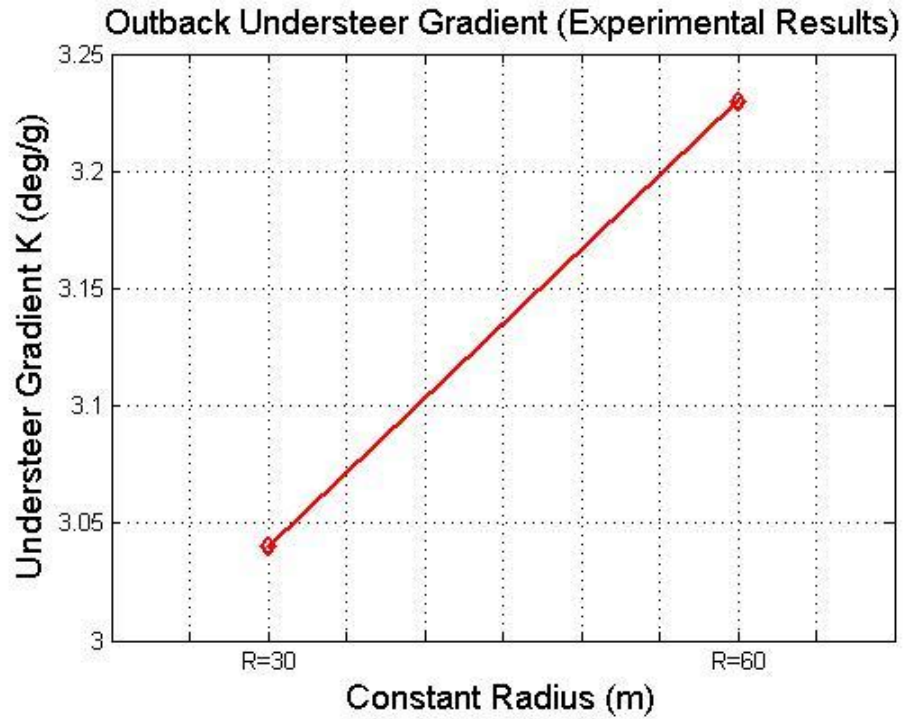


Figure 71: Outback Understeer Gradient (Constant Radius Test)

Figure (71) shows the understeer gradient values for the field test data for the constant radius test. These plots are for Subaru Outback. This plot agrees with the simulation results in showing that the value of understeer gradient increases with an increase in radius.

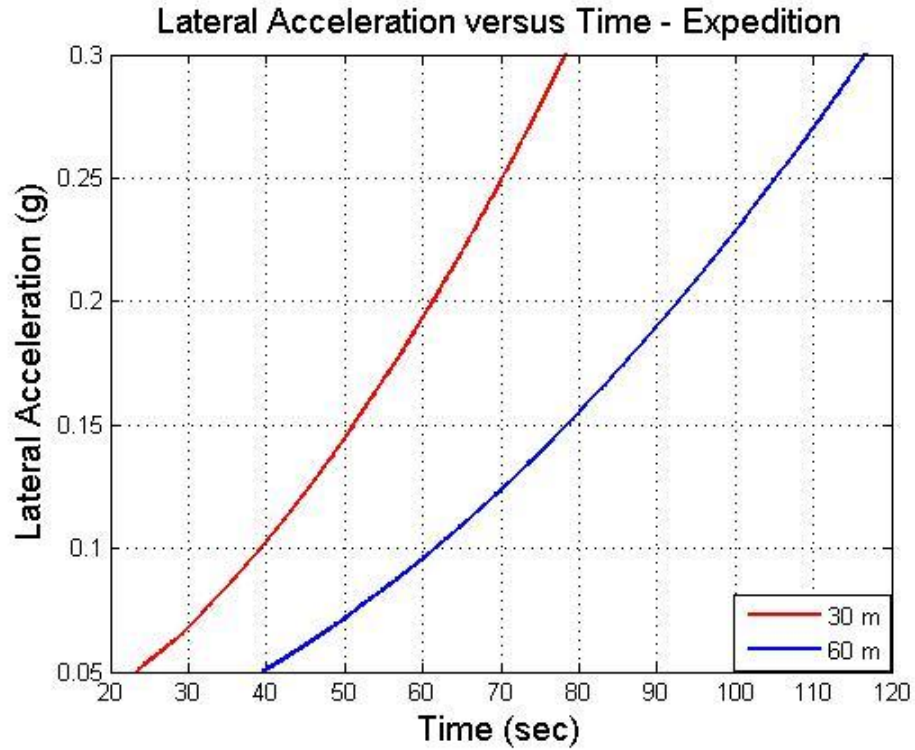


Figure 72: Lateral Acceleration versus Time Comparison (Constant Radius Test)

Figure (72) shows the comparison of lateral acceleration versus time for different radius values for Ford Expedition. The rate of build up of lateral acceleration is observed to be higher for the smaller radius.

6.3 Constant Steer Test

The results from the simulation and experimental constant steer test are discussed in this section.

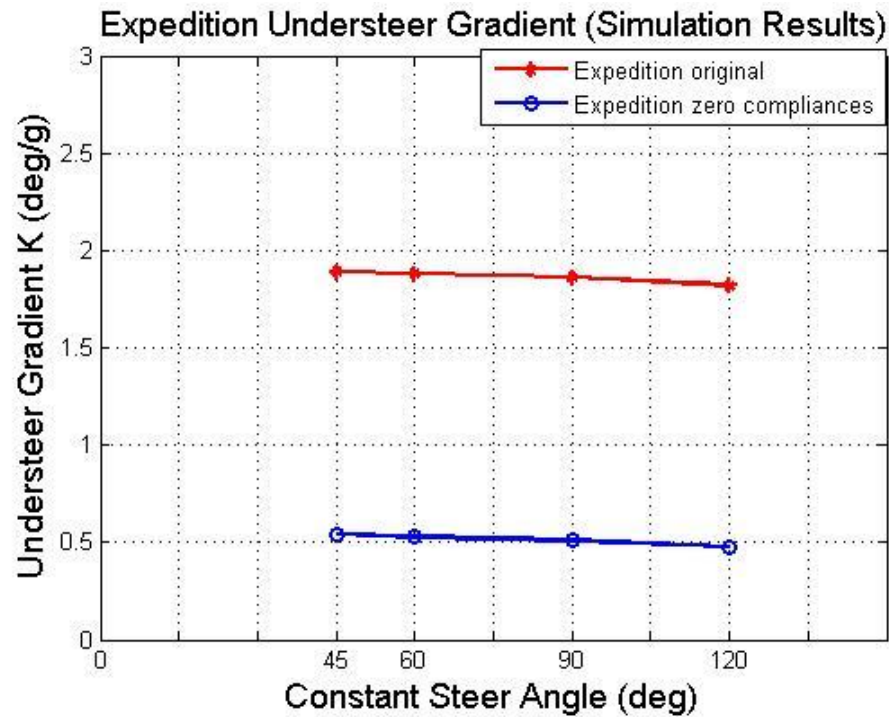


Figure 73: Expedition Understeer Gradient (Constant Steer Test)

Figure (73) shows the understeer gradient values from the simulation of constant steer test for the Ford Expedition. The results are for constant steering wheel angle values of 45 degree, 60 degree, 90 degree and 120 degree. The first set of data shown in the plot represents the understeer values obtained for the original Expedition model. The second set of data represents the results obtained for the Expedition model with zero

compliances. Both these sets show that the understeer gradient value decreases as the value of steer angle increases.

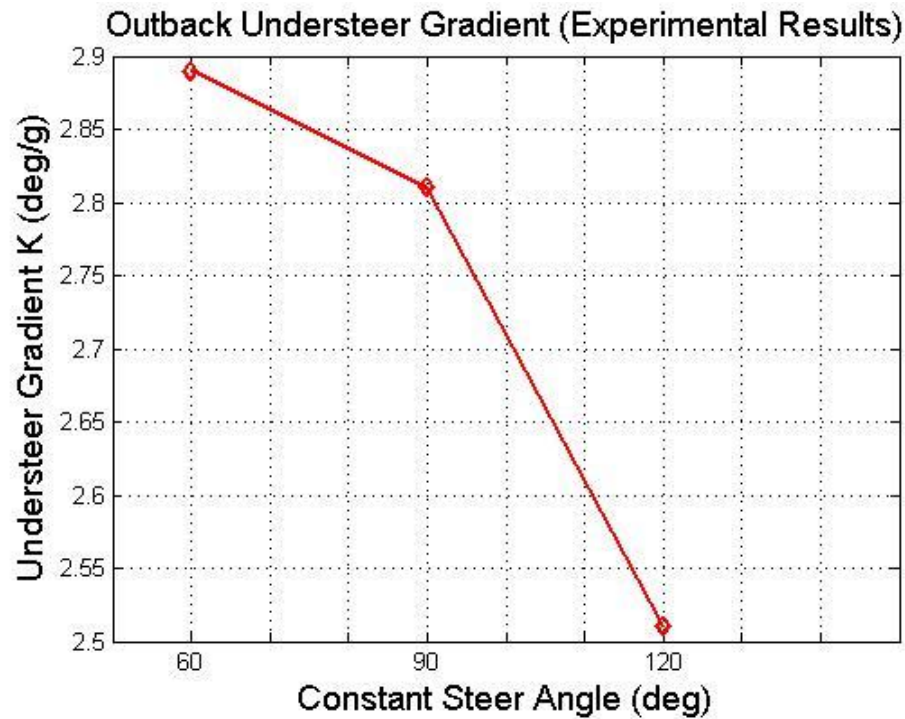


Figure 74: Outback Understeer Gradient (Constant Steer Test)

Figure (74) shows the results from the constant steer test field test data for Subaru Outback. The plot shows the understeer gradient values for constant radii of 60 degree, 90 degree and 120 degree. The understeer gradient values for the field test data are also seen to decrease with an increase in steer angles.

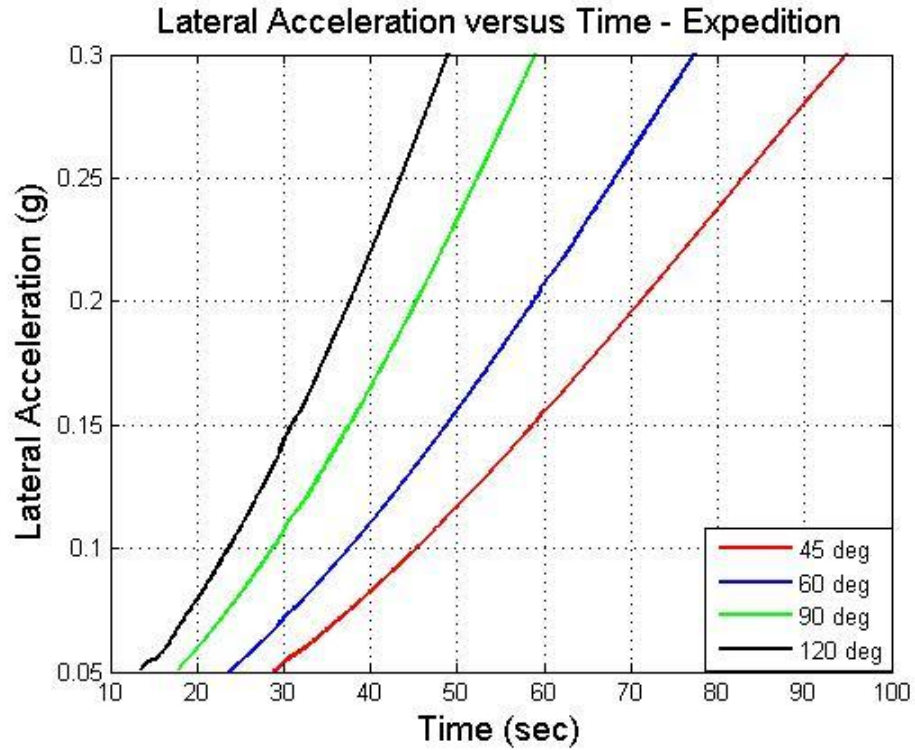


Figure 75: Lateral Acceleration versus Time Comparison (Constant Steer Test)

Figure (75) shows the comparison of rate of lateral acceleration for different steer angle values for Ford Expedition. The rate of lateral acceleration increases as the steer angle increases.

6.4 Constant Speed Test

This section discusses the results obtained from simulation and experimental constant speed tests.

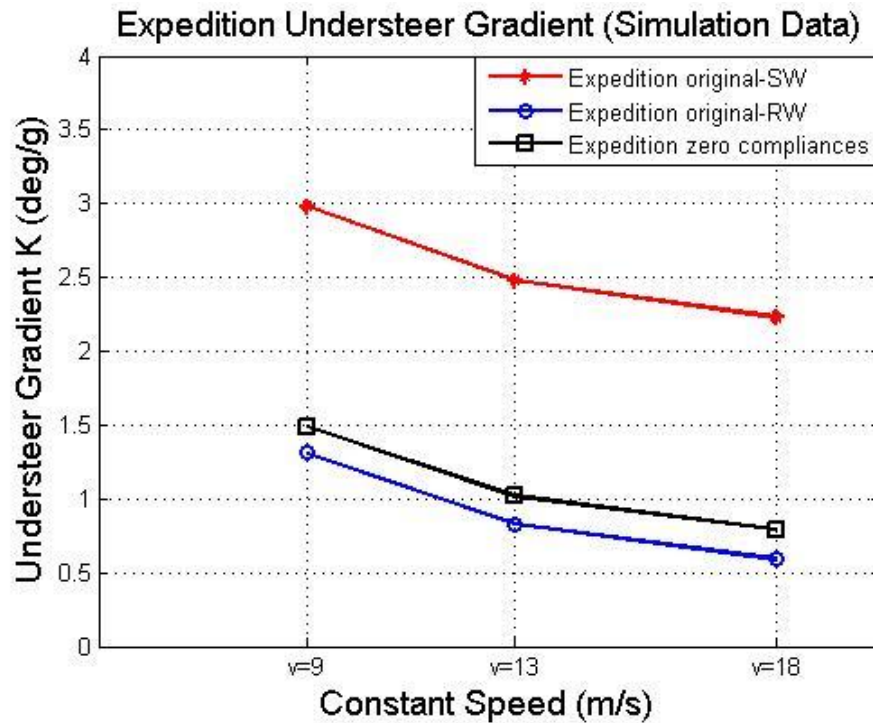


Figure 76: Expedition Understeer Gradient (Constant Speed Test)

Figure (76) shows the results for the constant speed test simulation for Ford expedition. The plot shows the understeer gradient values for constant speed values of 9 m/s, 13 m/s and 18 m/s. There are three sets of data shown in the plot. The first set represents the understeer gradient values calculated from steering wheel angle for the original Expedition model. The second set represents understeer values calculated from

the road wheel angles for the original Expedition. The third set represents understeer values obtained for the zero compliance Expedition model. The understeer gradient values are seen to decrease with an increase in speed.

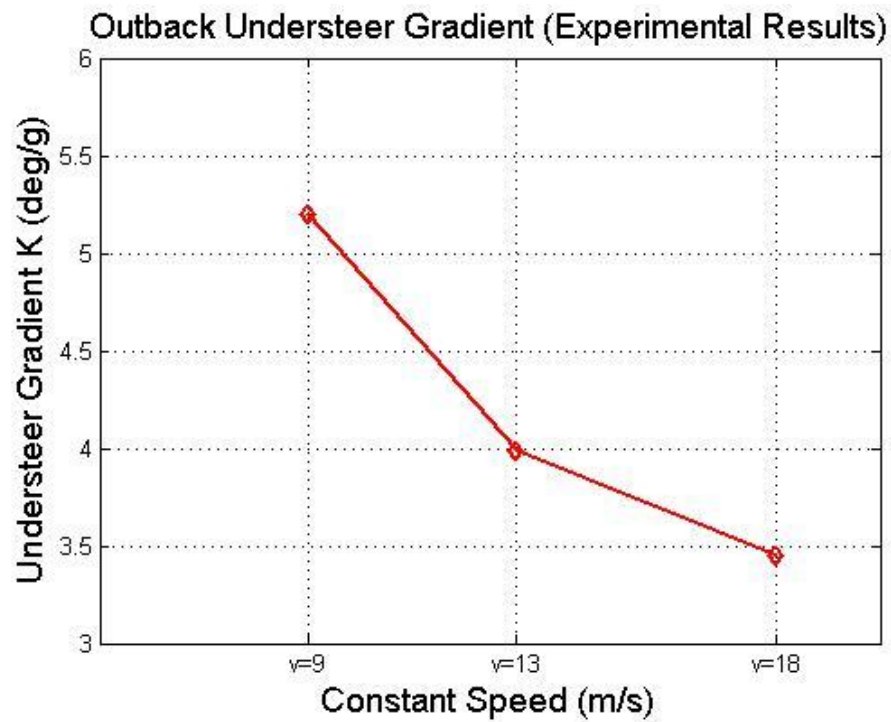


Figure 77: Outback Understeer Gradient (Constant Speed Test)

Figure (77) shows the understeer values for the Subaru Outback constant speed test. The understeer gradient value is seen to decrease with an increase in speed. This trend agrees with the simulation results.

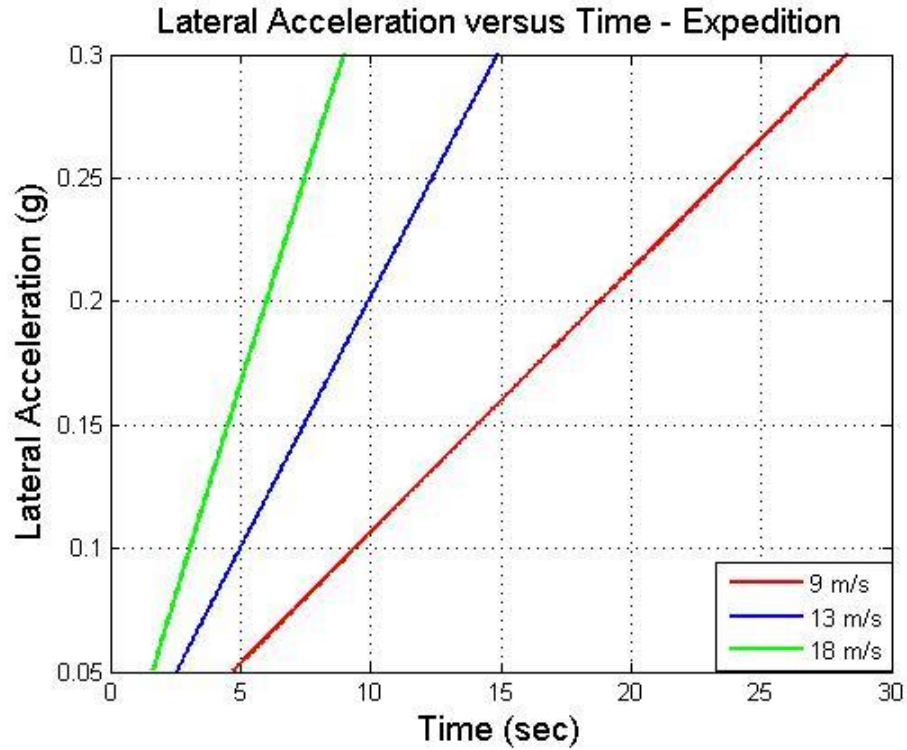


Figure 78: Lateral Acceleration versus Time Comparison (Constant Speed Test)

Figure (78) shows a plot of lateral acceleration versus time for different values of speeds. The rate of lateral acceleration is seen to increase with increase in speed.

6.5 Effect of system compliances

System compliances have a significant influence on the understeer gradient values. This was observed from the large difference in understeer gradient values derived from steering wheel angles and that derived from road wheel angles. To investigate this further, the existing Expedition vehicle model was modified to eliminate the effect of system compliances. From the tests for this modified model, it was observed that the

understeer gradient values obtained for the zero compliance Expedition model differ largely from the understeer gradient values obtained for the original Expedition model.

The constant steer angle test is executed by locking the steering wheel at a constant steer angle value. This causes the steering wheel angle to remain constant for the test but due to the effect of system compliances, the road wheel angles are not constant through the test. The road wheel angles were measured for the CarSim simulation for constant steer test. Simulation gives the values of road wheel steer angles whereas it is not possible to measure road wheel angle data for field tests. The SAE J266 formulation for calculating the understeer gradient for the constant steer test does not involve the steer angles and thus this effect is not reflected in the understeer gradient values. However, the understeer gradient values for the zero compliance Expedition model are different from the ones for the original model thus showing the system compliance effects. There is a need for better formulation and procedure for measuring the understeer gradient for the constant steer test. The system compliance effects need to be taken into consideration while formulating the procedure for measuring understeer gradient.

6.6 Overall Understeer Gradient Value

For both the experimental and simulation constant radius method, the understeer gradient is lower for the radius of 30 m i.e. the smaller radius. The rate of lateral acceleration for this radius is higher. Therefore, for the constant radius method, the understeer gradient value decreases as the rate of lateral acceleration increases.

For both the experimental and simulation constant steer method, the understeer gradient is lowest for the steer angle of 120 degrees. The rate of increase of lateral acceleration is highest for this condition. Therefore, for the constant steer method, the understeer gradient value decreases as the rate of lateral acceleration increases.

For both the experimental and simulation constant speed method, the understeer gradient is lowest for the speed of 18 m/s. The rate of increase of lateral acceleration is highest for this speed. From both the experimental and simulation results of all three understeer methods, it is observed that a lower understeer gradient value is obtained if the rate of build up of lateral acceleration is higher. Therefore, the value of understeer gradient decreases as the rate of lateral acceleration increases.

Figure (79) shows the understeer values for the Subaru Outback from the field test calculations. It gives the individual test values average value for each test method. Figure (80) shows the understeer values for Ford Expedition from the CarSim test calculations. For the plot, 'CRT' refers to constant radius test, 'CST' refers to constant steer test and 'CSP' refers to constant speed test.

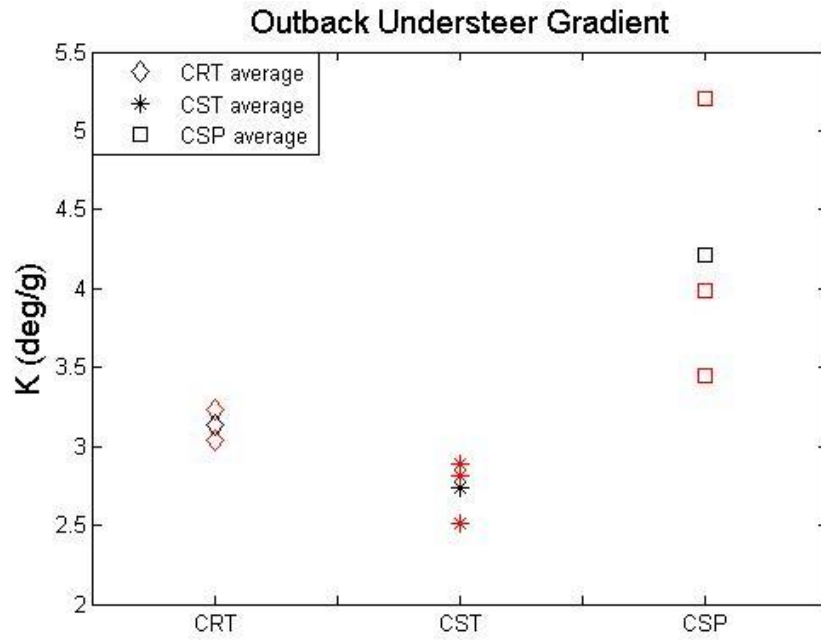


Figure 79: Understeer Gradient (Experimental)

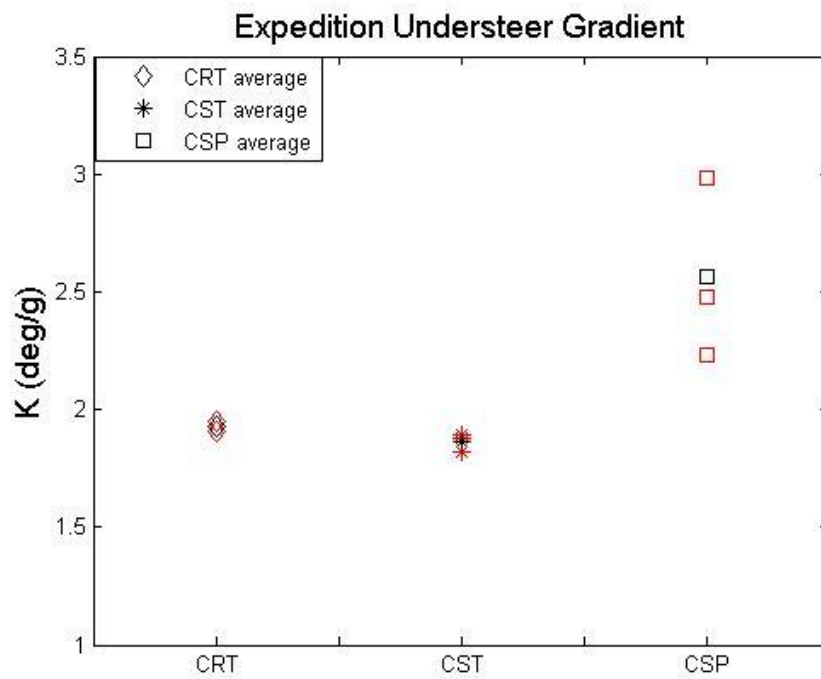


Figure 80: Understeer Gradient (Simulation)

For both the experimental and simulation results, the understeer gradient values obtained for the constant speed method are the highest and have the most scatter. The understeer gradient values for the constant radius method and constant steer method are close to each other, and the constant steer method gives the lowest values.

From the results for the three tests, it can be observed that the understeer values for the Expedition with zero compliances lie between the values obtained for original Expedition steer wheel angle values and original Expedition road wheel values.

6.7 Contributions to Understeer Gradient from Different Sources

Equation (6.1) gives the theoretical contributions to understeer gradient from seven different sources. These sources are described by Equation (6.1) and their definitions are provided below [1].

$$K = K_{tire} + K_{lt} + K_{rs} + K_{lcs} + K_{camber} + K_{strg} + K_{at} \quad (6.1)$$

For the 2003 Ford Expedition, these components were calculated from the vehicle parameter measurement data. The following section shows the contribution of each component.

Tire Cornering Stiffness:

$$K_{tire} = \frac{W_f}{C_{\alpha f}} - \frac{W_r}{C_{\alpha r}} \quad (6.2)$$

Where W_f - Load on front axle = 2895.70 lb
 W_r - Load on rear axle = 2818.30
 $C_{\alpha f}$ - Front cornering stiffness = 405.20 lb/deg
 $C_{\alpha r}$ - Rear cornering stiffness = 410.68 lb/deg

From the measurement data for Ford Expedition, we have the values for the necessary parameters [4].

Therefore,

$$K_{tire} = 0.28 \text{deg/ } g$$

Lateral Load Transfer:

$$K_{lt} = \frac{W_f}{C_{\alpha f}} 2b \frac{\Delta F_{zf}^2}{C_{\alpha f}} - \frac{W_r}{C_{\alpha r}} 2b \frac{\Delta F_{zr}^2}{C_{\alpha r}} \quad (6.3)$$

Where

$$F_{yf} = [C_{\alpha f} - 2b\Delta F_{zf}^2] \alpha_f \quad (6.4)$$

And

$$F_{yr} = [C_{\alpha r} - 2b\Delta F_{zr}^2] \alpha_r \quad (6.5)$$

Where ΔF_{zf} - front lateral load transfer = 188.24 lb

ΔF_{zr} - rear lateral load transfer = 145.27 lb

b - second coefficient in the cornering stiffness polynomial = 7.67 e-05 1/lb-deg

F_{yf} - front lateral force lb

F_{yr} - rear lateral force lb

Therefore,

$$K_{lt} = 0.01 \text{deg/ } g$$

Roll Steer:

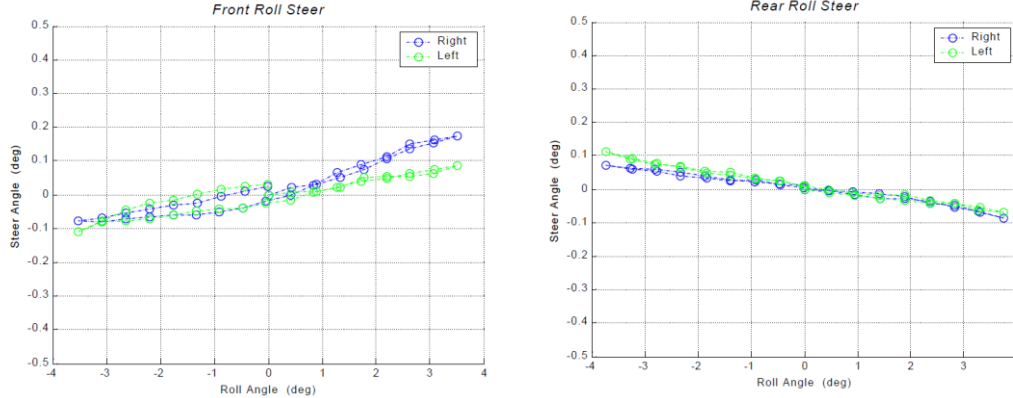


Figure 81: Roll Steer Characteristics

$$K_{rs} = (\varepsilon_f - \varepsilon_r) \frac{d\phi}{da_y} \quad (6.6)$$

Where ε_f - front roll steer coefficient = 0.033 deg/deg

ε_r - rear roll steer coefficient = -0.025 deg/deg

$\frac{\partial\phi}{\partial a_y}$ - roll angle gradient with respect to lateral acceleration = 4.56 deg/g

Therefore,

$$K_{rs} = 0.26 \text{ deg/g}$$

Lateral Force Compliance Steer:

$$K_{lfs} = A_f W_f - A_r W_r \quad (6.7)$$

Where A_f - front lateral force compliance steer coefficient = 5.42 e-04

A_r - rear lateral force compliance steer coefficient = -1.67 e-04

Therefore,

$$K_{lfs} = 1.02 \text{ deg/g}$$

Camber Thrust:

$$K_{camber} = \left(\frac{C_{\gamma f}}{C_{\alpha f}} \frac{\partial \gamma_f}{\partial \phi} - \frac{C_{\gamma r}}{C_{\alpha r}} \frac{\partial \gamma_r}{\partial \phi} \right) \frac{\partial \phi}{\partial a_y} \quad (6.8)$$

Where,

$$F_y = C_{\alpha} \alpha + C_{\gamma} \gamma$$

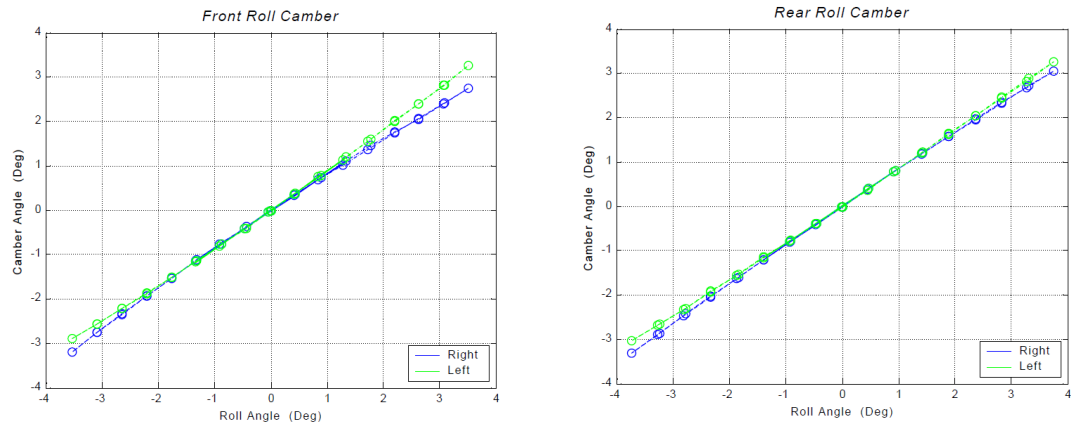


Figure 82: Camber Gradient

Where $C_{\gamma f}$ - front camber stiffness = 20.99 lb/deg

$C_{\gamma r}$ - rear camber stiffness = 20.46 lb/deg

$\frac{\partial \gamma_f}{\partial \phi}$ - front camber gradient = 0.8 deg/deg

$\frac{\partial \gamma_r}{\partial \phi}$ - rear camber gradient = 0.833 deg/deg

Therefore,

$$K_{camber} = -2.37e-004 \text{ deg/g}$$

Steering System Compliance:

$$K_{strg} = W_f \frac{rv + p}{K_{ss}} \quad (6.9)$$

Where r - wheel radius = 0.373 m
 ν - caster angle = 0.129 rad
 p - pneumatic trail = 0.048 m
 K_{ss} - steering stiffness = 3690 N-m/deg

Therefore,

$$K_{strg} = 0.34 \text{ deg/ } g$$

Aligning Torque:

$$K_{at} = W \frac{p}{L} \frac{C_{\alpha f} + C_{\alpha r}}{C_{\alpha f} C_{\alpha r}} \quad (6.10)$$

Therefore,

$$K_{at} = 0.28 \text{ deg/ } g$$

The addition of all these components gives the total understeer gradient

$$K = 0.28 + 0.01 + 0.26 + 1.02 + -2.37e-04 + 0.34 + 0.28$$

Therefore,

$$K = 2.19 \text{ deg/g}$$

From the calculations, we can observe that the contribution of the tire cornering stiffness to the understeer gradient is small compared to some other components.

The expression for understeer gradient for the bicycle model contains only the effect due to the tire cornering stiffness. This is because the bicycle model is only based on the tire slip angles and it does not include any compliances or kinematics effects. For a

full vehicle model, the contributions to the understeer gradient due to other sources become predominant due to the geometry and force balance for the real vehicle. Thus, the understeer gradient expression for bicycle model has simplifications and does not include the effect of system compliances which affect the real vehicle.

CHAPTER 7

7 CONCLUSIONS AND FUTURE WORK

7.1 Conclusions

The understeer gradient values obtained from the steering wheel angles divided by steering ratio were different than those obtained from the roadwheel angles. The effect of system compliances has a significant contribution to the total understeer gradient of the vehicle as seen from the understeer gradient calculations for the 2003 Ford Expedition.

The current procedure for executing the constant steer understeer test and the methodology used for calculation of understeer gradient does not take into account the change in the roadwheel angles during the course of the test. Hence, there is a need for a better procedure to address this issue. The understeer gradient calculation techniques based on the bicycle model have simplifications and do not take into account the effects of the system compliance.

From the comparison of the understeer gradient values obtained from all of the understeer tests, we note that the understeer gradient values obtained for the original Expedition model using roadwheel angles are different from those obtained for the zero

compliance Expedition model. To understand the cause of this difference, we need to investigate further the vehicle response and development of slip angles and lateral acceleration during these maneuvers.

From both the experimental and simulation results of all three understeer methods, it is observed that a lower understeer gradient value is obtained if the rate of build up of lateral acceleration is higher. The value of understeer gradient is seen to decrease as the rate of lateral acceleration increases.

7.2 Future Work

The work done for this thesis gave us a good idea about the different factors affecting understeer gradient. To gain more confidence in our findings it would be very helpful to have a comparison of experimental and simulation results for the same vehicle. Also, different types of vehicles can be tested to find a correlation between understeer gradient trends and type of vehicle for example SUV versus Sedan.

There is a strong correlation between the rate of change of lateral acceleration and change in the understeer gradient value. This can be studied further by comparing rates of lateral acceleration from different understeer tests.

The methods for calculation of understeer gradient based on bicycle model have simplifications. For a full vehicle model, the contributions to the understeer gradient due to other sources become predominant due to the geometry and force balance for the real vehicle. Further analysis is needed to investigate this effect. Another area worth studying would be the effect of vehicle roll motion on understeer gradient.

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